

Community Medicine

World of Revision

Marrow

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Instructions

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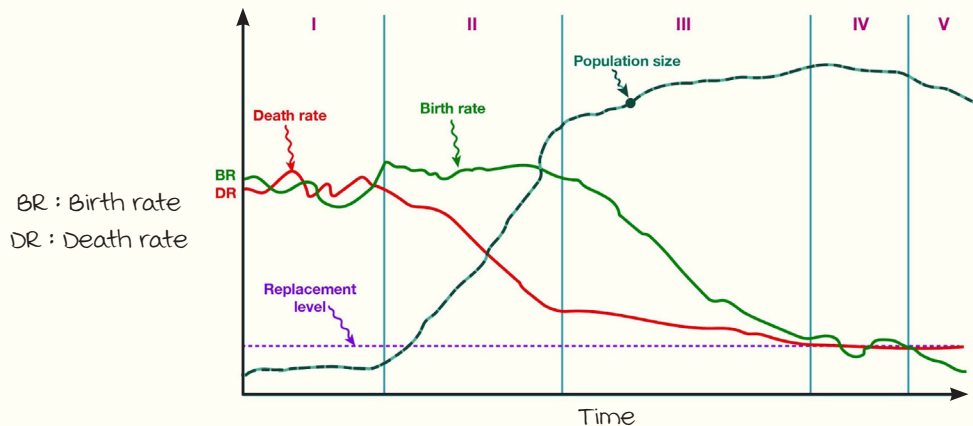
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DEMOGRAPHY

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Demographic Stages/Phases

00:00:45



Characteristics :

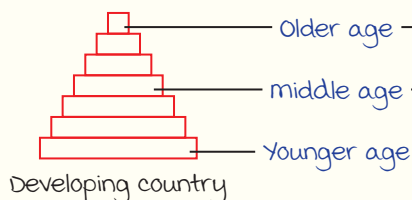
Stages	Phase	Birth rate (BR)	Death rate (DR)	Demographic gap (BR - DR)	Countries & features
I	High stationary	High	High	Narrow	-
II	Early expanding	High	Start declining	- Starts $\uparrow$ (D/t $\downarrow$ in DR) - max in late II	Demographic trap $\oplus$: $\uparrow$ BR $\rightarrow$ Hard to $\downarrow$ population
III	Late expanding	Start declining	Slow declining	Starts $\downarrow$ (D/t $\downarrow$ in BR)	- India, China - Demographic gift $\oplus$
IV	Low stationary	Slow declining	Slow declining	Low stable	- Demographic gift $\oplus$ - Developed countries
V	Declining	Slow declining	very low declining/stable	minimum, negative value, BR < DR	- Japan, Norway - Demographic liability $\oplus$

Age Pyramids :

X axis : Population ; Y axis : Age.

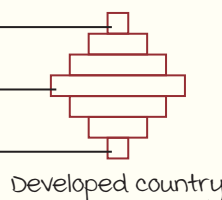
Broadbase pyramid :

$\uparrow$ BR



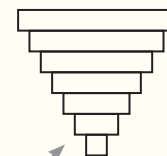
Broad belly/Constricting pyramid :

- $\downarrow$ BR.
- Economically productive.
- $\uparrow$ middle age group.
- max demographic gift/dividend, bonus.



Inverted pyramid :

- $\uparrow$ Older population : $\uparrow$ Dependency
- Demographic burden/liability
- $\downarrow$ DR



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Annual Growth Rate :

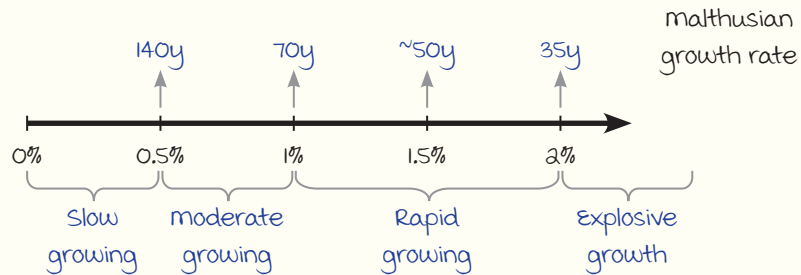
$$\frac{BR - DR}{10}$$

10

Population
doubling in :

Annual
growth rate :

Category :

Fertility Indicators

00:15:35

Fertility & Reproductive Rates 🌐 :

	Description
General fertility rate (GFR)	- No. of children a female bears during her reproductive years $\frac{\text{Total no. of live births}}{\text{WRA}} \times 1000$
Total fertility rate (TFR)	- No. of children a female bears during her reproductive years taking ASFR into consideration $5 \times \frac{\sum \text{ASFR}}{1000}$ - Best indicator for complete family size - Epidemiological, planning, budget indicator - Target : < 2.1
Gross reproduction rate (GRR)	- No. of daughters a female will bear during her reproductive years in ASFR $5 \times \frac{\sum \text{ASFR for female live births}}{1000}$
Net reproduction rate (NRR) : Best indicator	- No. of daughters a newborn child will bear during her reproductive years in ASFR and ASMR - Best indicator for impact of national family planning programme - Target : 1 (Replacement level)

WRA : Women in reproductive age group (15 to 49 years) in same area & same year.

ASFR : Age specific fertility rate.

ASMR : Age specific mortality rate.

Couple Protection Rate (CPR) :

- Contraceptive methods to be provided for eligible couples.
- Field/operational indicator.

Eligible couples : Couples with women in reproductive age (15 - 49 yrs).

Targets of national family planning/welfare program :

$CPR > 60\% \rightarrow TFR < 2.1 \rightarrow NRR = 1 \rightarrow$ Stable population.

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Survey Methods

00:25:55

Civil Registration System (CRS) :

- Birth & death :
Register within 21 days.
- under ministry of Home Affairs (moHA).

National Family Health Survey (NFHS) :

- Conducted by moHFW (ministry of Health & Family Welfare).
- Evaluated every 5 years.
- Latest NFHS : NFHS-5, 2019 - 2020.

Sample Registration System (SRS) :

- under ministry of Home Affairs (moHA).
- Annually reporting.
- most sensitive.

Uses :

- Prevalence of diseases :
 - Anemia.
 - DM.
 - HTN.
 - malnutrition.
- Sex ratios.
- utilization rates :
 - ANC coverage.
 - Immunisation coverage.
- Literacy rates.

Components evaluated :

- Fertility : Crude birth rate (CBR).
- mortality :
 - Crude death rate (CDR).
 - Infant mortality rate.
 - maternal mortality ratio.
- Natural growth rate of country/state.

Census :

- Evaluates demographic structure.
- Evaluated every 10 yrs.

1st census : 1881 (Completed & officially documented).

Last census : 2011.

Next census : 2026.

- methods :
 - Defacto : Based on as & where a person is present.
 - Dejure (m/c) : Based on jurisdiction.
- Completely online.
- Also includes : Homeless, unorganised section.

Demographic Variables & Statistics

00:33:35

Demographic Variables :

- Births & deaths.
- migrations.
- Age at marriage.
- Female literacy.
- **Social mobility** : movement across socio - economic classes.

Demographic Statistics of India :

	Current status	Reference
CBR	18.4/1000 mYP	SRS Sept 2025
CDR	6.4/1000 mYP	
Natural growth rate (CBR - CDR)	12	
Total dependency ratio	48.17%	Park Ed. 28
Sex ratio	1020 (Skewed)	NFHS-5
	917	Child sex ratio at birth
Eligible couple rate	15 - 18%	2011 census
Literacy rate	72 - 85%	NFHS-5
Couple protection rate	67%	NFHS-5
Total fertility rate	1.9	-

} Achieved targets

mYP : midyear population

Dependency ratio :

$$\frac{\text{Total dependent}}{\text{Total independent}} = \frac{\text{No. of persons aged } < 14 \text{ \& } > 65 \text{ years}}{\text{No. of persons aged } 15 - 64 \text{ years}} \times 100$$

Potential support ratio :

$$\frac{\text{No. of working age population (15 - 64 yrs)}}{\text{Dependent population}}$$

Population Statistics :

- Population size.
- Sex ratio :
No. of females for every 1000 males.
- **Dependency ratio.**
- **Density.**

Vital Statistics :

- Birth, death rate.
- Natural growth rate.
- Life expectancy at birth.
- mortality.
- Fertility rates.

CONTRACEPTION AND FAMILY WELFARE PROGRAM

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Family Planning Methods (FPM)/ Contraception

00:00:36

Eligible Couple vs. Target Couple 🌐 :

	Eligible couple	Target couple
Definition	Female : Reproductive age (15 - 49 yrs)	Completed family + atleast 1 live child + adopts permanent FPM
Ideal FPM	Contraceptives	Sterilization



National family planning programme

Failure Rate 🌐 :

Life table analysis :

Gold standard.

Pearl's index (PI) :

• m/c.

$$\bullet \text{ PI} = \frac{\text{No. of accidental pregnancies}}{\text{Total women years of exposure}} \times 100$$

$$= \frac{\text{Total no. of accidental pregnancies}}{\text{Total women months of exposure}} \times 100 \times 12.$$

Contraception	PI (Per 100 women years) 🌐
Condoms	2 - 20
IUD	0.5 - 1.5
OCP	< 0.1
Implants	< 0.05

Types of Contraceptives :

Reversible/temporary methods :

- Natural methods.
- Barriers.
- IUD.
- OCP.
- Implants.
- Gossypol.
- Centchroman.
- Emergency contraception.

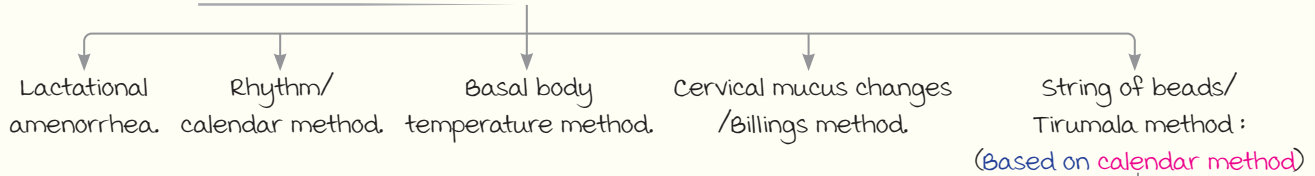
Irreversible/permanent methods :

- male sterilization (vasectomy).
- Female sterilization (Tubectomy).

----- Active space -----

Natural Methods & Barriers

00:07:03

Natural methods 🌐 :

Advantages :
used by illiterate females.

Disadvantages :
Not useful in :

- Irregular cycles.
- Short cycles (< 26 days).
- Long cycles (> 32 days).

Barriers :

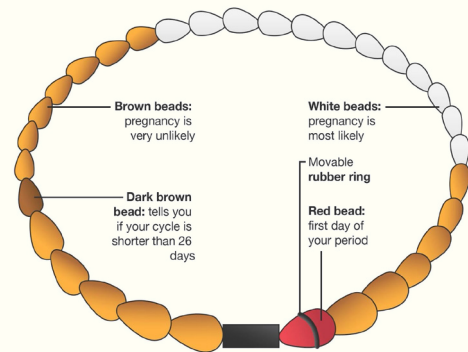
	male condom	Female condom
material	Latex	• Latex • Polyurethane (Earlier)
more common use	Contraception	STI prevention



Female condom



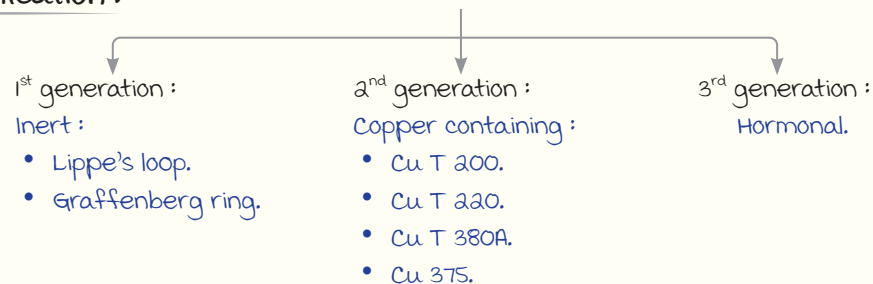
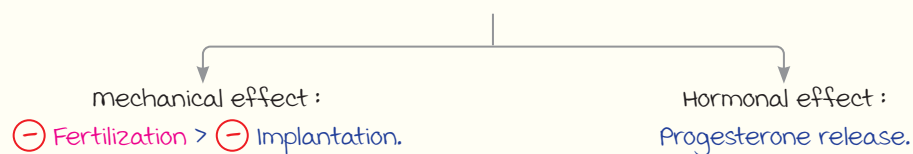
male condom



String of beads

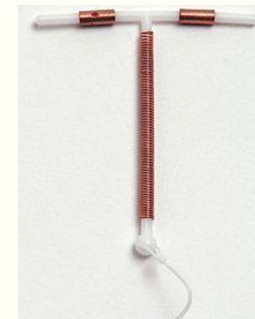
Intrauterine Devices

00:12:11

Classification :**mode of Action 🌐 :**

Copper IUD 🌐 :

	Cu T-380A	Cu 375 multiload device
Shape	T shaped	Inverted U shaped
Copper content	• 380 mm² • ⊕ on arms (A)	• 375 mm² • ⊖ on arms
Thread	Polyethylene	monofilament nylon (blue green)
Life span	10 years	3 - 5 years



CuT-380A

Hormonal IUD :

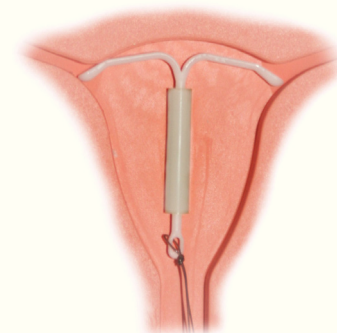
	Progestasert	mirena
Content	Natural progestin	Levonorgestrel
Life span	1 year	5 years
Total hormone load	38 mg	52 mg
Rate of secretion	65 mcg/day	20 mcg/day
Characteristic	m/c a/w ectopic pregnancy	Least side effects



Cu 375 multiload device

IUD Insertion 🌐 :

Time of insertion	Description
Intra-Cesarean	During C-section
Post placental	within 10 minutes of delivery
Post partum	10 mins - 48 hrs post delivery
Interval	6 weeks post delivery
Post surgical abortion	12 - 15 days post Sx
Non-pregnant female	within 10 days of start of menses (Ideal : First 5 days)



mirena

Contraindications :

Absolute contraindications :

- Suspected pregnancy.
- Pelvic inflammatory disease.
- vaginal bleed of undiagnosed etiology.
- Cancer cervix/uterus/adnexa or other pelvic tumors.

C/i for hormonal IUD : Previous ectopic pregnancy (Relative / general contraindication.)

----- Active space -----

Oral Contraceptive Pills (OCPs)

00:24:37

	mala-N	mala-D
Content	Ethinylestradiol 0.03 mg + Levonorgestrel 0.15 mg	
Pills	21 hormonal pills + 7 ferrous fumarate tablets	
Cost & delivery	Free	Paid, home delivery



mala-N

s/e :

- metabolic :
 - ↑Body weight (m/c).
 - Dyslipidemia.
 - Headache, migraine.
- CVS :
 - Thromboembolism.
 - CAD.
 - Stroke.
 - HTN.
 - DVT.
- malignancy :
 - Breast cancer (↑Risk during use) :
BRCA-1 mutation : ↑Risk by 8 - 24%.
 - Cervical cancer.
 - Hepatocellular adenoma.
 - Gall bladder disease.

Other Temporary Methods

00:28:41

Injectables/Implants 🌐 :

Depot medroxy progesterone acetate (DMPA) :

- Dose : 150 mg Im, 3 monthly.
- ——— Injection ———
2 weeks due date 4 weeks
No UPT

- s/e : Amenorrhea (On prolonged use).
- Return of fertility :
5 - 6 months after last injection.

Centchroman :

manufactured at CDRI, Lucknow.

Content : Ormeloxifene.

Dose : 30 mg tablets biweekly x 3 months

↓ f/b
weekly.

MOA : Selective estrogen receptor modulator (SERM).

s/e : Amenorrhea, Hot flashes.

Norethisterone enanthate (NET-EN) :

Dose : 200 mg, 2 monthly.



Antara (DMPA)



Centchroman : Chhaya (Earlier : Saheli)

Emergency Contraceptives :

----- Active space -----

	LNG tablets	IU Cu device
Dose	1.5 mg levonorgestrel, single dose	-
Time window (Post unprotected intercourse)	72 hours	5 days



LNG tablet

Tubectomy & Vasectomy

00:35:22

	Tubectomy 🌐	Vasectomy
Technique	modified Pomeroy's	Non scalpel
Cost	Expensive	Cheaper
Sterility	Immediately	- After 12 - 15 weeks - After 30 ejaculations
Feature	m/c contraceptive used (36%)	-

S/e of vasectomy 🌐 :

- Operative/psychological (m/c).
- Sperm granules :
 - Residual sperms proximal to vasectomy site.
 - Self-limiting.
- Spontaneous recanalization (< 0.05%).
- Auto immune response : 2 - 3 years post sx $\xrightarrow{40-50\%}$ Antisperm antibody.

Contraceptive of Choice (C^oOC) 🌐

00:39:03

Criteria	C ^o OC
Unmarried females	- Barrier methods (C^oOC) - Barrier methods + OCP (Ideal)
Delay 1 st child	OCP > IUCD
Delay 2 nd child	IUCD > OCP
Occasional meeting, married couple in separate cities	IUCD > OCP > Antara
Breastfeeding females	- POP/Chhaya : Anytime after delivery - Antara : 6 weeks post delivery - OCP : 6 months post delivery
Post delivery non-lactating females	- POP/Antara : Anytime post delivery - OCP : 3 weeks post delivery
All females (Irrespective of breastfeeding/not)	- IUCD : Within 48 hrs/6 wks post delivery - Female sterilization : Within 1 week/after 6 weeks post delivery - Emergency contraceptives : Only 4 weeks post delivery

----- Active space -----

Recent Advances

00:45:44

Mission Parivar Vikas (MPV) :

- Promote use of contraception among eligible couples .
- Includes :
 - Injectable (Antara).
 - Chhaya.
 - IUCD.
 - Female sterilization.
 - Mala-N.
 - Ezy pill.
 - Nirodh.

**ASHA Incentives :**

	Criteria	Incentives
Planned family	Delay of 1 st child till 2 yrs after marriage	Rs. 500
	Gap of at least 3 yrs b/w 1 st & 2 nd child	Rs. 500
	Adoption of permanent family planning method after 2 nd child	Rs. 1000
Home delivery	Condoms	Rs. 1 - 5/3 condoms
	OCPs	Rs. 3 - 5/1 packet

National Family Welfare Programme Updates :

Nishchay Kits : UPT for field testing.

Fixed day approach for sterilization :

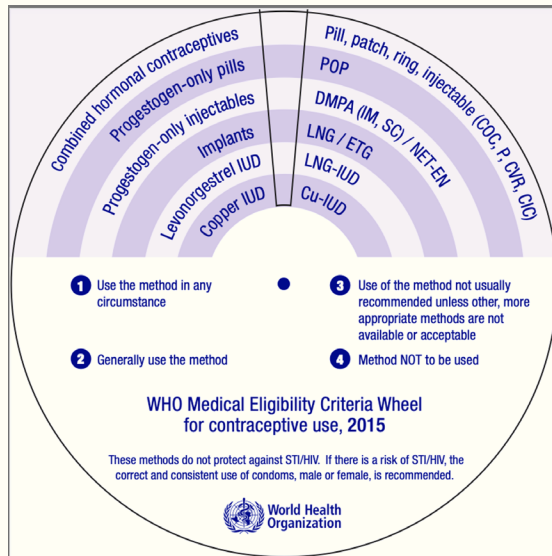
- PHC : monthly.
- CHC : Fortnightly.
- Sub-district hospitals : Weekly.
- District hospitals : Twice weekly.

**medical Eligibility Criteria (MEC) wheel**

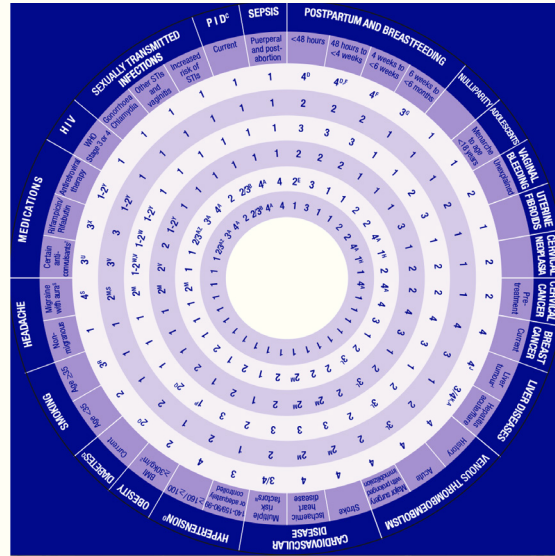
Given by WHO.

Inference :

Score	Inference
1	Can be used
2	Generally used with caution
3	Not used, may be used with caution
4	Not recommended



mEC wheel



----- Active space -----

miscellaneous :

- **Potential support ratio** : working population per 100 older population (9.9).
- Family size best given by **TFR**.
- m/c s/e of IUCD : Bleeding/spotting
- Lowest Pearl's index among IUCD : **Mirena**.
- Birth rate starts decreasing in Stage III.
- Death rate is lowest in Stage V.
- Demographic gift : **Stage III > Stage IV**.

----- Active space -----

HEALTH CARE SYSTEM IN INDIA

	NHm		ICDS (Integrated child development services)	CPHC (Comprehensive primary health care)
	NRHM (National rural health mission)	NUHM (National urban health mission)		
Under	MoHFW	MoHFW	MoWCD	Ayushman Bharat
Date	2005	2013	1975	2018

National Health Mission

00:01:43

Pillars :

1. Communitization.
2. Flexible financing.
3. Innovations in human resources.
4. Improve manpower capacity.
5. Setting standards & norms with monitoring.

NRHM :

Level	Per population		manpower (NUHM & NRHM)	No. of total staff
	• Hilly • Tribal • Forested	Plains		
District hospital	1 Every district		All clinical specialties : • Lab : Biochemistry, pathology, microbiology • Nurses • Blood banks • Lab & OT technicians • Programme manager	-
CHC	80,000	1,20,000	• Specialists (≥ 6) : medicine, OBG, surgery, paediatrics, ophthalmology, anesthesia • OT & X-ray technicians • Pharmacists • ANM • Nurses	• Basic : 45 - 46 • Desirable : 50 - 52

----- Active space -----

Level	Per population		manpower (NUHM & NRHM)	No. of total staff
	- Hilly - Tribal - Forested	Plains		
PHC	20,000	30,000	- medical officer (MO) - AYUSH MO - ANM - Lab technicians - Pharmacist - Health assistant (LHV)	- Basic : 13 - 14 - Desirable : 18 - 21
Subhealth centre	3000	5000	mPW → male → Female	3 - 4
ASHA	1000		1 : 1000	-

● under state govt.

● under central govt.

mPW : multipurpose worker

ASHA : Accredited social health activist

ANM : Auxiliary nurse midwife

HA : Health assistant

LHV : Lady health visitor (Obsolete term)

mowCD :

ministry of women & child development

moHFW : ministry of health & family welfare

ASHA :

Functions :

1. mother & child care (MCH) :

- Home visits.
- Counsel women on reproductive & child health.

2. Facilitate care to pregnant, malnourished, special needs individuals.

3. Primary basic medical care to all.

4. Depot holder : ORS, antibiotics, IFA tablets, FP methods.

5. Inform : Births, deaths (Registration by mPW, ANM).

6. Total sanitation campaign : Household toilets.

7. member of VHSN committee (Village health sanitation & nutrition) :

Organising monthly meets.

Selection criteria :

- Gender : Female.
- Age : 25 - 45 yrs.
- marriage ⊕ : Present/past.
- Education : ≥ 10th pass.
- Family : 1 - 2 children only.
- Permanent resident of the village.

----- Active space -----

Multipurpose Workers :

	male	Female
Functions	• NVBDCP : - malaria (Dx, Rx, slides) - Insecticide spray • Water quality : Chlorination • maintain cold chain • Records & registration maintenance	mCH : • Delivery • Immunization • Immediate newborn care • ANC, PNC • Family planning & related records
	• OPD	• Survey

NUHM :

- manpower & management norm same as in NRHM.
- **No sub-centres** seen.

Level	Population
District hospital	Every district
Urban CHC	2.5 L : Cities
	5 L : metro cities
Urban PHC	50,000
ANM centre (mCH care)	10,000
USHA	1000 - 2500 : 200 - 500 households

Types of Centres

00:20:40

SC, PHC, CHC :

	Subcentre (SC)		PHC		CHC		
	A	B	A (Low load)	B (High load)	Rural		Urban
					Non-FRU	FRU	FRU
Birth facility	⊖	⊕	< 20 deliveries/ month	> 20 deliveries/ month	⊕	⊕	⊕
No. of beds	0 - 1	≥ 2 - 3	6 - 8	≥ 8 - 10	30 - 50	50 - 100	50 - 100

First referral unit (FRU) :

- Promote emergency obstetric care.
- Safe & skilled delivery services.

District Hospital :

min : 1 bed/1000 population.

Ideal : 2 beds/1000 population.

Integrated Child Development Services Scheme

00:27:24

----- Active space -----

Under : **mowCD.**

Beneficiaries :

1. Children : 0 - 6 yrs.
2. Adolescent girls : 10 - 19 yrs.
3. Women of reproductive age : 20 - 49 yrs.
4. Pregnant & lactating women.



Organisation :

Levels	Population
CDPO : Block level	1,00,000
Anganwadi supervisor	25,000
Anganwadi centre	500 - 800
mini anganwadi : Hard to reach areas	150 - 300

CDPO : Child development project officer

Functions :

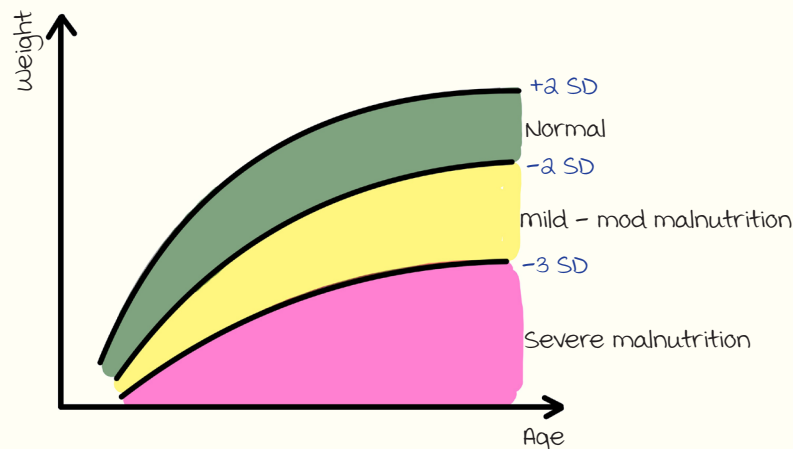
1. Supplementary nutrition : 1 meal/day
2. Non formal education.
3. Health care & health education.
4. Adolescent education.
5. MCH care.
6. **vitamin A prophylaxis.**
7. Facilitating immunization.

Beneficiary	Calories	Proteins
Child	500 KCal	12 - 15 g
Pregnant female	600 KCal	15 - 18 g
malnourished child	800 KCal	20 - 25g

Growth Charts :

AKA : mother & child protection card (MCP)/ICDS card.

Weight for age :



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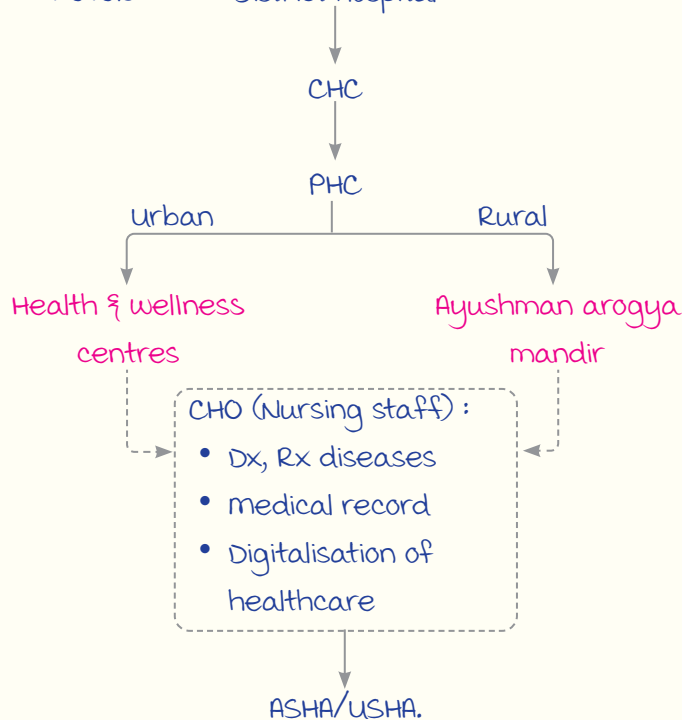
Ayushman Bharat

00:34:44

New IPHS Standards :

Comprehensive primary health care :

- Upgraded subcentre.
- Levels : District hospital



Polyclinics :

- Upgraded PHC.
- Specialist care in PHC in urban areas.
- Only OPD care.
- Population : 2 L - 5 L.

Continuum of care :

- Diagnostic.
- Therapeutic.
- Preventive.
- Promotive.

Waiting period : ≤ 30 mins.

Digitalization of health records.

ABHA ID : Ayushman Bharat ID.

(CHO : Community health officer).

Miscellaneous

00:39:24

Health Personnel :

	Per population norms
Nurses	1 : 5000
Health worker (MPW)	1 : 3000 - 5000
Health assistant	1 : 30,000
Pharmacists	1 : 10,000
Lab technicians	1 : 10,000
ASHA	1 : 1000

eHEALTH/Digital Healthcare :

Portal	Services	
eHospital	OPD services	
eSanjeevani	Telemedicine	
eRakt Kosh	Blood bank logistics	
eVIN	Vaccine logistics	
Kilkari	Immunization promotion	
NHP	Health guidelines provision	
mera aspataal	Hospital management	
m-Diabetes	Diabetes	
m-Cessation	Smoking cessation helpline	
NIKSHAY	TB notification	
NIKUSHT	Leprosy notification	
DAKSHATA	MCH	Training modules
UDISHA	AW workers	
SAKSHAM	Skill development & health personnel	

Availability of Kits :

Kit No.	Available
A, B, C	Subcentre
D	PHC
E - P	CHC

----- Active space -----

eVIN : Electronic vaccine intelligence network.

NHP : National health portal.

SAKSHAM : Stimulating advanced knowledge for sustainable health management.

Misc. Calculations :

ANC registration calculation :

- Pregnant women = LB + 10% of LB (Pregnancy wastage).
- Total no. of registration : > 50% of annual expected.

vaccines/drugs to procure :

- Number of vaccines = Beneficiary x dose x vaccine wastage factor (VWF).
- **VWF (multiplier) :**
 - BCG : 2.
 - MR : 1.3.
 - Other : 1.1.

PREVENTIVE OBSTETRICS

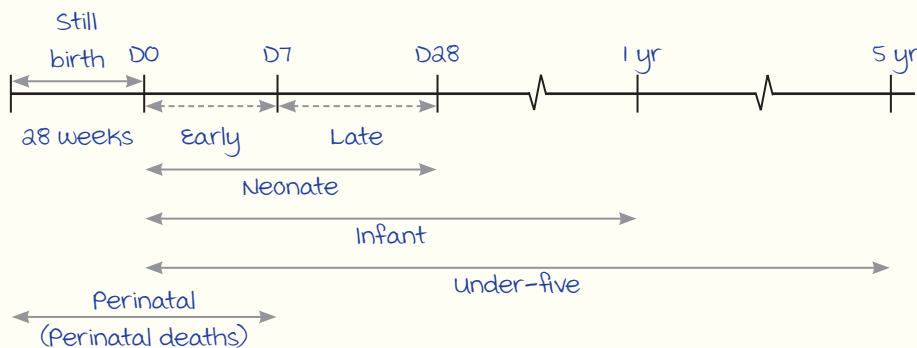
MCH Indicators

00:01:10

Indicators & their Significance :

- Perinatal mortality rate : PNMR → Accessibility, utilisation of health services (PNMR > IMR).
- Infant mortality rate : IMR → Overall development of a country.
- Neonatal mortality rate : NNMR.
- Still birth rate : SBR.
- Child death rate : CDR.
- Under-five mortality rate : USMR → Overall development of a country as per the **UN**.
- maternal mortality **ratio** : mMR → Quality of maternal/obstetric care

Chronological Nomenclature :



Still birth (SB) :

Late fetal death :

- > 28 weeks gestation (or)
- > 35 cm crown heel length (or)
- > 1 kg birth wt.

Important MCH Indicators 🌐 :

	Calculation	Current	Target (2030)	m/c cause
IMR	$\frac{ID}{LB} \times 1000$	25	-	• LBW • Prematurity
NNMR	$\frac{NnD}{LB} \times 1000$	19	< 12	
SBR	$\frac{SB}{\text{Total births}} \times 1000$	5	-	Antepartum hemorrhage
PNMR	$\frac{(SB + \text{ENnD})}{\text{Total births}} \times 1000$	-	-	Fetal + maternal factors
USMR	$\frac{USD}{LB} \times 1000$	29	< 25	• LBW • Prematurity
mMR	$\frac{mD}{LB} \times 1,00,000$	88 per lakh LB	< 70	PPH (uterine atony), anemia, unskilled delivery

ENnD : Early NnD
 ID : Infant death
 NnD : Neonatal death
 LB : Live births
 USD : Under-5 death
 mD : maternal death

	maternal mortality ratio	maternal mortality rate
Calculation	$\frac{MD}{LB} \times 1,00,000$	$\frac{MD \times 1,00,000}{\text{Women in reproductive age (15 - 49 yr)}}$

----- Active space -----

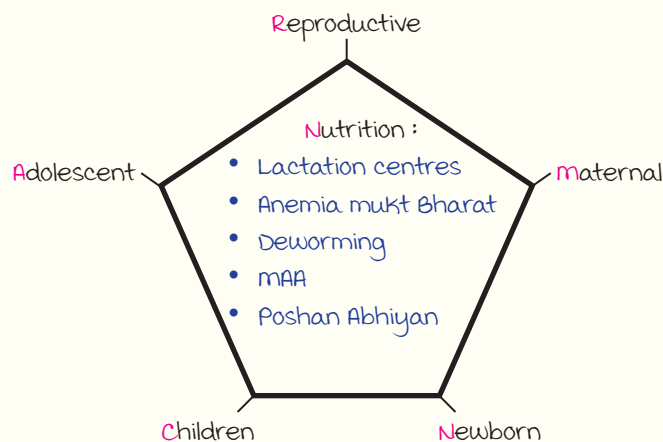
maternal death : Intrapartum or ≤ 42 days post partum
(except d/t accidents/electrocution).

$$\text{Lifetime maternal risk} : 1 - \left(1 - \frac{MMR}{1 \text{ lakh}}\right)^{35}$$

$$LB : \frac{CBR \times \text{population}}{1000}$$

RMNCAH+N

00:18:40



Visits

Recommended ANC : 13 - 14 visits

Period of gestation	Visits
0 - 7 months	monthly
8 th months	Fortnightly
9 th months	weekly

	No. of visits
Minimum ANC	4
Recommended ANC	13 - 14
WHO care model	8

WHO care model :

Trimester	visits
1 st	1
2 nd	2
3 rd	5

National Health mission :

Visits : 4	min ANC (Weeks POG)	PNC
1 st	ASAP/ ≤ 12	D1
2 nd	14 - 26	D3
3 rd	28 - 34	D7
4 th	≥ 36 weeks POG - term	D42 (6 weeks)

----- Active space -----

ANC Guidelines :

Nutrition 🌐 :

Pregnancy : +350 Kcal/d (Avg.).

Trimester	Energy (kcal/day)		Protein	Folic acid	Iron
	10 Kg GWG	12 Kg GWG			
1 st	70	85	-	570 µg/day	27 mg/day
2 nd	230	280	+9.5 g/day		
3 rd	390	470	+22 g/day		
Avg (T ₂ , T ₃)	310	375	-		

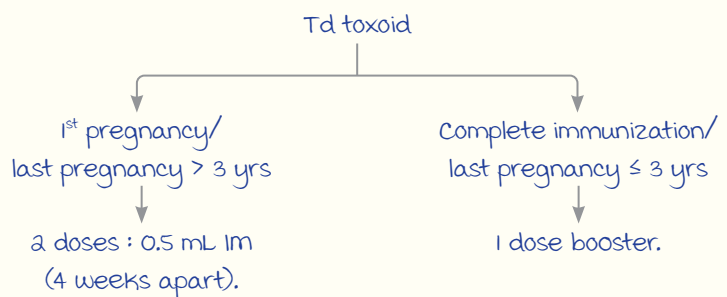
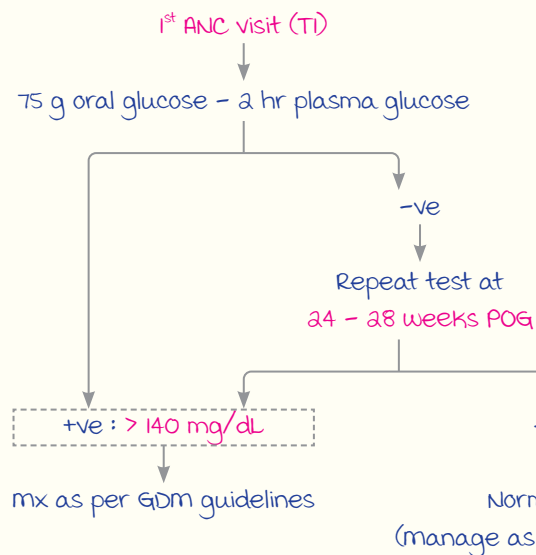
GWG : Gestation weight gain

Screening for diseases :

Test	Rural	Urban
Urine	- Protein - Sugar	
Serology	Hemoglobin	
	malaria test (Endemic)	- HIV - VDRL - Hepatitis B - Test for GDM

One step OGTT (IOC for GDM) 🌐 :

vaccine schedule :



Maternal Health Interventions

00:30:10

----- Active space -----

Home Based Newborn Care (HBNC) :

Conducted by : ASHA.

Beneficiary : mother & newborn.

Visits : 6 (minimum).

Home Based Care for Young Child (HBYC) :

Conducted by : ASHA.

Beneficiary : mother & child

(3 - 15 months age).

Visits : 5 (Every 3 months).

Visit	Day of delivery	Vaginal delivery	Home delivery	LSCS
1 st	D1	⊖	↑ ⊕ ↓	⊖
2 nd	D3	↑ ⊕ ↓		⊖
3 rd	D7			↑ ⊕ ↓
4 th	D14			
5 th	D21			
6 th	D28			
7 th	D42			

Visit	Age of child
1 st	3 months
2 nd	6 months
3 rd	9 months
4 th	12 months
5 th	15 months

Janani Suraksha Yojana (JSY) :

Sponsor : MoHFW (Central).

Beneficiary : mother (> 19 yrs age),
upto 2 live births.

Use : Promote institutional delivery.

Scheme : Incentive based (MoHFW).

₹ 1000 - 2000 —→ mother
—→ ASHA

↓

₹ 400/- (Urban) ₹ 600/- (Rural)

Janani Shishu Suraksha Karyakram (JSSK) :

Beneficiary : Pregnant women & sick infants (< 1 year).

Use : Promote public health facility utilization.

Scheme : Free benefit.

- OPD/IPD.
- Food.
- IX.
- Drugs.
- Transport (102, 108).

Pradhan Mantri Surakshit Matritva Abhiyaan (PMSMA)

Scheme : Free ANC check up at PMSMA clinics on 9th of every month.

Sticker colour	Implication
Green	No risk factor
Red	High risk pregnancy
Blue	Pregnancy induced hypertension
Yellow	Co-morbid conditions (DM, hypothyroid, STD)



----- Active space -----

Surakshit Matritva Aashwasan (SUMAN) 🌐 :

Scheme :

- Free ANC, delivery, PNC.
- Free Rx for sick neonates.
- Free transport & zero expense delivery.
- **No tolerance to denial.**
- **Ensure respectful care :**
 - Privacy & dignity.
 - Support for breastfeeding.
 - Zero dose vaccination.

Follows : SUMAN service guarantee charter.

Beneficiaries : Pregnant women & newborns.

Prime minister Matru Vandana Yojna :

Sponsor : MoWCD.

Under : mission Shakti.

Beneficiary : **mother only** (1st born child/and born girl child).

Scheme : Incentive based (WCDM).



	Installments	Prerequisite
First born	1 st : ₹ 3000/-	• Registration of pregnancy • ≥ 1 ANC, ≤ 6 m of LMP
	2 nd : ₹ 2000/-	• Registration of birth of child • Child received 1st dose of vaccines (BCG, OPV, DPT, Hepatitis B)
Second born (Girl)	Single : ₹ 6000/-	- → update.

mother's Absolute Affection (MAA) :

Indian adaptation of BFHI from UNICEF.

Use : Promote breastfeeding (BF).

- Exclusive BF x 6 months.
- Early BF start :
 - Vaginal delivery : < 1 hr (ASAP).
 - LSCS : < 4 hrs.

Labour Room Quality Improvement Initiative (LaQshya) :

Improve labour room standards.



Nayjat Shishu Suraksha Karyakram (NSSK) :

----- Active space -----

- Training of health workers.
- Beneficiary : Neonates.

Rashtriya Bal Swasthya Karyakram (RBSK) :

- Integrated with Ayushman bharat scheme.
- Beneficiary : Children (0 - 18 yrs age).

Aim : Screening & Rx of children (32 conditions).

RBSK
RASHTRIYA BAL SWASTHYA KARYAKRAM
राष्ट्रीय बाल स्वास्थ्य कार्यक्रम

(Congenital) Defects
Deficiencies
Developmental delays
Diseases of childhood

Conducted in :

- School.
- Juvenile homes.
- Anganwadi centres.

Rashtriya Kishor Swasthya Karyakram (RKSK) :

Scheme :

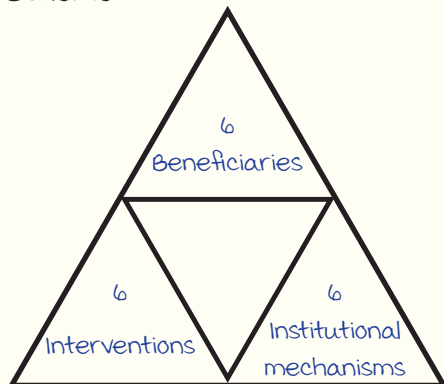
- AFHC (Adolescent friendly health clinic).
- WIFS (Weekly iron folic acid supplements).
- menstrual hygiene.
- Peer education initiative (Saathiya approach).

Beneficiary : Adolescents (10 - 19 yrs).

Anemia Mukh Bharat 🌐 :

6 × 6 × 6 strategy to combat anemia.

Scheme :



Key interventions :

- Prophylactic IFA supplementation.
- Deworming : Tab Albendazole [200 mg (1 - 2 yr) ; 400 mg (> 2 yr)].
 - National deworming day : 10 Feb & 10 Aug (Biannual).
 - Rx of hookworm infections.
- Behavior change communication : Solid body, smart mind.

IFA prophylaxis for anemia 🌐 :

Beneficiaries	Iron (mg)	FA (µg)	Frequency	Tablet color
1. Children : 6 - 59 months age	20	100	2 times/week	-
2. Children : 5 - 9 yrs age	45	400	Weekly	Pink
3. Adolescents : 10 - 19 yrs age	60	500	Weekly	Blue
4. Reproductive age women : 20 - 49 yrs (Non pregnant)	60	500	Weekly	Red
5. Pregnant women	60	500	Daily	Red
6. Lactating women				

→ Liquid bottle with auto dispenser

→ 4 m gestation to 6 m in lactation

----- Active space -----

Rx for anemia :

- mild to moderate anemia :

	Gest. age < 34 wks	Gest. age > 34 wks
Tolerant to IFA	(IFA) x Double dose	• IV Iron sucrose • IV Ferric carboxymaltose
Not tolerant to IFA	• IV iron sucrose • IV Ferric carboxymaltose (Not Im.)	

- Severe anemia :

	Hb : 5 - 7 g/dL	Hb : < 5g/dL
T1, T2	• IV Iron sucrose • IV Ferric carboxymaltose	Hospitalization + IV infusion and/or blood transfusion
T3	Hospitalization + IV infusion and/or blood transfusion	

Baby Friendly Hospital Initiative (BFHI) :

Started by : UNICEF.

Use : Promote child survival & breastfeeding.

Concept : **GOBI 3F**.

G : Growth monitoring

O : ORS

B : Breast feeding

I : Immunisation

F : Female literacy

F : Food supplements

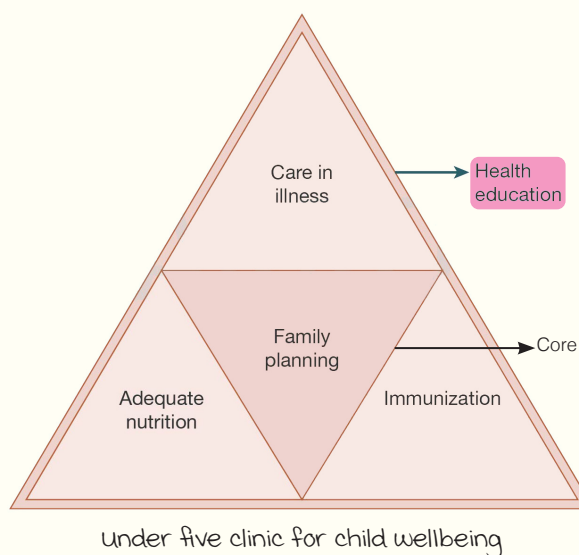
F : Family Planning



Under - Five Clinic :

Initiative by : UN.

Use : Promote child survival.



PREVENTIVE PAEDIATRICS

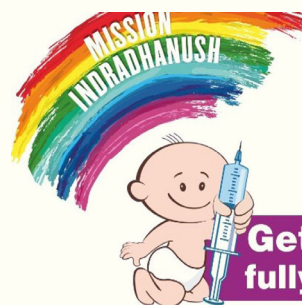
----- Active space -----

Mission Indradhanush

00:00:24

National Immunization Schedule 🌐 :

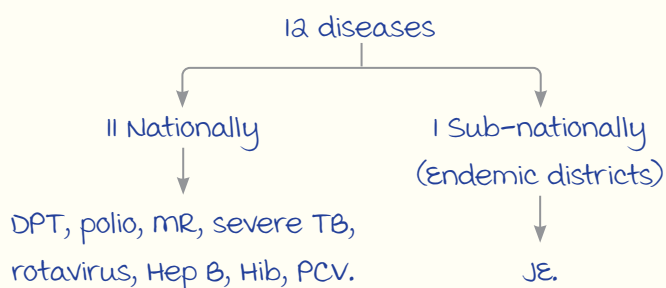
Age	Vaccination schedule
At birth	BCG, OPV-0 dose, Hep B-birth dose
6 weeks	OPV-1, Rota-1, Pentavalent-1, fIPV-1, PCV-1
10 weeks	OPV-2, Rota-2, Pentavalent-2,
14 weeks	OPV-3, Rota-3, Pentavalent-3, fIPV-2, PCV-2
9 months	measles/mR-1, vit A, PCV booster, fIPV-3, JE-1
16 - 24 months	measles/mR-2, DPT 1 st booster, OPV booster, JE-2
5 - 6 years	DPT 2 nd booster
10 & 16 years	Td
For pregnant women	- Td-1 : Early in pregnancy - Td-2 : 4 weeks after Td-1 - Td booster : If pregnancy occurs within 3 years of last pregnancy and 2 Td doses were received



Be Wise!
Get your child fully immunized

Logo of mission Indradhanush

Diseases Covered :



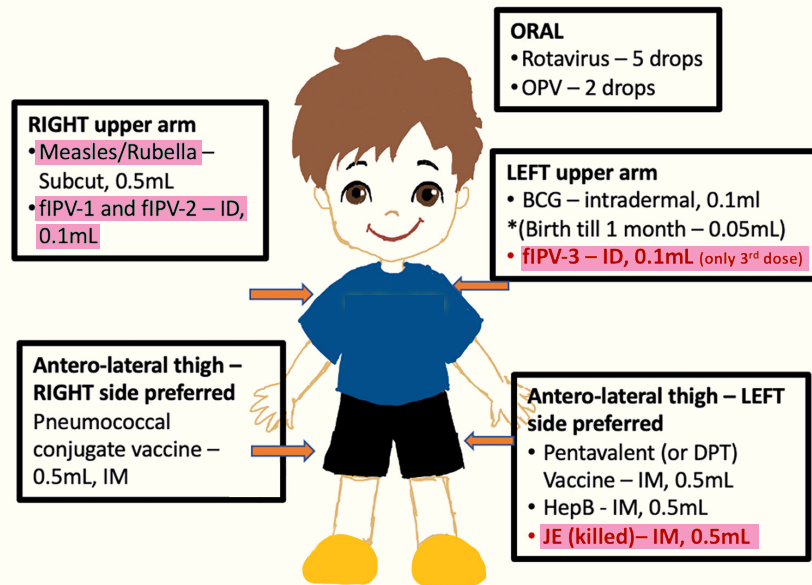
- Hib → meningitis.
- PCV → Pneumonia.

Upper Age Limit for Vaccine 🌐 :

Vaccine	Upper age limit
Hep B birth dose	within 24 hours
OPV-0 dose	within 15 days
BCG, Rotavirus, Pentavalent, fIPV, PCV	until 1 year of age
OPV/measles/mR	up to 5 years
DPT	up to 7 years
JE	up to 15 years

Prevents : mother
↓
Vertical transmission
↓
Child.

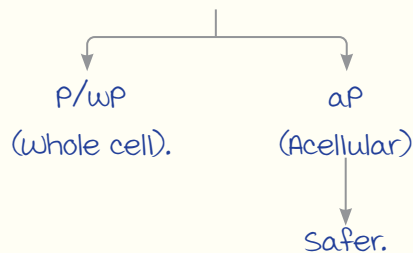
----- Active space -----

Vaccine Administration 🌐:**Pentavalent vaccine** 🌐:

- Diphtheria → Toxoid
- Pertussis → Killed (whole cell)
- Tetanus → Toxoid
- Haemophilus influenzae type b (Hib) → Conjugate
- Hepatitis B → Recombinant

} mixed vaccine.

Pertussis vaccine :



s/e d/t pertussis component :

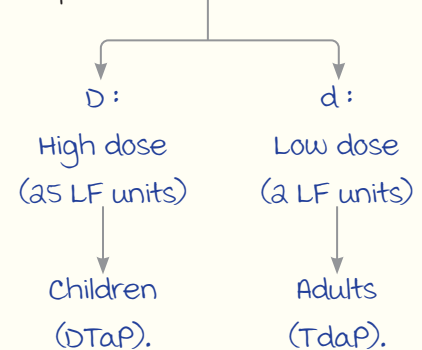
- Unconsolable cry (> 3 hours).
- Neurological deficits.

c/i :

- High grade fever (> 40°C)
- Visibly sick child
- Convulsions/epilepsy
- Anaphylaxis to previous dose
- Neurological defects :
 - Progressive
 - Non-progressive : Relative.
(E.g. : Cerebral palsy).

} Absolute.

Diphtheria vaccine :



LF : Limits of flocculation

Side effects of vaccines:

----- Active space -----

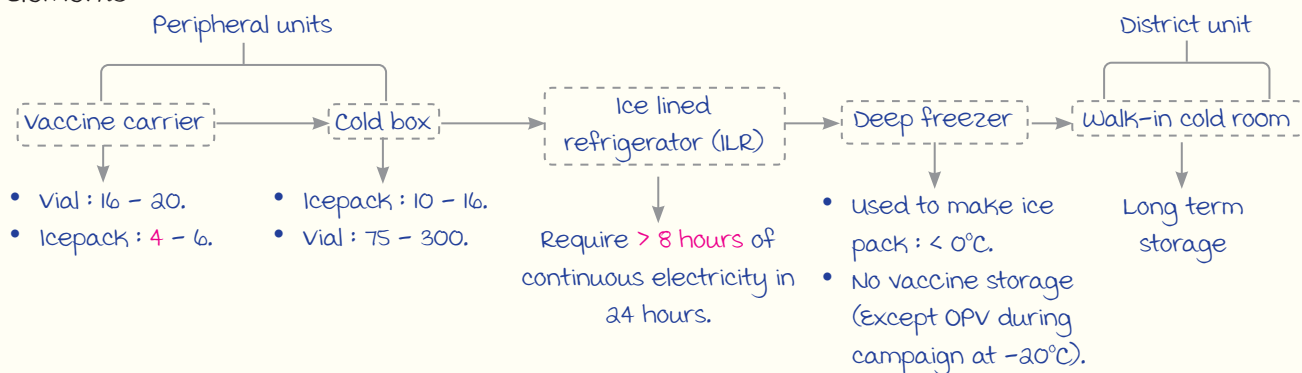
Vaccine	s/e
BCG	Lymphadenitis
OPV	Vaccine associated paralytic polio (VAPP)
Pentavalent	Inconsolable cry, neurological deficits
Rotavirus vaccine	Intussusception
MR vaccine	measles like rash (5 - 7 days post vaccination)
Td vaccine	Brachial neuritis

Diluents :

Vaccine		Diluent
BCG		Normal saline
measles/Rubella		Distilled water
JE	Live vaccine	Phosphate buffer
	Killed (Kolar)	No diluent required

Vaccine Storage & Viability

00:20:30

Cold Chain :Elements :

Temperature : +2 to +8°C.



Vaccine carriers

----- Active space ----- ILR :

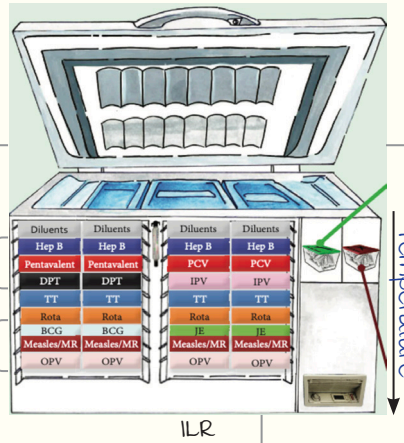
Vaccine sensitivity :

- Freeze sensitive :

Hep B > Pentavalent > PCV

- Heat sensitive :

Reconstituted BCG > OPV > mR > BCG



Temperature monitoring :

- Dial thermometer : Twice/day
- Location : Center of ILR

Storage :

Heat sensitive (Top to bottom order in ILR)	Live vaccines	Killed vaccines
Don't Hurt people In This Room Because Japanese manners Outshine	Cool Boys Love CRIME Types	P3 Killed In Cold Rabbit Jelly
• Diluents • Hep B • Pentavalent, PCV • IPV • Td, TT • Rotavirus • BCG • JE • measles, Rubella • OPV	• COVID (viral vector) • BCG • Live • Chicken pox • Rubella • Influenza • measles, mumps • JE (Old) • Typhoid oral (Tyd)	• Plague • Polio • Pertussis • Killed HFD • Influenza • Covid (Covaxin) • Rabies • JE

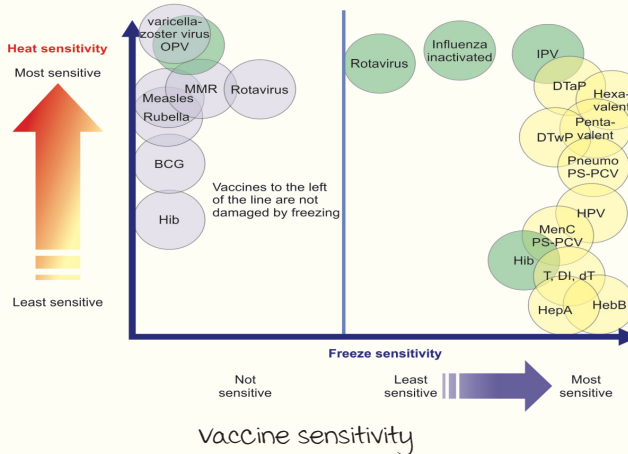
Vaccine sensitivity:

Most FREEZE sensitive

Most Heat sensitive

: Hep B > DPT, DT, TT

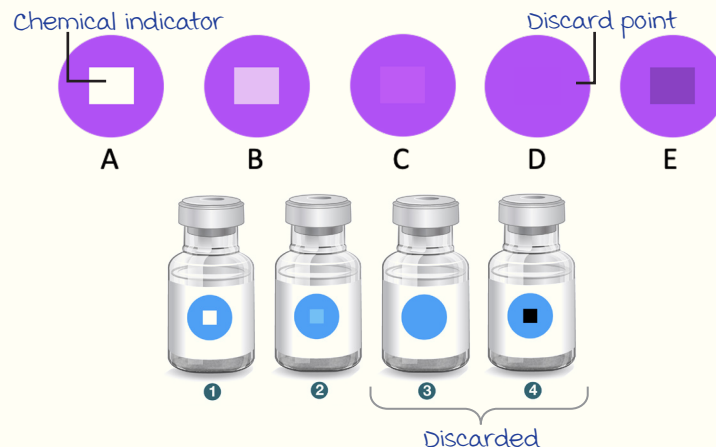
: rBCG > VZV, OPV > rMeasles



Vaccine sensitivity

vaccine vial monitor (vvm) 🌐 :

- used to check heat stability of vaccines.
- Discard point : Inner square same color/darker than outer circle.



Open Vial Policy :

- All vaccine used up to 28 days after opening if :
 - vvm located on body (Reusable).
 - Not expired.
- vvm on lid/neck (Non-reusable) :
 - BCG, MR.
 - used < 4 - 6 hrs after reconstitution.

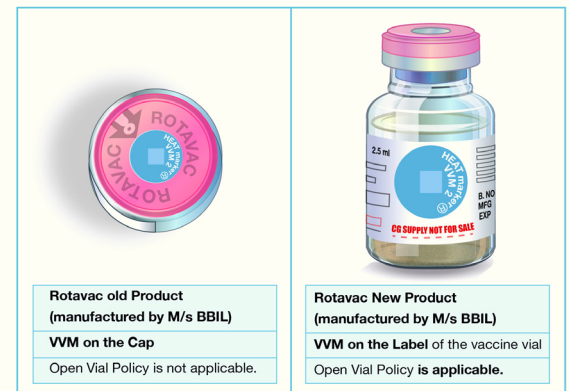
Shake test :

Principle : Precipitation of adjuvant $[\text{Al}(\text{OH})_3]$ in freeze sensitive vaccine.

Procedure :



Rotavirus under open vial policy :



Immunization status

00:35:35

Unimmunized Child :

Age	vaccines that can be administered
11 months	BCG, Pentavalent, fIPV, PCV, MR, JE, OPV, Rotavirus, vit A ↓ Preferred vaccine
24 months	MR, OPV, DPT, vit A, JE (if applicable)
6 years	DPT, JE-1
8 years	JE-1
10 years	Td

----- Active space -----

Universal Immunization Program Evaluation :

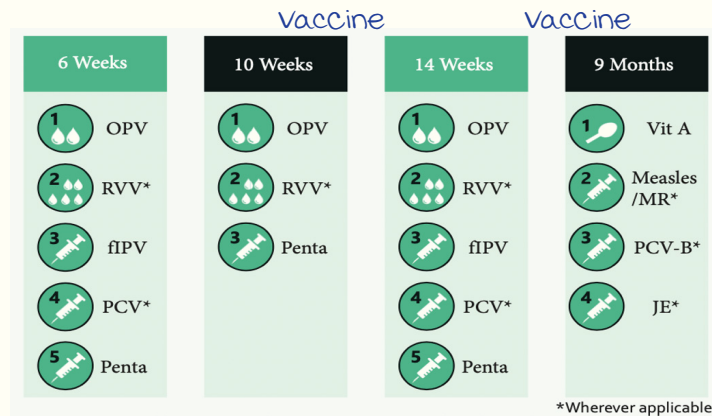
1. Fully immunized child : Better indicator.

- Received all due vaccines till 1st year of life.
- Age group : 12 - 24 months.
- Cluster sampling : 210 children (District → 30 clusters Each cluster → 7 children).

2. Completely immunized child : Received all due vaccines till 5 yrs of age.

Sequence of Vaccine Administration :

Oral vaccine → Intradermal → Intramuscular → Pentavalent.

Malnutrition Indicators

00:45:10

markers :

Weight for age (most sensitive) :

General malnutrition/combined malnutrition (Acute & chronic).

Weight for height : Acute malnutrition.

Height for age : Chronic malnutrition.

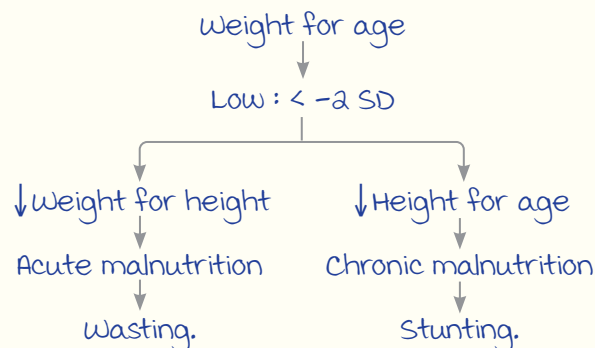
Classification :

1. Waterlow's classification :

Differentiates b/w acute & chronic malnutrition.

2. Welcome trust classification :

used in patients with edema (Kwashiorkor).



Instruments for measurement :

----- Active space -----

1. Shakir's tape :

measures MUAC (mid upper arm circumference).

Colour	MUAC	Interpretation
Red	< 11.5 cm	Severe malnutrition
Yellow	11.5 – 12.5 cm	mild to moderate malnutrition
Green	12.5 cm onwards	Normal



Shakir tape

MUAC : most sensitive **age independent** marker for malnutrition.

2. Salter's weighing scale :

- Age group : < 2 years.
- Sensitivity : 100 gms.



Salter scale

Nutritional Rehabilitation Centres (NRC)

00:52:50

Severe Acute malnutrition (SAM) :

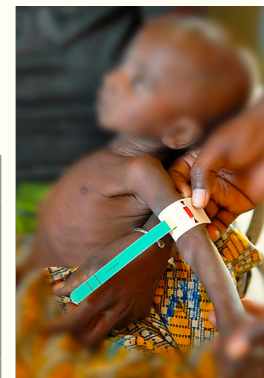
Criteria :

1. Weight for height/length : < -3 SD.
2. mid upper arm circumference : < 11.5 cm.
3. Presence of **nutritional edema**.

management of SAM : Refer to NRCs.

Phases at NRC :

Phase	Dietary intervention
1. Stabilization phase : 1 - 2 days	Starter diet (F-75) + micronutrients (C/i : Fe) : • 75 kcal/100 gms • 0.9 g of protein
2. Transition phase : 2 - 3 days • Oral diet • Signs of improvement ⊕	Catch up diet (F-100) : • 100 kcal/100 gms • 2.9 g of protein
3. Maintenance/rehabilitation phase : Child can complete > 90% of the expected feed	Home based balanced diet



Severe acute malnutrition

----- Active space -----

Growth Charts

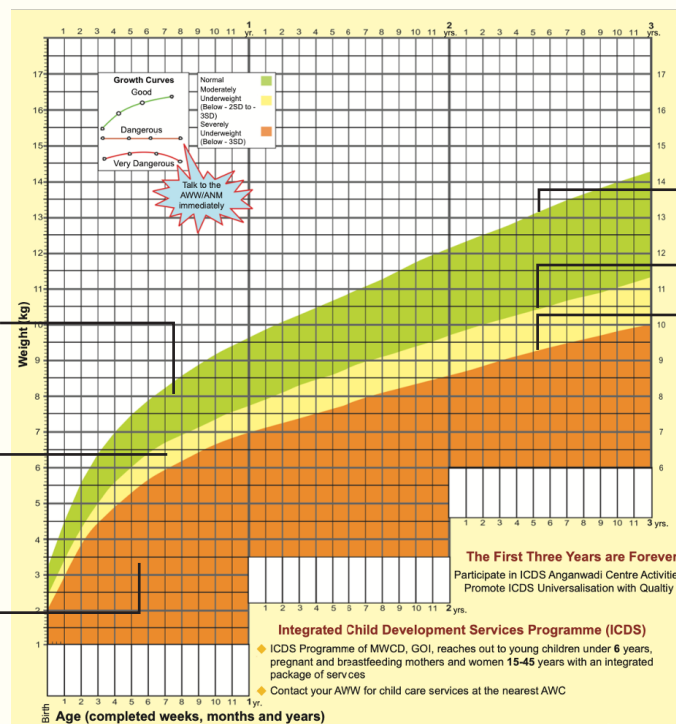
00:56:42

Integrated child development services/ICDS growth charts :

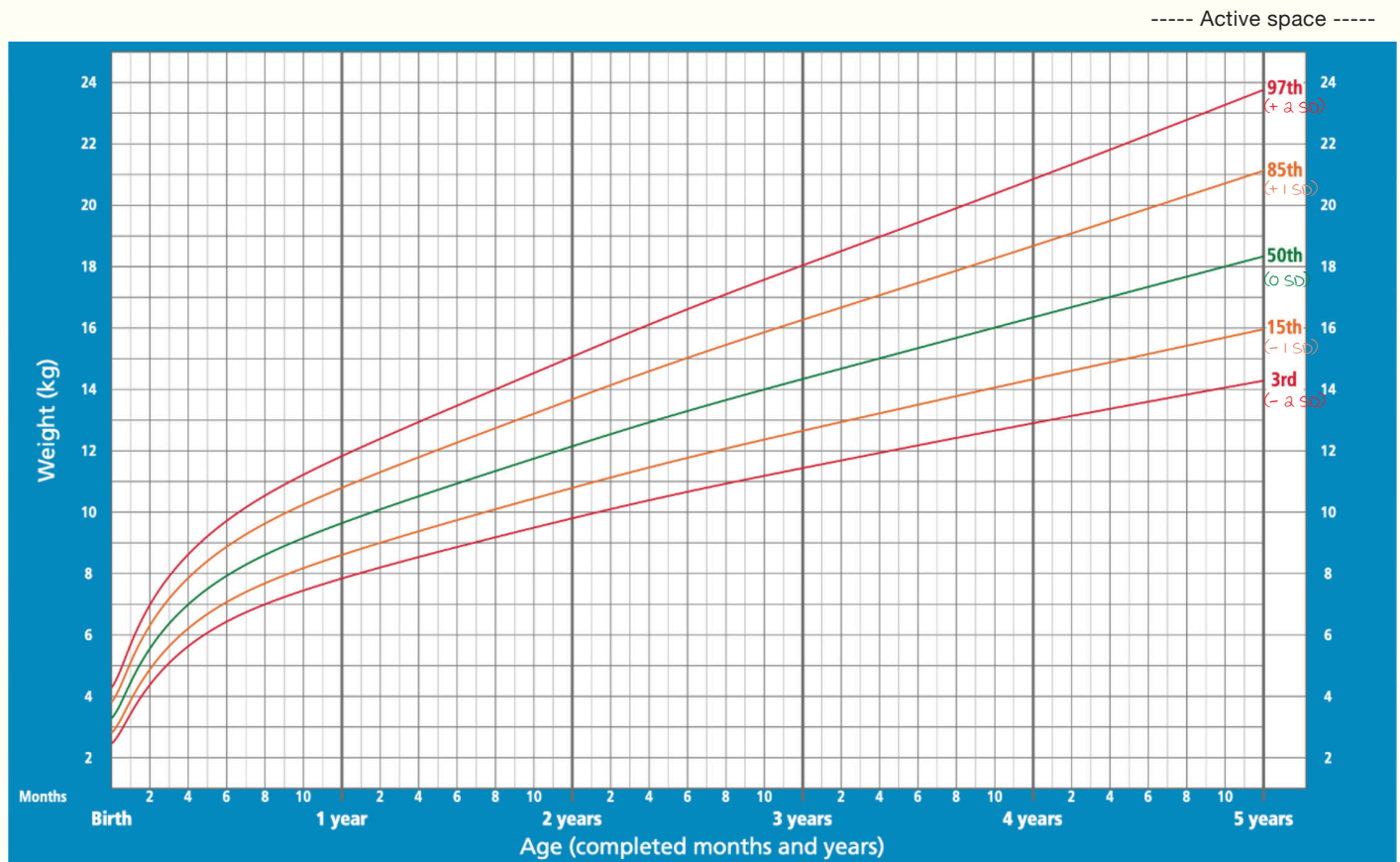
- Use :
 - mother & child protection card (MCP).
 - On field assessment.
- Assess weight for age :
 - Trend of growth : most important.
 - Location plot on graph : Not important.

Zones :

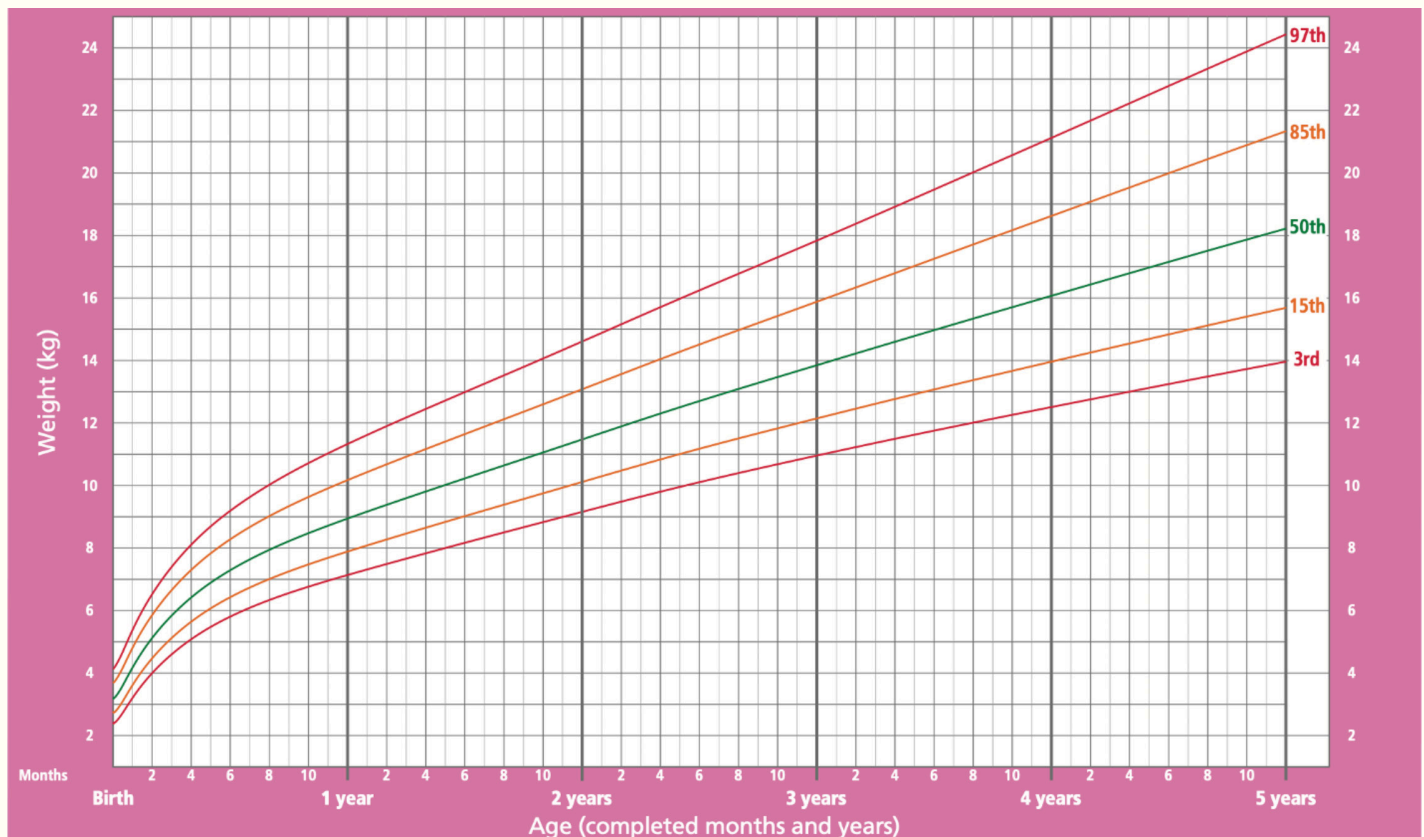
- B/w +2 SD to -2 SD
↓
Normal nutrition
- B/w -2 SD to -3 SD
↓
mild to moderate malnutrition
- Below 3rd curve
↓
Severe malnutrition



ICDS Growth charts



WHO growth charts : weight for age in boys (Birth to 5 yrs)



WHO growth charts : weight for age in girls (Birth to 5 yrs)

----- Active space -----

Mother & Child Services

01:01:15

National Quality Assurance Programme (NQAP) :

Launch : 2018.

Kayakalp awards : Better hospital administration & patient satisfaction.

Swachh Swasth Sarvatra : Clean & safe drinking water.

LaQshya : Quality of care during delivery & post partum.

mera Aspataal : IT platform for feedback of OPD services
(Capture patients voice → Improve quality).

AEFI surveillance platform.

musQan 🌐 :

- Child friendly public health care facilities.
- Maternity & morbidity in child < 12 yrs.

SaQushal :

Safety & quality assessment for healthcare facilities.



Kangaroo mother care (KMC) : Elements (SAANS).

- Skin to skin contact (Prevention of hypothermia).
- Airway disease prevention (Pneumonia).
- Ambulatory support.
- Nutritional support (Exclusive breastfeeding).
- Support (Emotional).

Important Dates :

- 14th April : Ayushman Bharat - health & wellness centre day.
- 7th April : World health day.
- April last week : World immunization week.
- 28th May - 8th June : Intensified diarrhea control fortnight.
- August 1st week : World breast-feeding week.
- September 1st week : National nutrition week.
- 15th-21st November : Newborn week.
- 10th Feb & Aug (Biannual) : Deworming day → T. Albendazole administered.

NTEP AND NACP

----- Active space -----

NTEP : Key Components

00:00:10

World TB day : 24 march.

Theme (2025) :

Yes! We can end TB : Commit, invest, deliver.

Targets :

	Initiative	End TB by
WHO	End TB	2035
UN	Sustainable development goals	2030
India	TB mukt Bharat	2025 > 2027



TB mukt Bharat :

Baseline : 2010 numbers

- ↓ No. of TB deaths : > 90%.
- ↓ TB incidence rate : > 80%.
- ↓ Catastrophic costs d/t TB : Zero.

Case Definitions :

Presumptive TB case :

≥ 1 of the following for > 14 days :

- Fever.
- Cough.
- Night sweats.
- Weight loss.

Drug sensitive TB :

TB patient sensitive to 1st line drugs :

- Isoniazid (INH).
- Rifampicin.
- Pyrazinamide.
- Ethambutol.

Drug resistant TB (DRTB) :

	Resistant to			
	INH	Rifampicin	Fluoroquinolone (Not Lfx, mx)	Group A drug (Lfx, Bdq, Lzd, mx)
H-mono/Poly DRTB	R	Sn	-	-
Rifampicin resistant TB	-	R	-	-
MDR TB	R	R	-	-
Pre-XDR TB	R	R	R	Sn
XDR TB	R	R	R	R

R : Resistant

Sn : Sensitive

Lfx : Levofloxacin

mx : moxifloxacin

Bdq : Bedaquiline

Lzd : Linezolid

RR : Rifampicin
resistant

----- Active space -----

Diagnosis under NTEP

00:08:35

	Sensitivity (Sn)	Specificity (Sp)	uses
CXR	⊖		Not diagnostic
Sputum for AFB	↑↑↑	↑↑	• Fastest • Screening test of choice
CBNAAT	↑↑	↑↑↑	Diagnostic test of choice
Line probe assay (Liquid sample)	-		Drug sensitivity testing
Culture (Liquid > Solid)	-		Gold standard

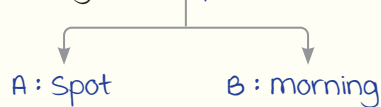
Sputum microscopy :

Quantity : > 5 mL expectorated.

Stains :

- ZN stain.
- Special stain : 1. Auramine
2. Rhodamine } Fluorescent microscopy.

Validity : 2 samples (< 10% squamous epithelium).



Turn around time : 30 mins (Fastest).

Reporting time : 24 hrs.

CBNAAT/Gene Xpert :

Universal test.

Sample :

- Body tissue, sputum, urine.
- Not accepted : Blood, feces.

Turn around time : 90 mins.

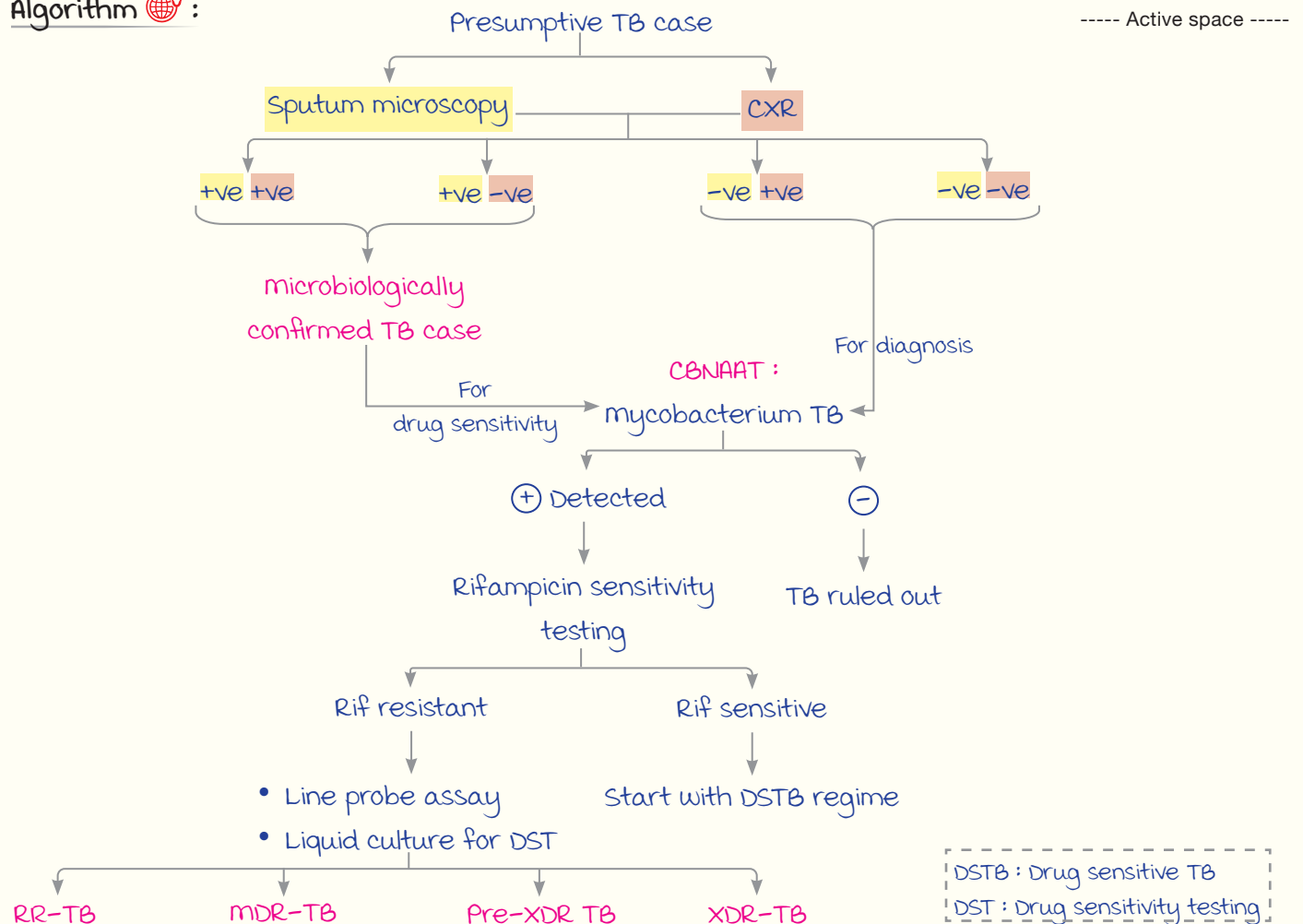
Advantages :

- Short time.
- Detects rifampicin Sn status.

Tru-NAAT :

- Similar Sn & Sp to CBNAAT.
- ↓ Cost.

Algorithm 🌐:



Management under NTEP

00:18:43

Regimes 🌐:

		Intensive phase	Continuation phase	Indications
DSTB	NTEP	(2 m) HRZE : OD • 56 doses	(4 m) HRE : OD • 16 w ~ 112 doses	- Adults : 5 - Children : 6 Weight bands for FDC
	2025 CDC recommendation (Not adopted by NTEP yet)	(2 m) HPZm	(2 m) HPM	- Age > 12 yrs - Weight > 40 kg - PLHIV on ART (or) ART to be started : CD4 > 100 cells/μL
		(2 m) HRZE	(4 m) HRE • 18 weeks	Newly diagnosed H & R susceptible TB case
DRTB	H mono/poly	(6 m) ZERO		-
	Shorter MDR	(4 - 6 m) CHOZEEen	(5 m) COZE	Single gene INH mutation
	Longer MDR	(18 - 20 m) LCLCy (6 m) B		Any gene INH mutation (Single/double)

----- Active space -----

BPaLM regime : MDR TB

6 months regime (Approved with restrictions under trial)

Rx : Bdq, Pretomanid, Lzd, mx

Role of pyridoxine :

Given with INH to prevent peripheral neuropathy.

- Adult : 25 - 50 mg/day.
- Children : 10 mg/day.
- Peripheral neuropathy : 100 mg/day (max).

Z : Pyrazinamide	E : Ethambutol
R : Rifampicin	En : Ethionamide
O : Levofloxacin	P : Rifapentine
H : Isoniazid (INH)	C : Clofazimine
B : Bedaquiline	Cy : Cycloserine
m : moxifloxacin	FDC : Fixed dose combination
PLHIV : People living with HIV	

Outcomes 🌐 :

	Definitions
Cured	Rx completed → Sputum -ve report
Rx completed	• Rx completed • No sputum report at end of Rx
Failure of Rx	• Sputum -ve $\xrightarrow{\text{Later}}$ Sputum +ve (or) • Sputum +ve : 5th month of Rx
Loss to follow-up	Started on ATT + continuous interruption of Rx (≥ 4 wks)
Recurrent TB	Rx completed → Anytime sputum +ve

Prophylaxis/TPT (TB Preventative Therapy) :

Indications :

- PLHIV : Rule out TB before Rx.
- Household contacts.
- Other risk groups
 - Immunosuppressive Rx.
 - Silicosis.
 - Dialysis.
 - Transplant surgery.

Standard regime : (6 m) H + pyridoxine <
Daily

New regimes

- (3) HP weekly (12 doses)
- (1) HP daily : Only in PLHIV/CLHIV (28 doses) (> 13 yrs)

TB surveillance & case finding :

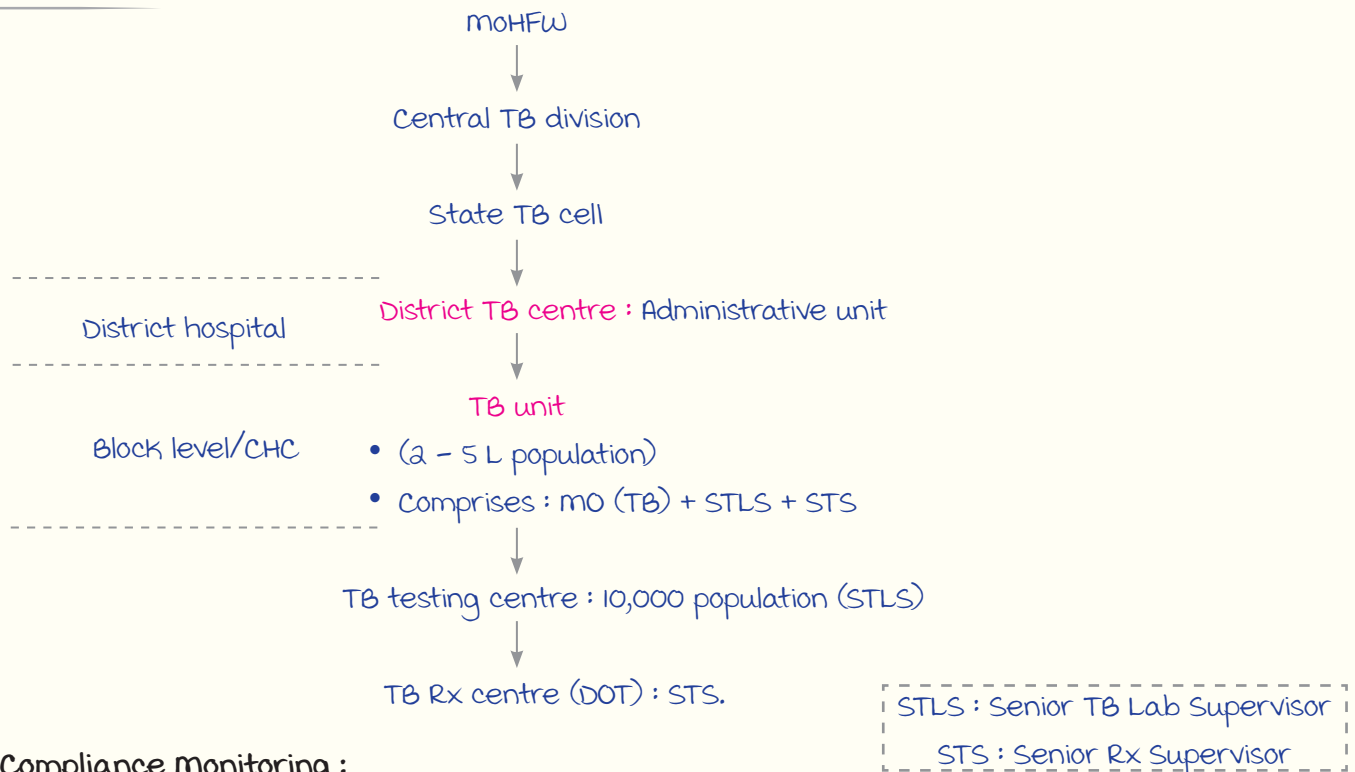
----- Active space -----

- **Passive** case finding : Case in hospitals.
- **Intensified case** finding : TB screening in people with other comorbidities (HIV).
- **Active** case finding : Screen for cases in high - risk groups.
 - BMI < 18.
 - Construction workers.
 - migrant workers.

NTEP : Framework

00:34:23

Structure :



Compliance monitoring :

99 dots : mobile phone based system.

mERM : medicine event reminder management system.

Nikshay software : Generates TB notifications.

Incentives :

Eligibility	Cash Incentives
Notification of TB (Informant)	Rs. 500
Private provider incentive	Rs. 500
TB Rx completion	DSTB : Rs. 1000
	DRTB : Rs. 5000
Nikshay Poshan Yojana (Patient)	Rs. 3000 every 3 m (2 installments)

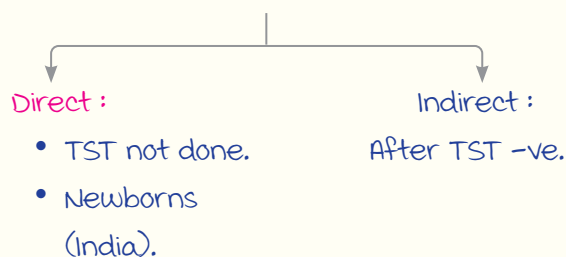
----- Active space -----

New TB Initiatives
- Nikshay Poshan Yojana : 1000/- incentive. - BNS Section 272 : Neglect in TB case notification (within 30 days) - New NTEP Action Plan : 2026 - 2031. - Adult BCG vaccination in Indian trial (ICMR) in : - Treated for TB $\leq$ 5 yrs. - Close contacts of TB case. - BMI < 18. - Age > 60 yrs (High risk d/t comorbidity).

Tuberculin Skin Testing & BCG Vaccination

00:39:21

BCG Vaccination :



TST :

Test : 1U of PPD RT-23 given $\xrightarrow{72 \text{ hrs}}$ Results.

Interpretation :

Wheal size on forearm	+ve in
> 5 mm	- HIV - Severe immunosuppression - History of TB - Close contact of TB
> 10 mm	- Recent immigrants - IV drug users - Age < 4 yrs - High risk occupation - Adolescents : High risk groups
> 15 mm	Those with no risk factors

NACO : Key Components

00:43:59

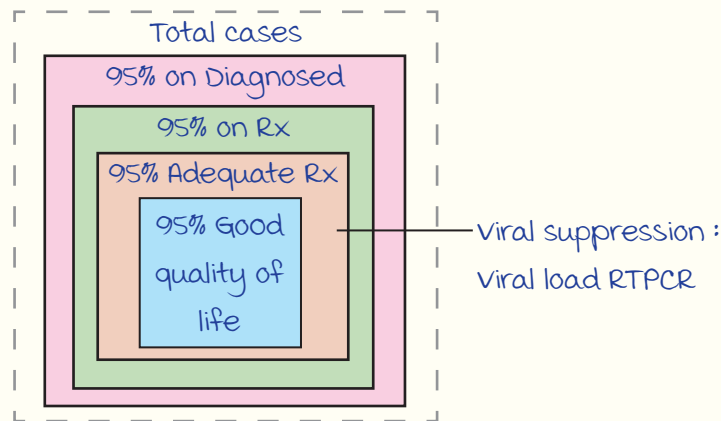
World AIDS day : 01 December.

Targets of NACP :

Universal ART : Start ART irrespective of CD4 count.

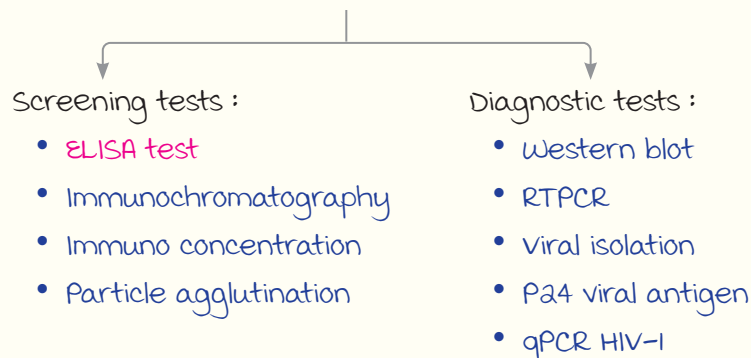


National AIDS Control Organisation
 India's Voice against AIDS
 Ministry of Health & Family Welfare, Government of India
www.naco.gov.in



Diagnosis under NACO

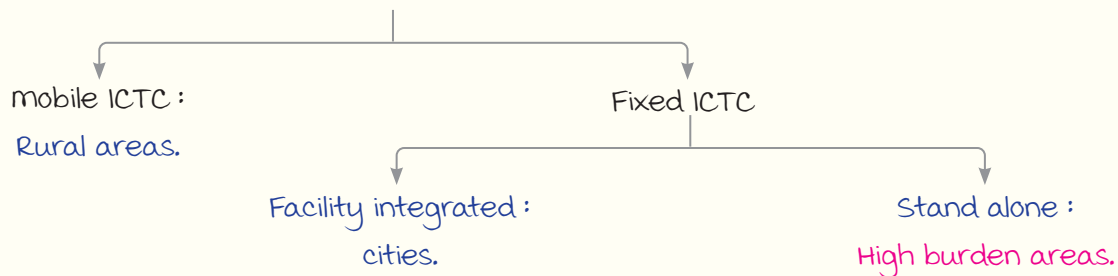
00:46:39



Note :

To assess response to ART : Viral load > CD4 count.

Integrated Counselling & Testing Centre (ICTC) :



Screening :

Techniques :

- ELISA test : most sensitive.
- Rapid test : Immuno concentration.
- Spot test : Dried blood spot.

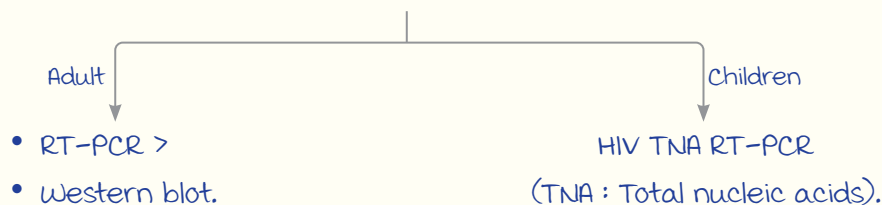
----- Active space -----

Indications :

	Probable HIV criteria	Protocol
All blood donations	Any 1 of 3 samples +ve	Discard blood unit (unlinked anonymous testing)
Symptomatic individuals	Any 2 of 3 samples +ve	Refer for confirmation
Asymptomatic individuals	All 3 samples +ve	- Refer for confirmation - ANC women : Opt out testing

Diagnosis :

IOC :

Treatment under NACO

00:52:53

Regimes 🌐 :

Age	Weight	Regime	Update
< 6 yrs	< 20 kg	Abacavir + Lamivudine + Lopinavir/Ritonavir (AL+LPV/r)	Raltegravir (ALR)
6 - 10 yrs	20 - 30 kg	Abacavir + Lamivudine + Dolutegravir (ALD) (or) Zidovudine	-
> 10 yrs	> 30 kg	Tenofovir + Lamivudine + Dolutegravir (TLD)	-

Prophylaxis :Post exposure prophylaxis :

- Indication : Needle stick injury.
- Regime : T (300 mg) + L (300 mg) + D (50 mg) × 28 days OD.
- Administration : ≤ 72 hrs of exposure (Ideal : < 2 hrs).

Pre exposure prophylaxis :

- Indication :
 - Ongoing continuous HIV exposure.
 - High risk IV drug users.
 - Sexual behaviour ↑
- Regime : TDF + Emtricitabine.

Pneumocystis jirovecii Prophylaxis :Infective at : CD4 count < 200 cells/mm³.

Regime : Cotrimoxazole (ds : Double strength)

= Sulphamethoxazole (800 mg) + Trimethoprim (160 mg).

Administration :

Start : $CD4 < 350 \text{ cells/mm}^3$



Stop : $CD4 > 350 \text{ cells/mm}^3$ on 2 occasions ($> 6 \text{ months apart}$) + $\uparrow$ CD4 count + no WHO stage 3, 4.

----- Active space -----

Prevention of mother to Child Transmission (PMTCT) 🌐 :

Baby exposed to HIV.

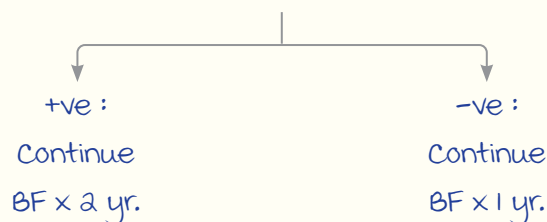
		Transmission	
		Low risk	High risk
maternal	viral load	$< 1000 \text{ copies/mL}$	$> 1000 \text{ copies/mL}$
	ART course	$\geq 4 \text{ wks}$	$< 4 \text{ wks}$
HIV prophylaxis		Nevirapine (NVP)	NVP + Zidovudine
Duration		6 weeks	Breastfeeding : 12 weeks
			Not breastfeeding : 6 wks

Vaccines :

- All given to child as per schedule.
- **Exceptions** for live vaccines :
 - If $CD4 < 200/\text{mm}^3$.
 - If relative $CD4 < 15\%$.
 - Symptomatic for HIV.
 - Stage 3, 4 HIV.

Breastfeeding (BF) :

- Encouraged.
- Exclusive BF x 6 months.
- If child HIV status



TB & HIV Coinfection :

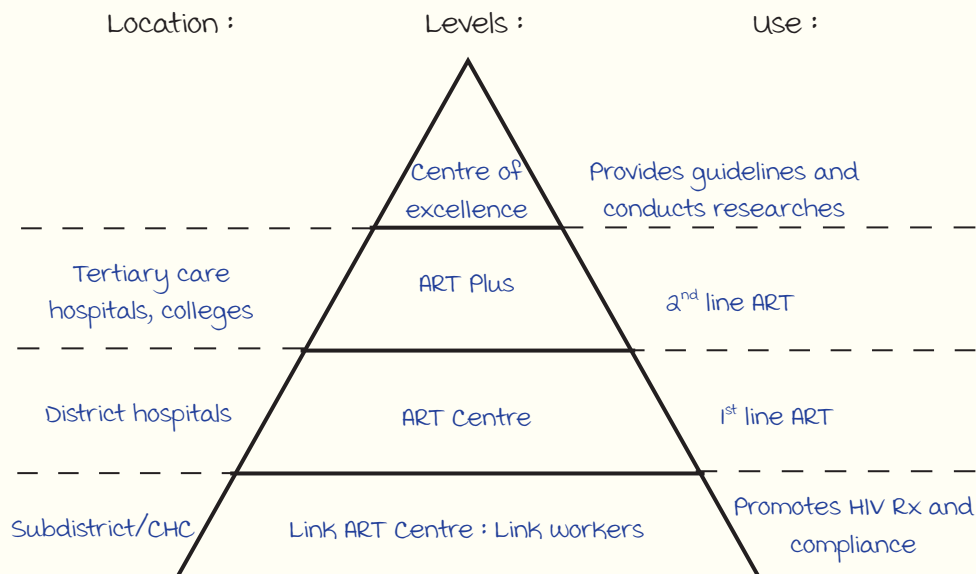
- **CBNAAT** (IOC) : Rule out TB $\rightarrow$ **TPT** to all HIV cases.
- Rifampicin replaced : **Rifabutin**.
- ATT $\xrightarrow{2 \text{ weeks}}$ ART (D/t risk of **IRIS**)
 - Exception : **TB meningitis** (ATT $\xrightarrow{8 \text{ weeks}}$ ART)

IRIS : Immune Reconstitution Inflammatory Syndrome.

----- Active space -----

NACO : Framework

01:05:46

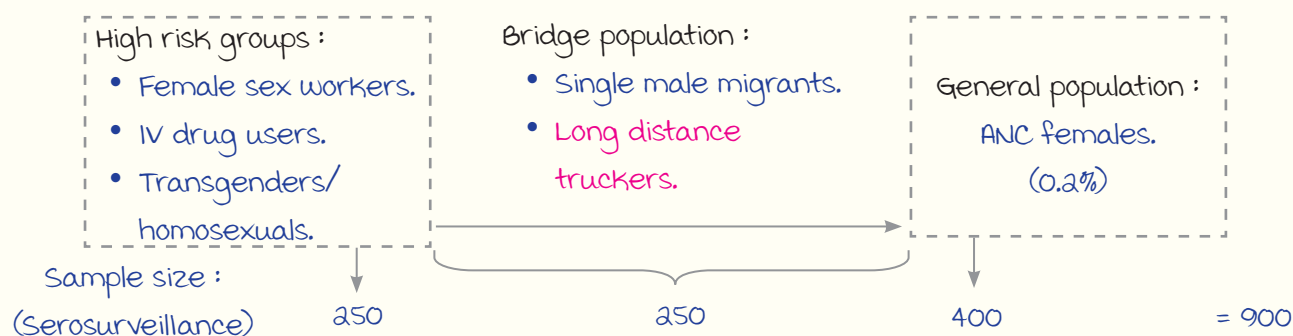
Structure :HIV Sentinel Surveillance :

Aim : Find missing cases.

Conducted at : RTI/STI treatment centres (Suraksha clinics).

Syndromic/Symptomatic/Presumptive approach :

Kits	Colour	use
1	Grey	Cervicitis, cervical discharge, presumptive Dx
2	Green	Bacterial vaginosis
3	White	Genital ulcers
4	Blue	Genital ulcers : Penicillin allergy
5	Red	viral genital ulcer
6	Yellow	PID, abdominal pain
7	Black	Inguinal bubos
8	Brown	Anorectal discharge

Burden of Surveillance :

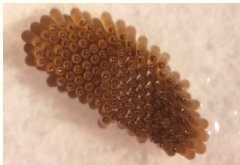
NATIONAL VECTOR BORNE DISEASE CONTROL PROGRAMME

----- Active space -----

Entomology 

00:00:17

Vectors :

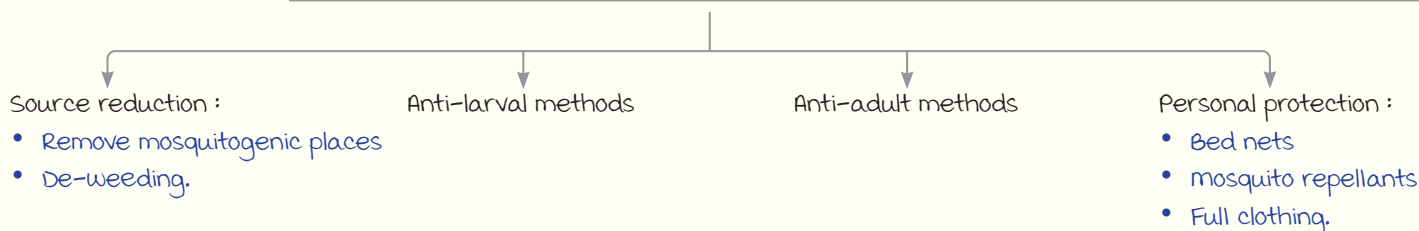
		Anopheles	Culex	Aedes/tiger mosquito	mansonia
Species		- stephensi - culicifacies - fluviatilis - dirus/sundaicus	- quinquefasciatus - vishnui	-	- annulifera - uniformis
Disease		malaria	- Japanese encephalitis - Lymphatic filariasis	- Dengue - Yellow fever - Chikungunya - Zika	Lymphatic filariasis
Breeding site		Clean, stagnant water	Dirty, polluted water	Artificial, stored water	Large water body with aquatic vegetations : Pistia plant, water hyacinth  Pistia plant
Eggs		- Boat shaped with lateral floats - Singular 	- Clusters - Rafts/sheets 	- Cigar shaped - Singular 	- Star shaped - Clusters 
Larva	Feeding	Surface	Bottom	Bottom	Bottom (Attach to roots of plant)
	Siphon tube	⊖	⊕	⊕	⊕
Adult	Biting time	Dawn, dusk (morning-evening)	midnight	Day (2 hrs after sunrise & before sunset)	Evening
	Resting place	Exophilic (Garden)	Exophilic & endophilic	Endophilic	Exophilic

----- Active space -----

		Anopheles	Culex	Aedes/tiger mosquito	mansonia
Adult	Flight range 🌐	2 - 3 km	11 - 13 km	200 m	2 - 3 km
	Resting position	Inclined/landing pose	Hunch back	Parallel to surface	Long legs, squatting
	Special features	Spots on wings	• Small body • Big wings • Brown colour	Stripes on body	Big body
	Appearance				

Integrated Vector Management

00:10:43



Anti-Adult methods :

Residual spray :

	Residual action	Spray frequency	Significance
malathion	3 - 4 mths	3 - 4 times/yr	m/c
DDT	6 m	Biannual	Biomagnification (Obsolete)

Space spray :

AKA : Fogging machine.

	Source	Potency	Cost
Pyrethrum	Chrysanthemum flower (Natural)	↑↑	↑↑
Synthetic pyrethroids : • Deltamethrin • Cyfluthrin	Artificial	↑↑	↓↓ (m/c used)

Anti-Larval methods :

Physical method : Larvicidal oils.

Chemical method :

- Paris green : Stomach poison.
 - malathion
 - Temephos
- } Contact poison

Biological method :

- Gambusia affinis
 - Poecilia reticulata
 - Bacillus thuringiensis.
- } Fish

NVBDCP : 2004

00:15:48

----- Active space -----

Diseases Included :

- **malaria** : National malaria control programme/NMCP (1953).
- Dengue.
- Japanese encephalitis.
- Chikungunya.
- Kala azar.
- Lymphatic filariasis.

mosquito

**Significant Dates 🌐 :**World neglected tropical disease day : 30th Jan.World malaria day : 25th April.World dengue day : 16th May.

Anti malaria month : June.

Anti dengue month : July.

Targets 🌐 :

1. Annual parasite incidence (API) : < 1/1000 population.
2. Annual blood examination rate (ABER) : > 10%.
3. microfilariasis rate : < 1% in children.
4. Control JE, Kala azar, dengue, chikungunya.

Malaria 🌐

00:18:42

Diagnosis :

Ix : Peripheral blood smear.

Smears → Thick : Identify parasite.
 → Thin : Identify species.

Stains → **JSB stain** (Jaswant Singh & Bhattacharjee).
 → Giemsa stain.

Vectors :

- **Plasmodia vivax** (m/c).
- **Plasmodia falciparum** (m/c in the northeast).
- Plasmodia ovale (Absent in India).
- Plasmodia malariae.
- Plasmodia knowlesi (Andaman & Nicobar).

Treatment :

1. Chloroquine (CQ)
2. ACT
3. Primaquine (PQ) : Prevents relapse of malaria.

C/I :

- **Pregnancy.**
- **Infants.**
- Glucose-6-phosphate dehydrogenase deficiency.

----- Active space -----

	Drugs/doses	Regime	
Plasmodium vivax (PVx)	- CQ : 25 mg/kg $\rightarrow D_1 : 10 \text{ mg}$ $\rightarrow D_a : 10 \text{ mg}$ $\rightarrow D_3 : 5 \text{ mg}$ - PQ : 0.25 mg/kg	CQ $\times$ 3 days + PQ $\times$ 14 days	
Plasmodium falciparum (PFI)	- PQ : 0.75 mg/kg (D_a : Single dose) - ACT-AL (NE states) : Artemether + Lumefantrine - ACT-SP (Other states) : Artesunate + Sulphadoxine + Pyrimethamine	NE states	Other states
		PQ D_a + ACT-AL (D_{1-3})	PQ D_a + ACT-SP (D_{1-3})
mixed infection (vivax + falciparum)	- PQ : 0.25 mg/kg - ACT-AL - ACT-SP	NE states	Other states
		PQ $\times$ 14 days + ACT-AL $\times$ 3 days	PQ $\times$ 14 days + ACT-SP $\times$ 3 days
uncomplicated falciparum in pregnancy	- Oral quinine salts - ACT-AL (NE states) - ACT-SP (Other states) - C/I : PQ	T_1 : Oral quinine salts $\times$ 3 days	
		T_a / T_3 : ACT $\rightarrow$ AL $\rightarrow$ SP	
vivax in pregnancy	- CQ - C/I : PQ	CQ $\times$ 3 days	

ACT : Artemisinin based combination therapy.

Prophylaxis :

	DOC	Regime	Frequency
Short term travel (< 6 wks)	Doxycycline	1 - 2 days (Before) $\xleftrightarrow{\text{Travel}}$ 4 weeks (After)	Daily
Long term travel (> 6 wks)	mefloquine	1 - 2 wks (Before) $\xleftrightarrow{\text{Travel}}$ 4 wks (After)	weekly

malarial Indicators 🌐:

----- Active space -----

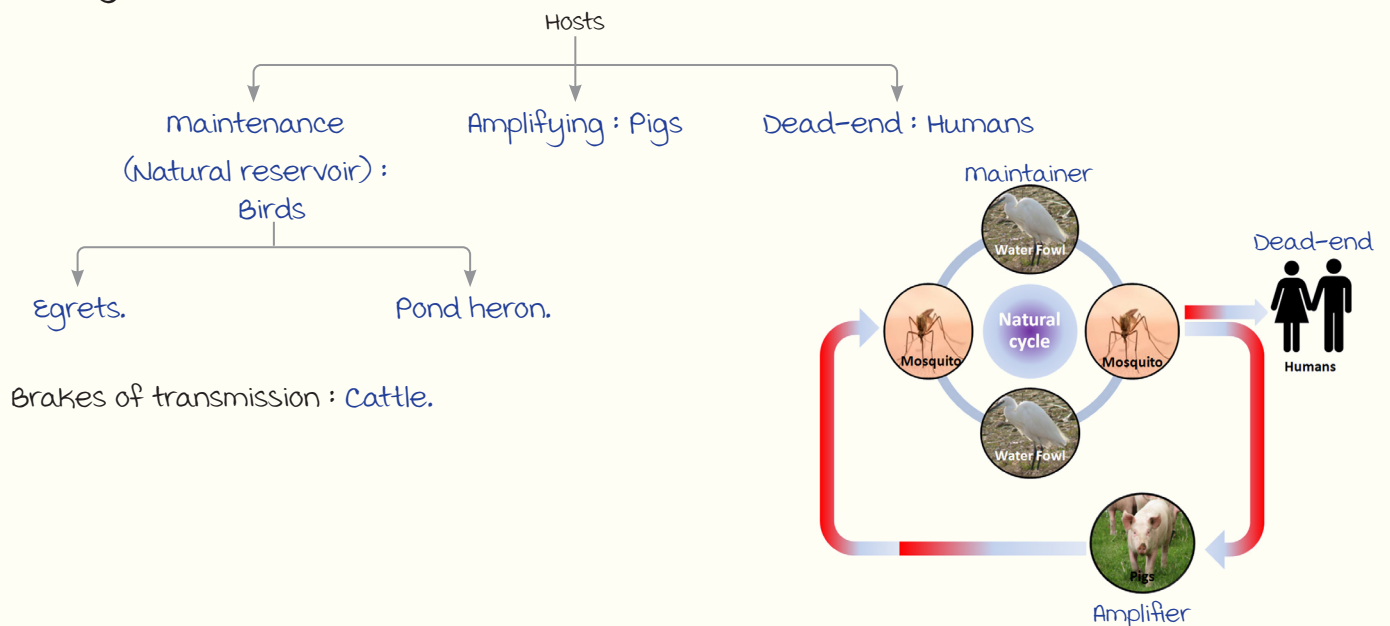
	Calculation	Significance
API (Incidence)	$\frac{\text{No. of confirmed cases in a year}}{\text{Population under surveillance}} \times 1000$	• malaria burden • Impact of malaria programme
ABER	$\frac{\text{No. of slides examined}}{\text{Population under surveillance}} \times 100$	• Operational indicator • Fever prevalence
Slide positivity rate (SPR)	$\frac{\text{No. of slides +ve}}{\text{No. of slides examined}} \times 100$	most sensitive outbreak indicator

Control Classification :

	API (State)	API (District)	Category
Intensified malaria control	> 1	-	3
Pre-elimination area	< 1	> 1 (Some districts)	2
Elimination area	< 1	< 1 (All districts)	1
Prevention of re-establishment	~ 0	-	0

Japanese Encephalitis 🌐

00:36:07

Life Cycle :

----- Active space -----

Affected Areas :

- Coastal areas.
- UP.
- Bihar.
- Orissa.
- Karnataka.
- Tamil Nadu.
- Telangana.

Clinical Features :

- High grade fever.
- Vomiting.
- Neurological features :
 - Encephalitis.
 - Convulsions.
 - Delirium.
 - Behavioural change (Long term).

Prevention :

JE vaccine (update) 🌐 :

- JENVAC vaccine.
- Type : Killed (Kolar strain).
- Dose : 0.5 mL (1m).
- Schedule : 9 m, 16 - 24 m (2 doses).
- Unimmunized child : Till 15 years.
- VVM : On body of vial.
- Open vial policy : ⊕.
- Freeze sensitive.
- No reconstitution.
- Site : Left anterolateral thigh 1m.

JE live vaccine : Obsolete.

Kala Azar

00:42:05

Vector :

Sandfly (Phlebotomus argentipes) :

- Nocturnal.
- Habitat : Cool damp places (Burrows, tree barks, cracks).
- Movement :
 - Can't fly : Lanceolate shaped hairy wings.
 - Hop : 50 ft range.
- Insecticide : Synthetic pyrethroids.

Lesion :

Painful bite, usually on lower limbs.

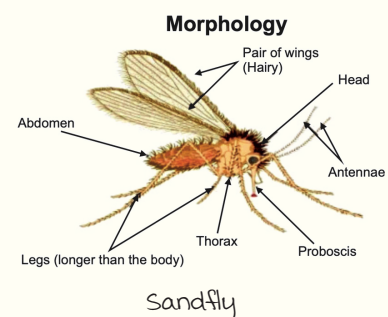
Investigations :

Screening test : RK 39 (Serological test).

Gold standard test : Parasite detection by microscopy in splenic aspirate.

Treatment :

Liposomal Amphotericin-B (DOC) > miltefosine, paromomycin.



Note :

Kala azar free status for India (2023).

----- Active space -----

Lymphatic Filariasis

00:45:14

AKA Elephantiasis.

Disease	Vector	Parasites
Bancroftian filariasis	Culex quinquefasciatus (Culex fatigans)	Wuchereria bancrofti
malayan filariasis	mansonoides	- Brugia malayi - Brugia timori

Epidemiology :

Infective form : Larval stage of microfilaria.

Portal of entry : Skin → Site of localization : Inguinoscrotal region (Lymphatics).

Reservoir : Person with **microfilaria in blood**
(microfilaremia $\propto$ Prevalence of MF in area).

Investigations :

Peripheral blood film (IOC) :

10 pm – 2 am (max activity of microfilaria).

DEC provocation test :

- Clinical IOC.
- 100 mg oral DEC $\xrightarrow{1 \text{ hr}}$ Blood test.

Concentration method :

- membrane filtration** (most sensitive).
- Knott's technique.

Treatment (DOC) :

Diethylcarbamazine (DEC) : 6 mg/kg/dose → 12 doses.

Prevention :

mass drug administration (update)  :

- Biannual**, 3 drugs :

- DEC : 6 mg/kg/day (2 times)
 - Albendazole : 400 mg ($> 2 \text{ y/o}$)
 - Ivermectin : 150 – 200 $\mu\text{g/kg}$ ($\geq 5 \text{ y/o}$)
- } $> 5 \text{ y/o}$

- Endemic areas :

- Given along with Albendazole on national deworming day (**10th February, 10th August**).
- DEC medicated salt : 1 – 4 g of DEC/kg of salt.

----- Active space -----

Elimination :

Integrated vector control management system.

Target : Eliminate lymphatic filariasis by 2027.

Dengue

00:52:20

Transmission :

Virus serotypes :

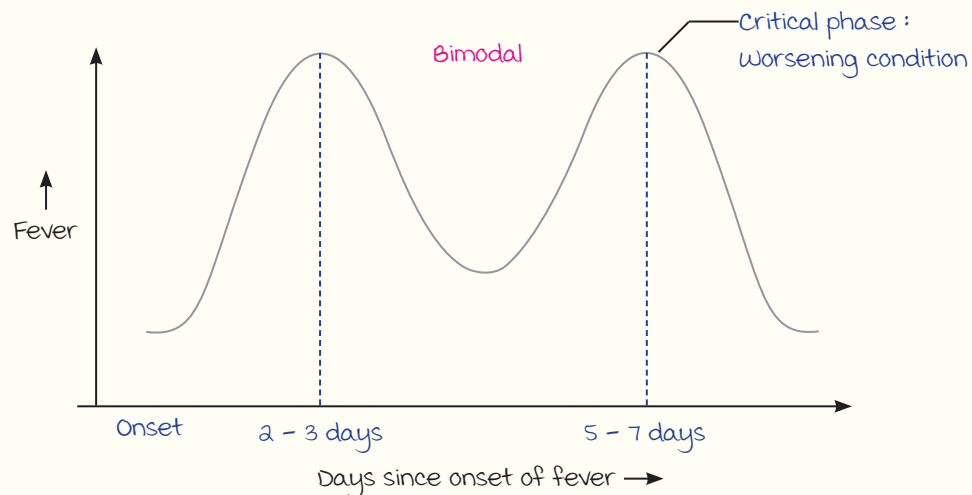
- DEN 1.
- DEN 2 (m/c).
- DEN 3.
- DEN 4.

Trend : 1 - 3 yr epidemic cycle.

Pathogenesis :

Dengue fever :

- High grade fever.
- Bimodal fever.
- A/w : Retro-orbital pain, body aches.



Dengue hemorrhagic fever 🌐 :

- Torniquet test :

BP cuff inflated midway b/w SBP & DBP

↓ 5 mins

Count petechiae

< 6 :

Definitely -ve

> 10 :

mostly +ve

> 20 :

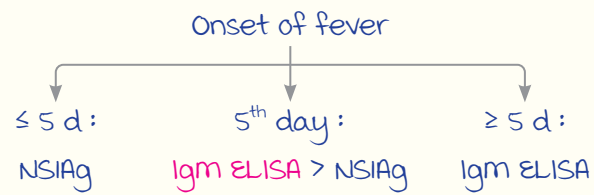
Definitely +ve

- Laboratory marker : ↑ in Hct > 20%.

Dengue shock syndrome : SBP < 90 mmHg.

Definitive Dengue Diagnosis :

----- Active space -----

Indicators for Dengue :

Breteau index :

No. of +ve containers per 100 houses inspected → Dengue outbreak potential.

Aedes aegypti index (AAI) :

- % of houses +ve for breeding of aedes larvae.
- AAI < 1 within 400 m of all international air/sea ports : yellow fever elimination (EYE strategy).

OTHER NATIONAL HEALTH PROGRAMS

National Polio Surveillance Program (NPSP)

00:01:40

Targets :

Polio free status in India : 27 march 2014.

Last case :

Strain eradicated	Date	Location
P1	13 January 2011	West Bengal
P2	24 October 1999	Uttar Pradesh
P3	22 October 2010	Jharkhand

End game strategic plan :

- Introduce fIPV (Safer) → Replace OPV.
- National switch day :
 - 25 April 2016.
 - Trivalent OPV (P1, P2, P3) → Bivalent OPV (P1 & P3).
- 3rd dose of fIPV introduced : 1 January 2023.

Strategy :

Polio case types :

Infection	% of all cases
Inapparent (Asymptomatic)	90 - 95
Abortive (Self-limiting) : Fever, vomiting, diarrhea	4 - 8
Non-paralytic aseptic meningitis	1
Paralytic polio	< 1

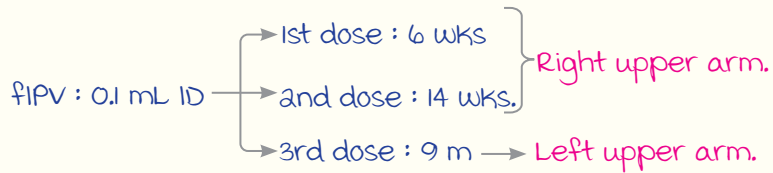
Associations of polio vaccines :

	vaccine derived polio virus (VDPV)	vaccine associated paralytic polio (VAPP)
Cause	New viral mutations	Aberrant immune reactions in host
Transmission in community	⊕	⊖
m/c associated vaccine strain	P2	P3

Polio vaccine :

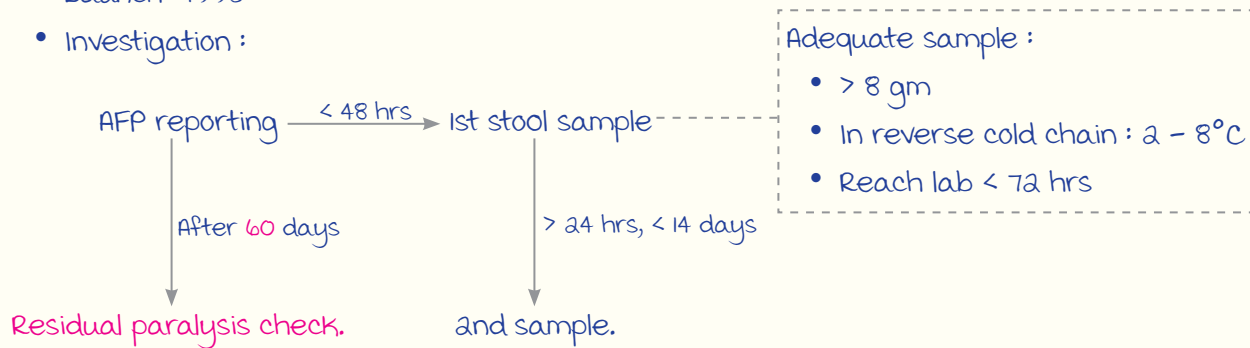
----- Active space -----

	OPV	IPV
Strain	Sabin	Salk
Type	Live	Killed : Fractional inactivated (fIPV)
Route	Oral	- fIPV : ID. - IPV : IM.
Immunity	- Gut (local) - Humoral	Humoral
use in epidemic	⊕	⊖



Acute flaccid paralysis (AFP) surveillance :

- Launch : 1995
- Investigation :



AFP D/d :

- Guillain - Barre syndrome.
- Traumatic neuritis.
- Transverse myelitis.
- Enterovirus 70/71.
- Poliomyelitis.

Indicators :

AFP surveillance indicators	Target
Sensitivity (m/d)	> 1 AFP/lakh/year in age < 15 years
Completeness of case ix	> 80% adequate stool sample collected
Operation efficacy	Sample collection $\xrightarrow{< 72 \text{ hrs}}$ Laboratory
Completeness of f/u	> 80% cases : Residual paralysis check

Organisation :

At district level.

----- Active space -----

National Program for Control of Blindness & Visual Impairment (NPCBVI)

00:13:40

Targets :

Blindness rate : $< 0.25\%$.

Strategy :

Visual impairment (VI) classification :



Logo of NPCBVI

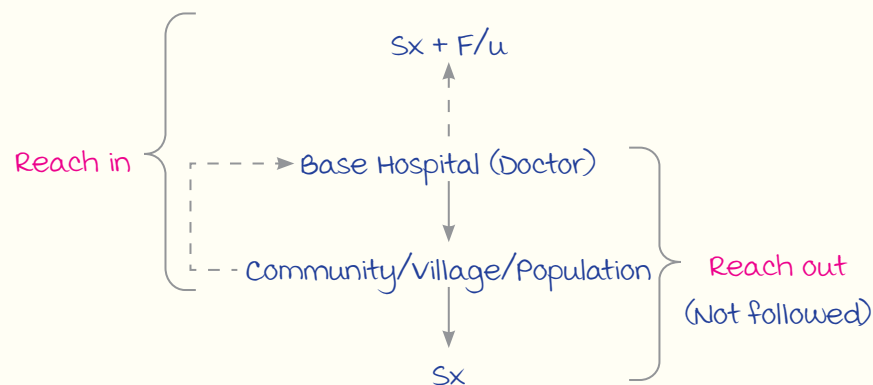
Categories	Presenting visual acuity in better eye with best possible correction
VI	$< 6/18$
Early VI	$< 6/12 - 6/18$
moderate VI	$< 6/18 - 6/60$
moderately severe VI	$< 6/18 - 3/60$
Severe VI	$< 6/60 - 3/60$
Blindness	$< 3/60$

m/c causes :

	VI	Blindness
< 50 years	Refractive errors	Non trachomatous corneal opacity : • Nutritional • Infections • Vit A deficiency
> 50 years	Cataract	

Prevalence of blindness : 0.36%

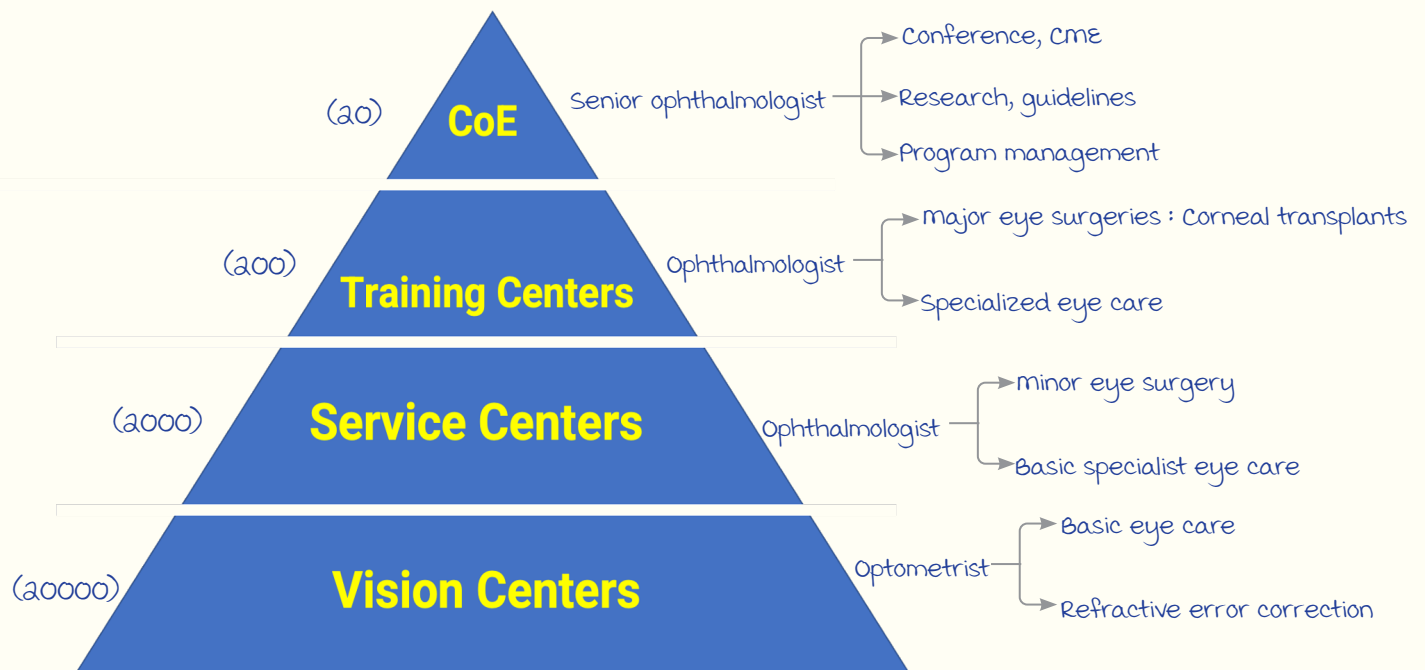
Cataract Sx :



Cataract surgery rate (CSR) : 3000/million/yr

Organisation :

----- Active space -----

**Netra Jyoti Abhiyan (2025) :**

- Eye health for all.
- Provide free spectacles.
- Reach in campaign intensified.
- Provide free cataract Sx in > 50 yrs.
- ↑ CSR > 3000/million.
- Intensified eye donation fortnight : 25 Aug - 8 Sept.
(↑ Corneal transplants)

Indicators :

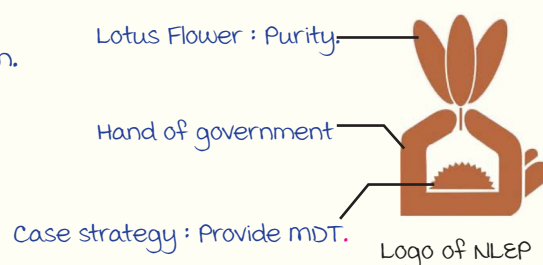
- Blindness rate : < 0.25%.
- CSR : > 3000/million/yr
- Bitot spots rate : < 0.5%.

National Leprosy Eradication Program (NLEP)

00:22:40

Launch :

- 1955 : National Leprosy Control Program.
- 1983 : NLEP.



----- Active space -----

Targets 🌐:Prevalence rate : $< 1/10,000$ population.Annual new case detection rate (ANCDR) : $< 10/\text{lakh}$ population.Grade 2 disability (GAD) rates : $< 1/\text{million}$ population.

(Best among general population)

} India achieved
in 2015.Cure rate : $> 85\%$.

Zero

→ Disabilities among new child cases.

→ Stigma & discrimination against person with leprosy.

Vision : Leprosy mukt Bharat by 2027.

Strategy:

Cardinal features of leprosy : Any 1 ⊕

- Hypopigmented/reddish skin lesion + **hypoanesthesia** (most specific).
- Involve peripheral nerves with definite thickening + loss of sensation and/or weakness.
- Demonstrate *m. leprae* in lesions.

Classification :

		Paucibacillary	multibacillary
Diagnostic criteria	Lesions	≤ 5	≥ 6
	Nerve involvement	⊖	≥ 1
	AFB from SSS	⊖	Any site
Duration of MDT		6 m	12 m

AFB : Acid fast bacilli

SSS : Slit skin smear

multidrug therapy (MDT) 🌐:

Drug	Dosage (mg)		Frequency
	Adult	Child (10 - 14 yr)	
Rifampicin	600	450	monthly (Supervised)
Dapsone	100	50	Daily
Clofazimine	300	150	monthly (Supervised)
	50	50	Daily/ alternate days in child

----- Active space -----

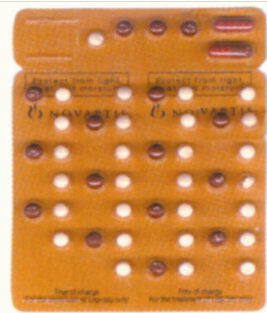


For adult (> 14 yrs)

**Rifampicin 600 mg
+
Clofazimine 300 mg
+
Dapsone 100 mg
(once a month)**

**Clofazimine 50 mg
+
Dapsone 100 mg
(Daily)**

For 12 months



For child (< 14 yrs)

{ Supervised
 { Unsupervised

Prophylaxis :

Single dose Rifampicin (Dose similar to Rx) :

Contacts	Sharing of
Household	Same house > 6 m in last yr (Except pregnant/child < 2 yr)
Social	Same closed space > 20 hrs/wk
Neighbourhood	3 houses on each side

Note :

Leprosy transmission : Nasal > Oral secretions.

Leprosy awareness 🌐 :

Best indicator : G2D rate.



NLEP mascot : Sapna



UN mascot : meena

New strategies :

- Welfare allowance : Rs 12000 for reconstructive Sx.
- ASHA based surveillance for leprosy suspects (ABSULS).
- National strategic plan and roadmap for leprosy (2023 - 2027).
- National guidelines for antimicrobial resistance (AMR) surveillance.
- Launch of Nikusth 2.0 : Integrated portal for reporting & case management of leprosy.

----- Active space -----

National Iodine Deficiency Disorder Control Program (NIDDCP) 00:37:30

Launch :

- 1962 : National goitre control program.
- 1992 : NIDDCP.

Targets :

Goitre rate < 5%.

Strategy :

Iodisation of salt → Production level : > 30 ppm.
 → Consumer level : > 15 ppm.

Organisation :

District goitre cell → Iodine deficiency disorder survey.

Indicators 🌐 :

Long term impact : **Goitre rate** (Grades 1 and 2).
 Principal impact/epidemiological :
urinary iodine excretion rate (≥ 100 mcg/L). } → To diagnose IDD.

Process/operational : Iodine in household salt rate.

Environmental iodine/Dietary deficiency : Neonatal hypothyroidism rate.

Goitre rate assessment :

Grades	Goitre mass	
	Visible	Palpable
0	⊖	⊖
1	⊖	⊕
2	⊕	⊕

Severity :

	mild (India)	moderate	Severe
Grade 1 & 2	> 5%	> 20%	$\geq 30\%$
median UIE (mcg/L)	50 - 99	20 - 49	< 20



Logo of NIDDCP

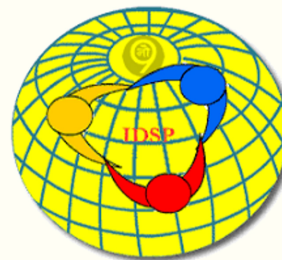
Integrated Disease Surveillance Program (IDSP)

00:43:25

----- Active space -----

Types of Surveillance :

	Done by	Interpretation
Syndromic	Health workers	Syndromes & field surveys
Presumptive	Doctors (Health center, hospitals)	History & clinical examination
Laboratory	Laboratories	Lab results



Logo of IDSP

Reporting : 52 weekly reports/yr.

Syndromes :

- Fever.
- Cough < 2 wks.
- Jaundice.
- Acute diarrhoea : ≥ 3 stool/day.
- Acute flaccid paralysis (< 15 yrs age).
- Unusual events causing death/hospitalization.

Integrated health intelligence platform (IHIP)

- New version of IDSP.
- Under Central Bureau of Health Intelligence (CBHI).

Non-Communicable Diseases (NCD) Program

00:46:10

- National program for prevention of NCD (Population based screening program).
- AKA previously : National program for prevention & control of cancer, diabetes, cardiovascular diseases & stroke (NPCDCS).
- Includes additional prevention of :
 - COPD
 - Rheumatic fever
 - CKD
 - Non alcoholic fatty liver disease (NAFLD)
- Integrated with : Ayush, NTEP.

Targets :

WHO Global action plan : Extended to 2027.

% Reduction	Risk factors/Outcome
30%	• Salt levels • Smoking
10%	• Alcohol • Physical inactivity
25%	• Hypertension • Premature deaths

----- Active space -----

Obesity Indicators :Body mass index (Bmi)/Quetelet index : m/c .

- $Bmi = \text{Weight (kg)} / \text{Height}^2 (m)$
- (N) in South Asians : 18.5 - 22.9

Broca's index (BI) :

- $BI = \text{Height (cm)} - 100$.
- Indicates ideal weight.

Corpulence index = Actual weight (kg)/Desirable weight (kg).

Criteria for obesity :

	males	Females
Waist circumference	> 102 cm	> 88 cm
Waist hip ratio	> 0.9	> 0.85
Skin fold thickness	> 40 mm	> 50 mm

Skin fold thickness at :

- mid triceps.
- Biceps.
- Subscapular.
- Supra iliac.

By Herpenden callipers.

Cancer Registry :

National Cancer Registry Program : Last survey in 2013 - 14.

Globocan (Global cancer observatory 2024) 🌐 :

- By UN & WHO.
- Incidence in India :
 - male : Lip/oral > Lung > Stomach/esophagus.
 - Female : Breast > Cervical > Ovarian.
 - Overall : Breast > Lip/oral > Cervical.

Highest mortality : Breast Ca (25%).

School Health Services

00:52:21

Launch : 1909 (Basic health services → National health services).

Incharge : medical officer of PHC.

Land area → Primary school : ≥ 1 acre.
 → Higher secondary : ≥ 1.5 acre.

Classrooms :

- Capacity/space : < 40 students/classroom.
- walls color : white.
- Source of light : Left sided.
- Desk : minus type.
- Toilets → Urinal : 1/60 students.
 → Latrine : 1/100 students.

POSHAN Programs

00:54:20

----- Active space -----

POSHAN Abhiyaan :

- Prime minister's overarching scheme for holistic nourishment.
- AKA National nutritional mission.

Governance :

- Nodal ministry : ministry of Health & Family Welfare (MOHFW).
- ministry of Women and Child Development (MOWCD).



सही पोषण - देश रोशन

Logo of Poshan Abhiyaan

Nutritional status improvement in :

- Children : 0 - 6 yrs.
- Adolescent girls.
- Pregnant women.
- Lactating mothers.

Target for 3 yrs :

- ↓ Low birth weight by 2%/year.
- ↓ Anemia prevalence by 3%/year.

Integrated with :

- Anemia mukt Bharat.
- National nutritional meals.
- Anganwadi programs.

POSHAN Scheme :

- Pm Poshan SHAKti Nirman.
- AKA previously : mid - day meal scheme.

Governance : ministry of Education.

Launch : 2021.

Goals → Improve nutrition for children.
→ ↑ Literacy rates.

Beneficiaries : Balvatika (Pre-school) + Class I - VIII.

Nutrition :

	Primary classes	upper primary classes
Calorie	450	700
Proteins (gms)	12	20

EPIDEMIOLOGY

----- Active space -----

Basics

00:00:16

Definition :

Given by Dr. John M Last.

Study of : $\left\{ \begin{array}{l} \text{Distribution} \\ \text{Determinants} \end{array} \right\}$ of disease/health related events $\xrightarrow{\text{?}}$ measures : Control/ prevent the same.

Historical Aspect :

		Scientists
Founder of epidemiology		Frascus Torius
Father of	Epidemiology	John Snow
	Evidence based medicine	David Sackett
	medical statistics	John Grant

Tools of measurement :

	Relation b/w numerator & denominator	Expression	Examples
Ratio	unrelated	-	- maternal mortality ratio - Dead/alive
Rate	Numerator is part of denominator	Time	- Incidence - maternal mortality rate
Proportion		%	- Prevalence - Secondary attack rate - Case fatality rate

$$\text{Case fatality rate} = \frac{\text{No. of deaths d/t a disease}}{\text{Total cases of same disease}} \times 100$$

- Indicates : virulence of disease (killing power).

$$\text{Proportional mortality rate} = \frac{\text{Deaths due to a disease}}{\text{Total deaths}} \times 100$$

measures of morbidity :

$$\text{Prevalence} = \text{Incidence} \times \text{Duration.}$$

----- Active space -----

	Calculation	multiplying Factor	measurement Tool
Incidence rate	$\frac{\text{No. of new cases of a disease during given time period}}{\text{Pop. at risk during that period}}$	$\times 1000$	Rate
Prevalence rate	$\frac{\text{No. of current cases (Old + new) of a disease at given point in time}}{\text{Estimated population at same point in time}}$	$\times 100$	Proportion > Rate, ratio

Standardization :

Comparing mortality rates in different population groups with different variables.

Types :

	Direct standardisation	Indirect standardisation
ASDR	Known	Unknown
Steps	Choose reference population ↓ Compare mortality rates	- Choose reference population ↓ Calculate : Standardized mortality ratio (SMR) ↓ Compare mortality rates $SMR = \frac{D_o}{D_e} \times 100$

ASDR : Age Specific Death Rate

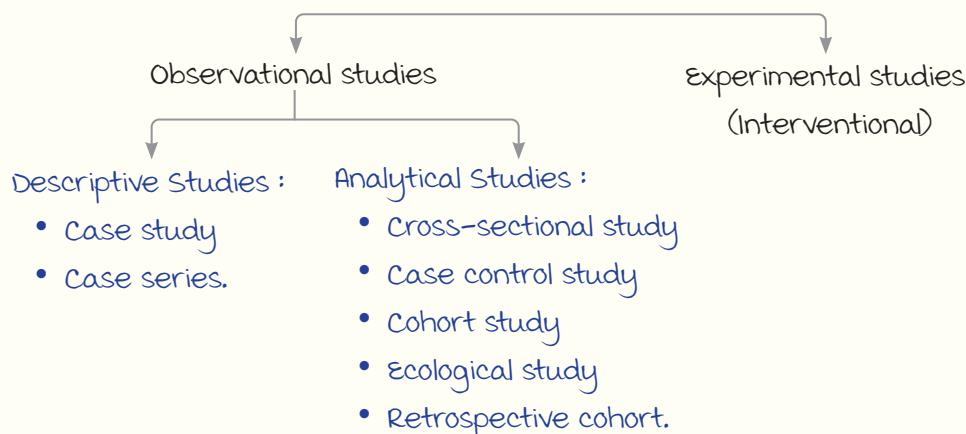
D_o : Deaths observed

D_e : Deaths expected

Observational Studies

00:15:30

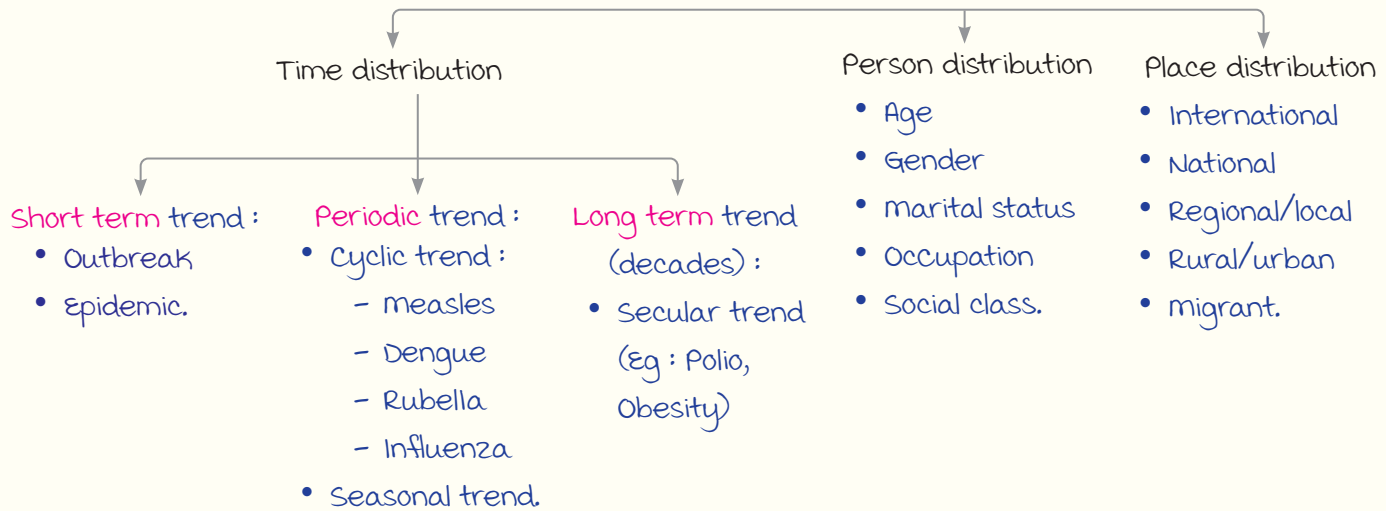
Epidemiological methods :



----- Active space -----

Descriptive Studies :

use : Formulation of hypothesis for unknown disease



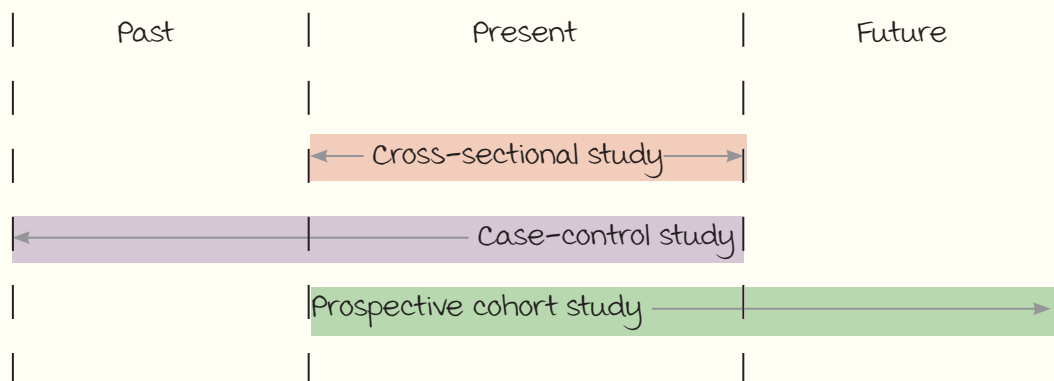
Important age distribution 🌐 :

Disease	Peak age
measles	6 mths – 3 yrs
Chicken pox	1 yr – 10 yrs
Rubella	3 – 10 yrs (Developing countries)
	> 15yrs (Developed countries)
Typhoid	10 – 19 yrs

Diphtheria	< 5 yrs
------------	---------

Analytical Studies 🌐 :

Aim : Testing for probable risk factors



----- Active space -----

	Cross-sectional study	Case-control study	Cohort study	Ecological study
AKA	- Snapshot of population - Transverse study	Retrospective study (↓ Time, ↓ cost)	- Prospective study - Follow-up design	Correlational study
Aim	Prevalence	- Association b/w cause & effect - multiple RF	- Incidence - Natural course - multiple outcomes of disease	Correlation b/w variables
For	Description/analysis	Rare diseases	Rare risk factors (RF)	Compare variables across populations
Unit of Study	Total/sample population ↓ Individuals	Individuals : - Diseased (D^+) : Case - Non-diseased (D^-) : Controls	- Individuals with : Common characteristics D^- → Exposed (E^+) Non-exposed (E^-)	Population
Steps	Survey of area ↓ Detect D^+ & prevalence	$D^+ \& D^-$ ↓ Interview ↓ Assess risk factors (Effect → cause)	D^- ↓ $E^+ \& E^-$ ↓ Follow up Detect D^+ (Cause → effect)	Collect data from records/third party ↓ Compare
Calculation	Odd's ratio		RR, AR, PAR	-

Retrospective cohort study :

Steps : Past medical reports
(Back in time)

Select $E^+ \& E^-$.

Nested case control study :

- Rare investigation analysis
- Type of cohort study.

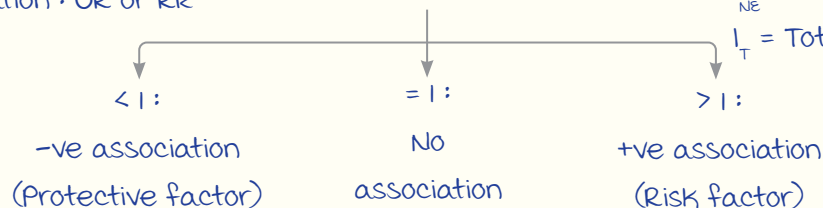
Relevant calculations  :

	D^+	D^-
E^+	a	b
E^-	c	d

----- Active space -----

	Odds ratio (OR)	Risk ratio (RR)	Attributable risk (AR)	Population AR (PAR)
Description	Cross product ratio	Relative risk	% of disease d/t RF among E^+ population	Change in D^+ in pop. if RF is removed
Calculation	$\frac{a \times d}{b \times c}$	$\frac{I_E}{I_{NE}} = \frac{a}{a+b} \bigg/ \frac{c}{c+d}$	$\frac{I_E - I_{NE}}{I_E}$	$\frac{I_T - I_{NE}}{I_T}$
Only use	Case-control study	Cohort study	Cohort study	Cohort study

Interpretation : OR or RR

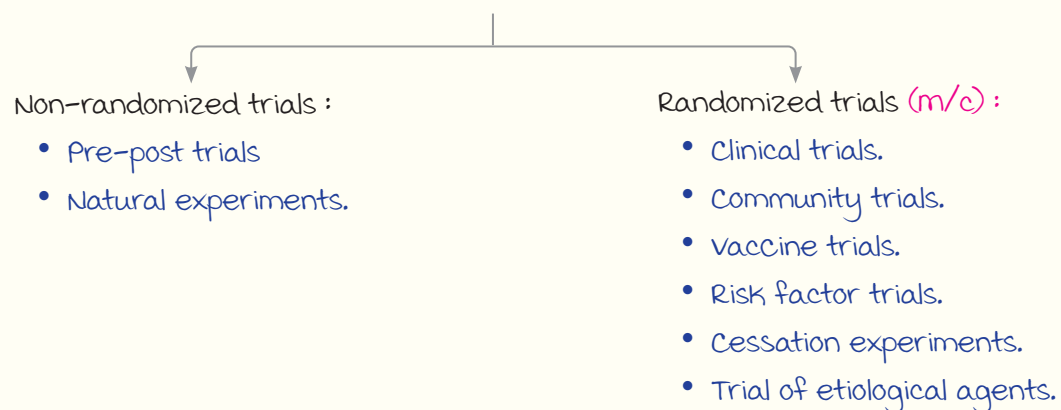
 I_E = Incidence (E^+) I_{NE} = Incidence (E^-) I_T = Total population

Experimental/Interventional Studies

00:42:09

Conducted to confirm hypotheses.

Classification :



Randomization :

Concept : Random allocation into groups.

Done at level of : Allocation.

Advantages :

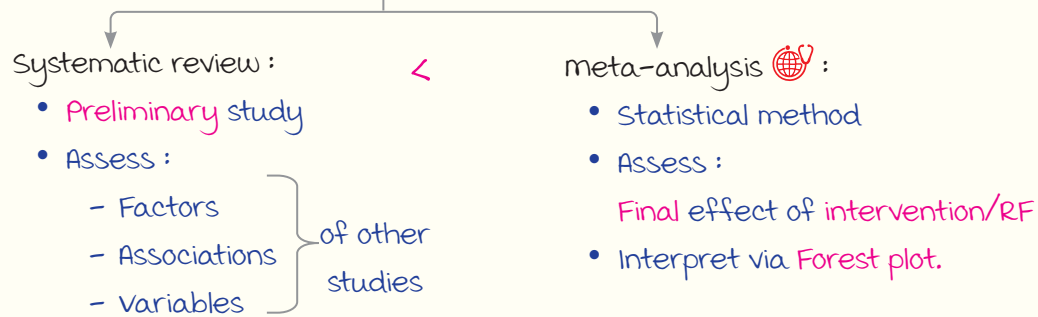
- Known & equal chance.
- Removes confounding & selection bias.

Evidence Based Medicine

00:45:16

----- Active space -----

Study of previously conducted research to confirm hypotheses.



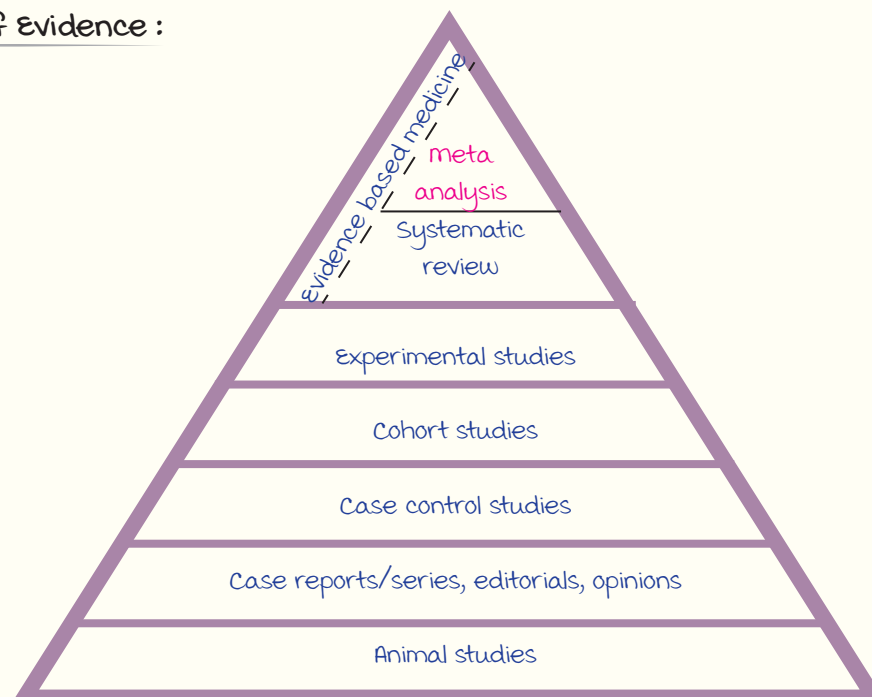
Clinical Trials, Levels of Evidence

00:47:44

Clinical Trials 🌐 :

Phases	Comprises	Participants	
0	microdosing (in humans at very low/ subtherapeutic doses)	-	
1	• Toxicity • Pharmacokinetics/pharmacodynamics	< 100	Approved for testing in humans
2	• Efficacy • Dose range	100 - 300	
3	• Therapeutic confirm trials • Confirms efficacy	1000s	Submitted for FDA approval
4	Post marketing surveillance	-	Approved
5	Pharmaco-epidemiology	multiple	

Levels of Evidence :



----- Active space -----

Errors in Epidemiology

00:49:40

**Bias** 🌐 :

Types of bias	Characteristics	Remarks
Neyman	Differential mortality pattern b/w groups	AKA : Incidence-prevalence bias
Berksonian	Differential hospital admission rates	Hospital based case-control study
Hawthorne effect	Change in attitude while under observation	Seen in cohort
Recall	Differential ability to recall b/w cases & controls	Seen in case-control
Attrition	Loss to follow up	Seen in : Cohort studies, RCT, trials
Selection	Differential selection into groups	-
Ecological fallacy	Results of ecological study when not applicable for individuals	-

Prevention : Blinding

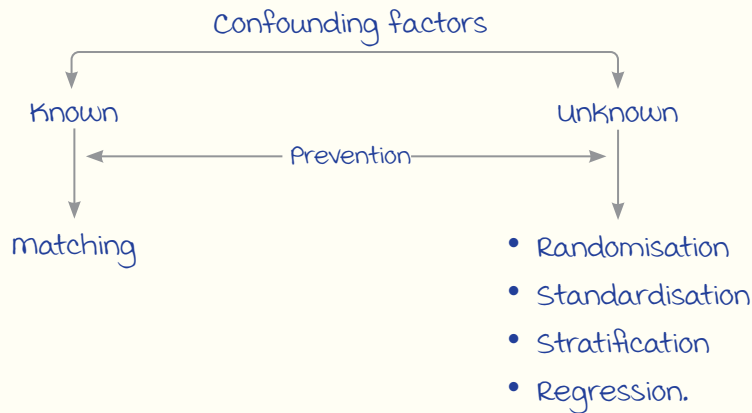
	Blinded from	Significance
Single blinding	Subject	-
Double blinding	Subject + Doctor	m/c
Triple blinding	Subject + Doctor + Statistician	Best

Confounding Factor 🌐 :

A third variable a/w :

- Risk factor.
- Disease.
- Directly & indirectly can lead to disease.

Types :



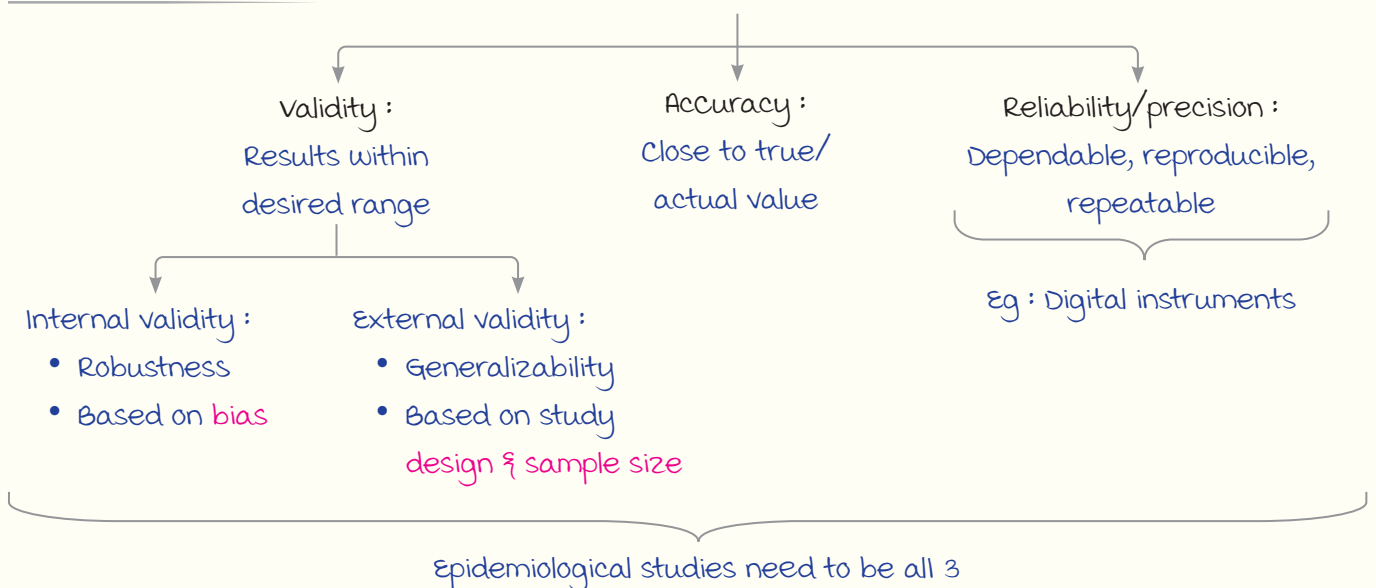
----- Active space -----

Miscellaneous

01:00:56

Hill's Criteria of Causal Association :

Criteria	Description	
1. Biological plausibility	Sound biological explanation required	
2. Specificity of association	Same RF → Same disease	Least imp & difficult (D/t multifactorial nature)
3. Coherence of association	RF $\xrightarrow{\text{consistently leads to}}$ Same disease	
4. Validity of association	OR, RR : +ve association	
5. Dose-response relationship	RF $\propto$ Disease	
6. Temporality	RF precedes disease	most essential

Assessment metrics :

SCREENING OF DISEASE

Types of Screening

00:01:10

Prescriptive Screening :

2° level of prevention : Early dx → Detect disease → Prevents complications.

Examples :

- Fecal occult blood test : CA colon.
- mammography : CA breast.
- Fasting blood glucose/HbA1c : Diabetes.
- Visual investigation with acetic acid/ PAP smear : Ca cervix.
- Urinary ALA levels : Plumbism ($\uparrow$ Pb).
- VDRL : Syphilis.

Presumptive/Prospective/Predictive Screening :

1° level of prevention → Prevent disease.

Examples :

- HIV screening
 - In mothers in ANC : Opt-out testing.
 - During blood donation : Unlinked anonymous testing.
- Pre-employment medical check (Eg : Chefs for typhoid).
- COVID screening in airports.
- Immigrant screening protocol.

Properties of Screening Tests

00:04:55

	Disease (D+)	No disease (D-)	
Test (+)	True positive (TP)	False positive (FP)	TP + FP
Test (-)	False negative (FN)	True negative (TN)	FN + TN
	TP + FN	FP + TN	

	Probability of having	Out of total	Formulae
Sensitivity (Sn)	Test +ve	Diseased	TP/(TP + FN) or TP/D+
Specificity (Sp)	Test -ve	Healthy	TN/(FP + TN) or TN/D-
Positive predictive value (PPV)	Disease	Tested +ve	TP/(TP + FP) or TP/T+
Negative predictive value (NPV)	No disease	Tested -ve	TN/(TN + FN) or TN/T-

Properties affecting PPV :

- Prevalence (most important).
- Sn.
- Sp.

↑ FP d/t :

- ↓ Sp.
- ↓ Prevalence.
- ↑ Sn.

----- Active space -----

Screening test : Sn used (Rule out disease).

Diagnostic test : Sp used (Rule in disease).

Likelihood ratio : Chance of having disease.

- For positive test : $TP/FP \sim \frac{Sn}{(1 - Sp)}$.
- For negative test : $FN/TN \sim \frac{(1 - Sn)}{Sp}$.

Bayes' formula :

To calculate :

- $PPV = \frac{Sn \times \text{Prevalence}}{(Sn \times \text{Prevalence}) + [(1 - Sp) \times (1 - \text{Prevalence})]}$
- $NPV = \frac{Sp \times (1 - \text{Prevalence})}{Sp \times (1 - \text{Prevalence}) + [(1 - Sn) \times \text{Prevalence}]}$

Combination of Tests :

Series : T_a only if T_1 is $\oplus$ and so on.

Parallel : All test occur simultaneously.

	Sensitivity	Specificity	NPV	PPV
Series	Low	High	Low	High
Parallel	High	Low	High	Low

BIOSTATISTICS

----- Active space -----

Data

00:00:58

Types :

Quantitative	Qualitative
Continuous data	Discrete data
measurable	Countable
E.g. : Height, weight, age, serum levels of various lab tests	E.g. : Gender, blood group, religion, type of disease

Scales :

	Nominal (N)	Ordinal (O)	Interval (I)	Ratio (R)
Features	Named data with no order	Data with order or ranks	No absolute zero/start point	Has absolute zero/start point
E.g.	Type of disease, blood groups, religion	Grade of disease, disease severity	°C or °F	Height, weight, age, hemoglobin levels, temperature in Kelvin scale

Types of ordinal scales :

- **Likert scale** : Agree/disagree continuum.
- **Guttman scale** : Cumulative/summative scale.

Measures

00:10:20

measures of Central Tendency :

	mean	median	mode
Definition	Average of all the values	Central value after arranging in ascending/descending order	most frequently occurring value
Feature	most affected by extreme values	Least affected by extreme values	Last to change with data variations (most robust value)

$$\text{mode} = 3(\text{median}) - 2(\text{mean}).$$

measures of variations :

Range : maximum value - minimum value.

Standard deviation (SD) :

$$\text{RMSD} : \text{Root of mean of Squared Deviation ; } SD = \sqrt{\frac{\sum(x - \bar{x})^2}{n}}$$

x : variable value. n : Sample size

$\bar{x}$: mean. Σ : Summation.

$x - \bar{x}$: Deviation.

Variance : mean of squared deviation (SD^2).

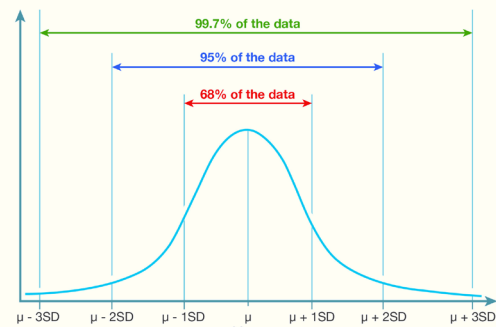
Distribution Curves

00:13:50

----- Active space -----

Normal Distribution Curve :

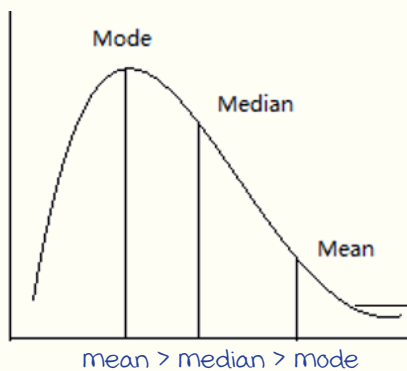
- AKA Gaussian distribution.
- B/L symmetrical, **bell-shaped** curve.
- Ends never touch baseline.
- **mean = median = mode** : Coincide at **centre**.
- $SD = 1$.
- $Variance = 1$.
- Area under curve (AUC) = 1 (100% of values).



Normal distribution curve

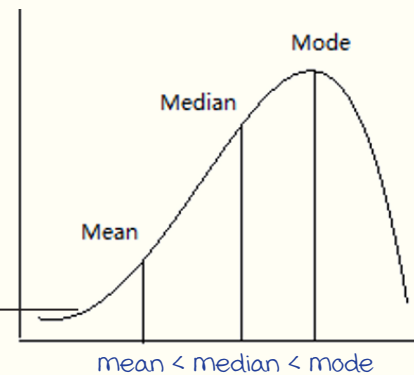
Skew :

Positive/right skew :



Skew direction :
Tail of graph

Negative/left skew :



P-value, Errors & Test of Significance

00:20:43

P-value (Probability) :

Probability of finding the effect.

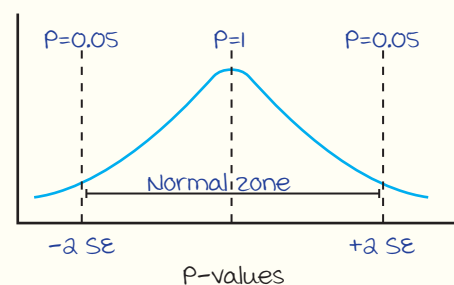
Terminologies :

- Population/values : Confidence interval (CI).
- $SD =$ Standard error (SE).

Range : 0 - 1.

Normal zone : $\pm 2 SD/SE = 95\% CI = 0.05$ p-value.

Types :



	Significant	Non-significant
P-value	≤ 0.05	> 0.05
P-value w.r.t. normal zone	Outside	Within
Variation among groups	Abnormal	(N)
Effect	$\oplus$	$\ominus$
$N^o H$	Rejected	Accepted

----- Active space -----

Errors :

Null hypothesis (H_0) : No difference b/w groups being compared.

Types :

		H_0	
		In reality	In research
Error	Type I	True (Effect $\ominus$)	Rejected (Effect $\oplus$)
	Type II (β)	False (Effect $\oplus$)	Accepted (Effect $\ominus$)

Tests of Significance :

mathematical formula to derive p-value.

	Parametric	Non-parametric
Type of data	Quantitative	Qualitative
Number of groups :		
1 group	Paired t-test	mc Nemar test
2 groups	Independent sample/unpaired t-test	Chi square test
≥ 3 groups	ANOVA	Kruskal Wallis test

ANOVA : Analysis of variance.

unpaired T- test : AKA student t-test.

Other Statistical Tests :

Kappa statistics :

- To check level of agreement b/w observers.
- Formula =
$$\frac{\text{Observed agreement} - \text{Expected agreement}}{1 - \text{Expected agreement}}$$

Kolmogorov Smirnov test : Normalcy of data.

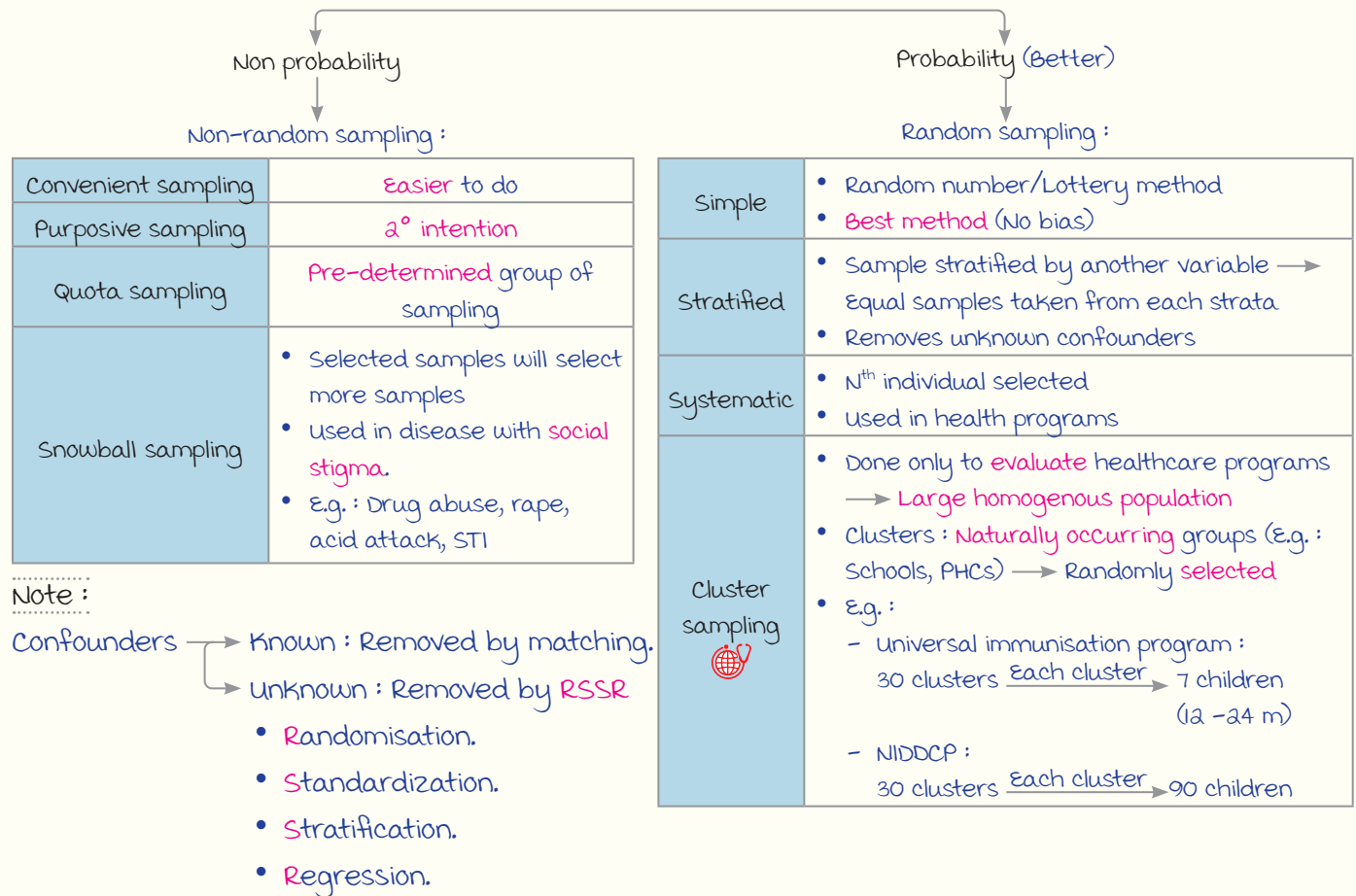
Kurtosis : Skewness of data.

Wilcoxon rank test : Ordinal data $\rightarrow$ Grouped : Sign test.
 $\rightarrow$ Ungrouped : Sum test.

Sampling Methods

00:37:07

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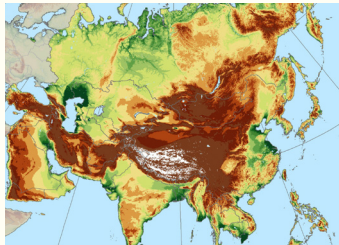
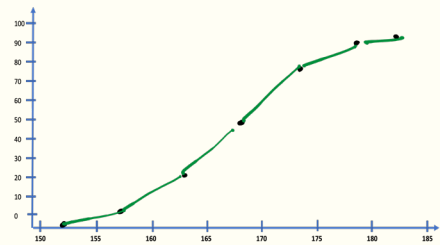
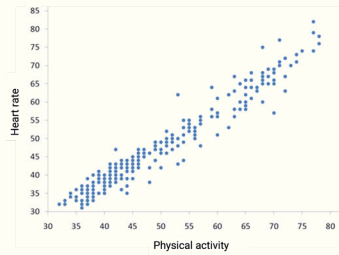
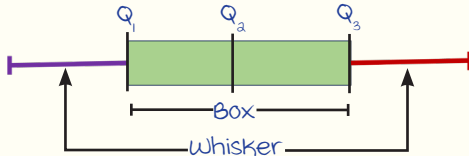
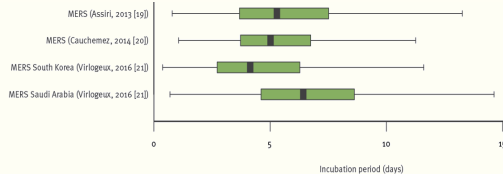
Graphical Presentations

00:43:15

Quantitative	Qualitative
- Histogram - Frequency polygon/chart - Scatter plot - Line diagram - Ogive	- Bar chart - Pie chart - Pictogram - Spot maps - Venn diagram - Choropleth

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Charts	Description	Image																																	
Bar chart	- Distantly placed column charts - used : - Qualitative data - Frequency on other axis	79% of new recovered cases in the 10 States & UTs <table><thead><tr><th>State/UT</th><th>Recovered Cases</th></tr></thead><tbody><tr><td>Maharashtra</td><td>11,060</td></tr><tr><td>Kerala</td><td>7,854</td></tr><tr><td>Delhi</td><td>6,121</td></tr><tr><td>West Bengal</td><td>4,283</td></tr><tr><td>Karnataka</td><td>2,701</td></tr><tr><td>Andhra Pradesh</td><td>2,452</td></tr><tr><td>Tamil Nadu</td><td>2,402</td></tr><tr><td>Uttar Pradesh</td><td>2,167</td></tr><tr><td>Rajasthan</td><td>1,810</td></tr><tr><td>Chhattisgarh</td><td>1,753</td></tr></tbody></table>	State/UT	Recovered Cases	Maharashtra	11,060	Kerala	7,854	Delhi	6,121	West Bengal	4,283	Karnataka	2,701	Andhra Pradesh	2,452	Tamil Nadu	2,402	Uttar Pradesh	2,167	Rajasthan	1,810	Chhattisgarh	1,753											
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Histogram	- Joined column chart - used : - Quantitative data - Frequency on another axis	<table><thead><tr><th>Height (feet)</th><th>Frequency</th></tr></thead><tbody><tr><td>60-65</td><td>3</td></tr><tr><td>65-70</td><td>3</td></tr><tr><td>70-75</td><td>8</td></tr><tr><td>75-80</td><td>10</td></tr><tr><td>80-85</td><td>5</td></tr><tr><td>85-90</td><td>2</td></tr></tbody></table>	Height (feet)	Frequency	60-65	3	65-70	3	70-75	8	75-80	10	80-85	5	85-90	2																			
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Frequency chart/ polygon curve	Formed by joining the midpoints of histogram	<table><thead><tr><th>Height (in cm)</th><th>Number of students</th></tr></thead><tbody><tr><td>135</td><td>0</td></tr><tr><td>140</td><td>80</td></tr><tr><td>145</td><td>100</td></tr><tr><td>150</td><td>160</td></tr><tr><td>155</td><td>140</td></tr><tr><td>160</td><td>120</td></tr><tr><td>165</td><td>20</td></tr><tr><td>170</td><td>0</td></tr><tr><td>175</td><td>0</td></tr><tr><td>180</td><td>0</td></tr></tbody></table>	Height (in cm)	Number of students	135	0	140	80	145	100	150	160	155	140	160	120	165	20	170	0	175	0	180	0											
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Line diagram or chart	- Always b/w time and frequency on the other axis - AKA time trend chart	MRSA bacteraemia rates April 08 - March 2017, per 100,000 bed days (acute Trust attributable / PIR apportioned) <table><thead><tr><th>Date</th><th>RSCH Rate</th><th>National Rate</th></tr></thead><tbody><tr><td>Apr 08</td><td>3.2</td><td>4.5</td></tr><tr><td>Apr 09</td><td>2.0</td><td>2.8</td></tr><tr><td>Apr 10</td><td>1.5</td><td>2.2</td></tr><tr><td>Apr 11</td><td>1.2</td><td>1.8</td></tr><tr><td>Apr 12</td><td>1.0</td><td>1.5</td></tr><tr><td>Apr 13</td><td>1.2</td><td>1.3</td></tr><tr><td>Apr 14</td><td>1.5</td><td>1.1</td></tr><tr><td>Apr 15</td><td>1.0</td><td>0.8</td></tr><tr><td>Apr 16</td><td>1.5</td><td>1.0</td></tr><tr><td>Apr 17</td><td>1.2</td><td>1.0</td></tr></tbody></table>	Date	RSCH Rate	National Rate	Apr 08	3.2	4.5	Apr 09	2.0	2.8	Apr 10	1.5	2.2	Apr 11	1.2	1.8	Apr 12	1.0	1.5	Apr 13	1.2	1.3	Apr 14	1.5	1.1	Apr 15	1.0	0.8	Apr 16	1.5	1.0	Apr 17	1.2	1.0
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Pie chart	- Circular/sectoral presentation - Always in percentage - Best for technical data - For qualitative data	<table><thead><tr><th>Cause</th><th>Percentage</th></tr></thead><tbody><tr><td>Perinatal causes</td><td>71%</td></tr><tr><td>Other causes</td><td>9%</td></tr><tr><td>Infectious & Parasitic Diseases</td><td>7%</td></tr><tr><td>Diseases of the Respiratory System</td><td>5%</td></tr><tr><td>Congenital malformations, deformations and chromosomal abnormalities</td><td>5%</td></tr><tr><td>Diseases of the Circulatory System</td><td>3%</td></tr></tbody></table>	Cause	Percentage	Perinatal causes	71%	Other causes	9%	Infectious & Parasitic Diseases	7%	Diseases of the Respiratory System	5%	Congenital malformations, deformations and chromosomal abnormalities	5%	Diseases of the Circulatory System	3%																			
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Venn diagram	For overlapping qualitative data																																		

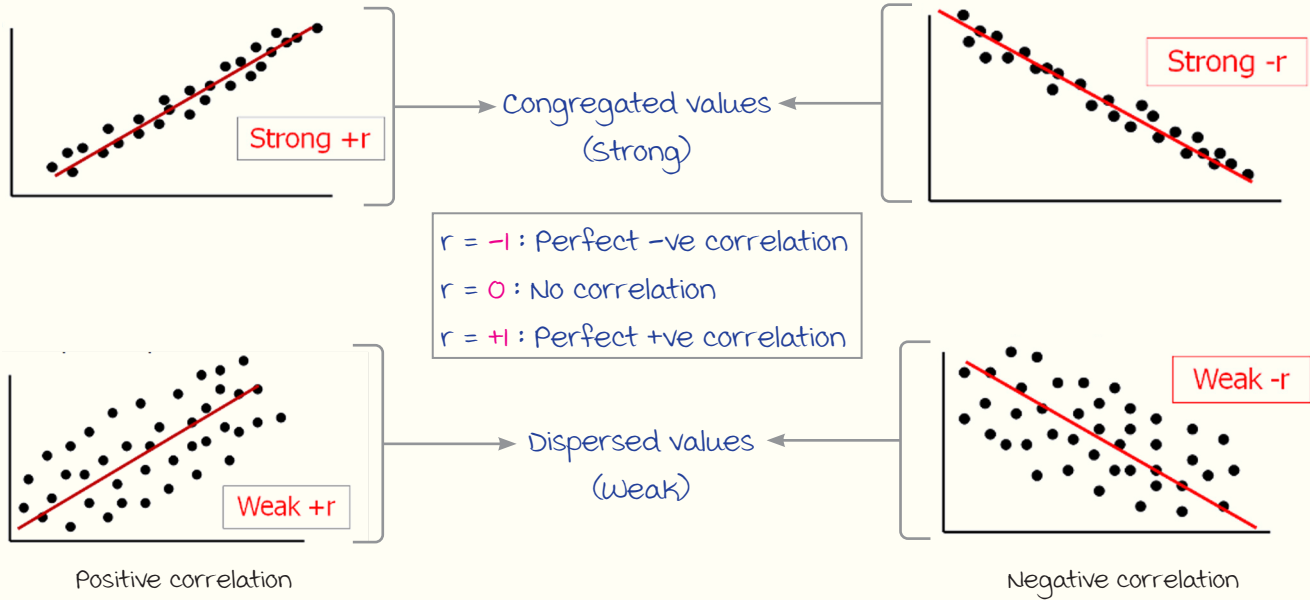
Charts	Description	Image
Spot map	Geographical location of data on a map	
Choropleth	Geographic intensity of data on a map	
Pictograph	- Symbolic presentation of data - Best for general population, to create mass awareness	
Ogive	- For cumulative frequency and variable - Curve always uprising - Helps to define : - Class interval - Cut off	
Scatter plot/ Correlation curves	- Always b/w 2 independent quantitative variables - Used to find the correlation b/w variables - Shows direction and strength or degree of relation	
Box and whisker plot	- used to find out data distribution - Based on quartile - Number of values in the box = 50% - Inter-quartile range : $Q_3 - Q_1$ (Size of box)	 

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Pearson's Correlation Coefficient (r) 🌐 :

used in scatter plot : To find correlation b/w variables.



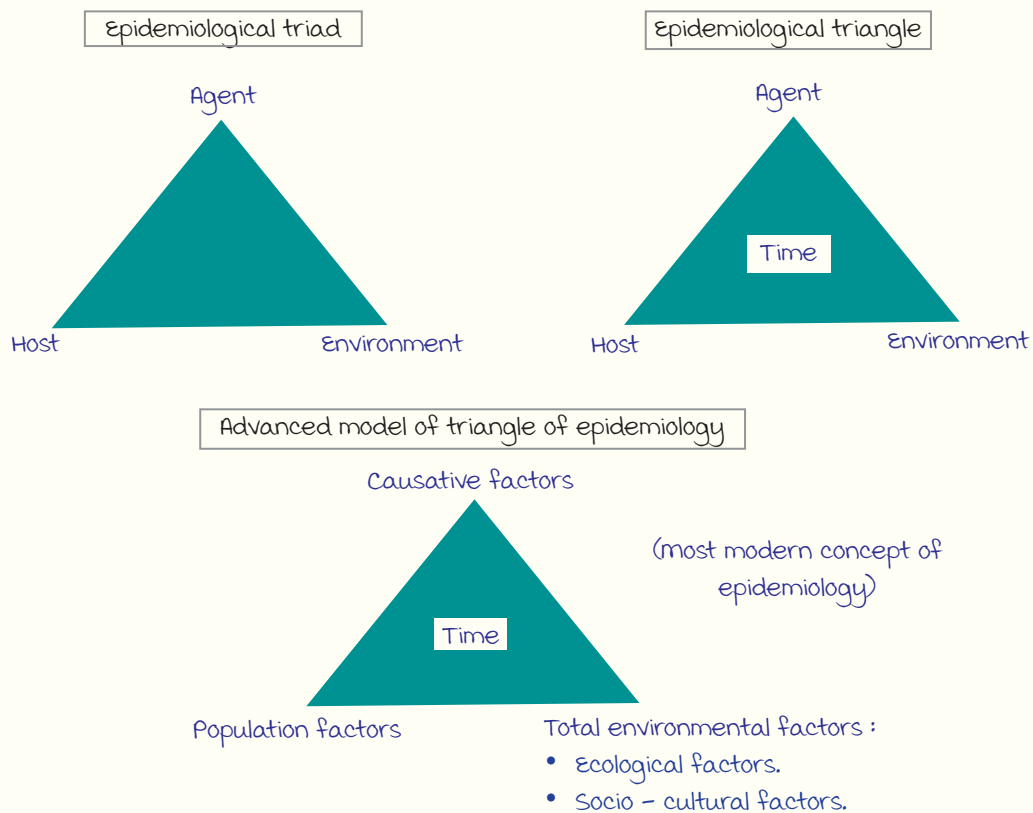
INFECTIOUS DISEASE EPIDEMIOLOGY

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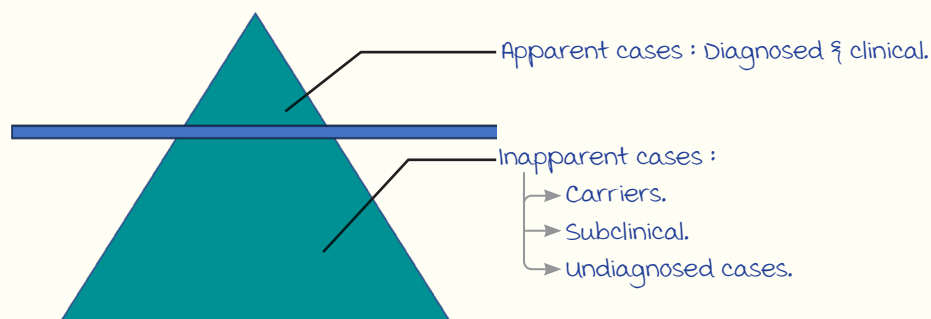
Basics of Epidemiology

00:00:40

Disease Causation :



Iceberg Phenomenon of Disease :



Diseases that do not show iceberg phenomenon :

Mnemonic : **mTR**

- **m**easles.
- **T**etanus.
- **R**abies.

----- Active space ----- **Levels of Prevention**  :

	Risk factor	Disease	Complication	Disability/death	Examples
Primordial (Ideal)	⊖	⊖	⊖	⊖	- Lifestyle modification - Health education - Safe water & sanitization - Food fortification
Primary (Best)	⊕	⊖	⊖	⊖	Specific protection : - Immunization - Pre exposure prophylaxis - Chemoprophylaxis - Pre employment checkup - Presumptive screening - Food fortification d/t deficiency ⊕ - Promote condoms among sexual workers - Awareness of leprosy in community
Secondary (m/c)	⊕	⊕	⊖	⊖	Early diagnosis → Prompt treatment: - Physical exercise to diabetic patient - Prescriptive screening - monitoring BP in HTN patients - Self breast examination
Tertiary (Delayed)	⊕	⊕	⊕	⊖	Rx disability : - Rehabilitation - Prosthesis - Pacemakers - Hearing aids - Transplants

Infection Time Variables

00:11:40

----- Active space -----

Time Intervals :

Incubation period (IP)  : Entry of organism $\rightarrow$ 1st sign & symptom.

Disease	Incubation period (Days)
Chicken pox	10 - 21
measles	8 - 16
Rubella	14 - 21
Hepatitis A	15 - 50
COVID-19	3 - 12
Influenza	1 - 3

Generation time : Entry of organism $\rightarrow$ maximum infectivity/communicability.

Serial interval :

- 1^o case $\rightarrow$ 2^o case.
- Proxy indicator of IP.

Latent period :

Risk factors onset $\rightarrow$ Signs & symptoms of **non-communicable** (Chronic) diseases.

Isolation vs. Quarantine :

	Isolation	Quarantine
Done for	Clinical cases	Apparently healthy individuals
maximum duration	Communicable period	Incubation period

Isolation periods  :

Disease	Duration of isolation
Chicken pox	until all lesions are crusted (About 6 days after onset of rash)
measles	7 days after onset of rash
Hepatitis A	3 weeks of onset
COVID-19	14 days
Tuberculosis	until 3 weeks of effective chemotherapy
meningococcal meningitis Streptococcal pharyngitis	until the first 6 hours of effective therapy are completed
Diphtheria	> 2 -ve culture reports after start of Rx (24 hours apart)

----- Active space -----

Source Variables

00:15:30

Natural habitat	Infectious agent
Reservoir	Lives and multiplies
Source	Lives, multiplies & transmits

Host :

		1°/Definitive	2°/Intermediate
maturation of organism		Sexual	Asexual
Examples	Filaria	man	Culex mosquito
	malaria	Anopheles mosquito	man
	Tapeworm	man	Pig (T. solium), cattle (T. saginata)
	Guinea worm	man	Cyclops
	Hydatid disease	Dog	Sheep, cattle, man
	Sleeping sickness	man	Tsetse fly
	Toxoplasmosis	Cat	Rodent, man

Types :

	Case	Subclinical case	Carrier
Risk factor	⊕	⊕	⊕
Pathology	⊕	⊕	⊕
Signs/symptoms	⊕	⊖	⊖
Transmission	⊕	⊖	⊕

Cases & Carriers

00:19:02

Cases :

Case	In community
Primary	1 st case
Secondary	Subsequent cases arising from 1° within 1 IP
Index	1 st observed case

Attack Rates :

Attack rate : AKA Special incidence rate.

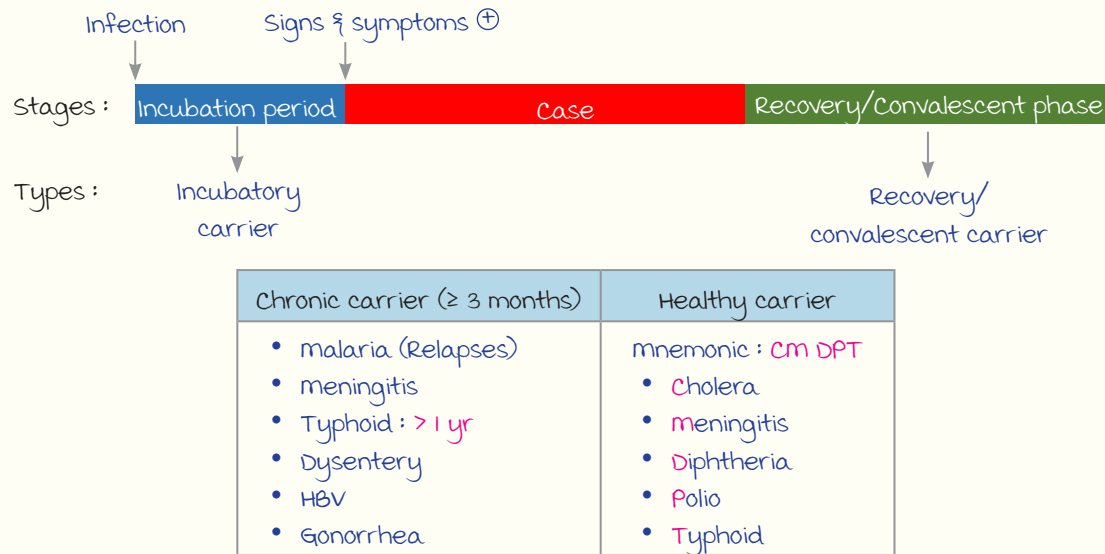
$$= \frac{\text{Total number of cases in defined area}}{\text{Total number of susceptible population}} \times 100$$

Secondary attack rate (SAR)  :

$$= \frac{\text{Total number of 2° cases}}{\text{Total number of susceptible population} - 1° \text{ cases}} \times 100 (\%)$$

Carriers :

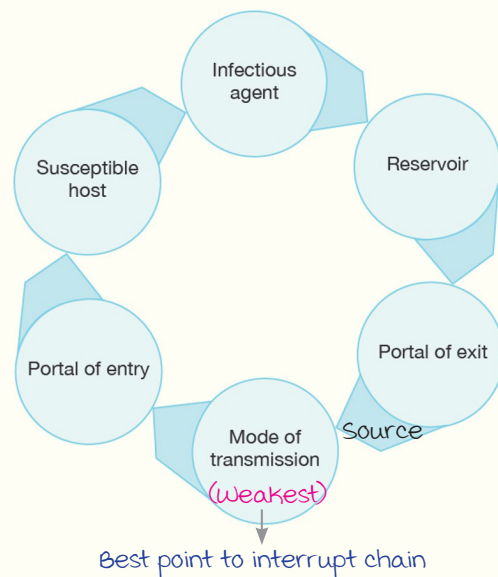
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Healthy carrier : From subclinical case/non-virulent strain.

Mode of Transmission & Precautions

00:26:10

Chain of Infection :modes of Transmission 🌐 :

Direct transmission	Indirect transmission
Droplet transmission	Fomite borne (Non-living object)
vertical/transplacental	Airborne (Dust/droplet nuclei)
Inoculation on skin/mucosa (Bite/needle injury)	Vehicle borne (Food/blood borne)
Direct contact	Unclean hands
Contact with soil : Helminthic infection	Vector borne → mechanical Biological
mnemonic : DTICS	mnemonic : FAV UV

----- Active space -----

Airborne mode :

Droplets :

- Range : 30 - 60 cm.
- **Direct** mode of transmission.

Droplet nuclei :

- **Dried residue** of fine droplets.
- Indirect mode of transmission.

**Airborne.**

Biological transmission :

	Change in	Examples
Propagative	Number	• Yellow fever in Aedes • Plague in rat flea
Cyclo-developmental	Shape, size, stage	microfilaria in Culex
Cyclo-propagative	Number, shape, size & stage	Plasmodia in Anopheles

Entomological diseases 🌐 :

Louse	Flea	Tick
mnemonic : RETIred • Relapsing fever (B. recurrentis) • Epidemic typhus (R. prowazekii) • Trench fever (Rickettsia quintana) • Dermatitis	• Plague • Endemic typhus • Chiggerosis	• Tick typhus • Tick paralysis • Tularemia • Viral hemorrhagic fever • Babesiosis

Level of Precautions :

Condition	Precautions
Abscess, diarrhoea, vomiting, hepatitis, pregnancy, general OPD	Contact/droplet
Rash, fever, cough	Droplet/airborne
TB, measles, COVID, influenza, chicken pox	Airborne

PPE :

- Droplet : Face mask.
- Airborne : N-95 mask.

Epidemic Variables

00:36:45

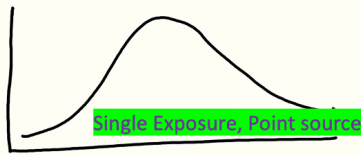
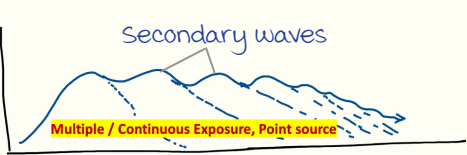
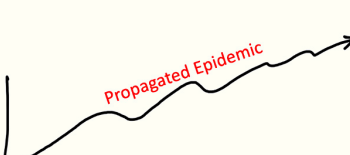
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Types :

	Sporadic	Endemic	Epidemic	Pandemic
Definition	Scattered number of cases.	Persistent presence of disease in a defined geographical area.	Sudden rise in number of cases : • $> +2 \text{ SD}$. • $> 80\%$ of expected frequency.	Spread of disease over 2 continents/ 2 WHO regions.
Diseases	- Tetanus - Anthrax - GBS	- malaria - Dengue - Typhoid - Cholera	-	- Plague - Influenza - Polio - COVID-19 - Yellow fever

Notifiable (priority) diseases : Reported to WHO within 24 hrs of diagnosis.

Epidemic Patterns 🌐 :

	Single/Point/Common source		multiple source
	Single exposure	multiple exposure	
AKA	Point epidemic	Continuous epidemic	Propagated epidemic
Frequency	Rapid rise → Rapid fall	Rapid rise → Persistent ↑ cases	↑ Cases, large outbreaks
Features	Cases within 1 IP	$> 1 \text{ IP} \rightarrow \text{Endemicity } \oplus$	- Person to person/animal transmission - multiple waves
Examples	- Disastrous explosion - Food poisoning	- Contaminated well - Hepatitis A/typhoid carrier chef	- Gonorrhoea - RTI/STI - COVID-19
Graphs			

Note :

Virulence of disease → By case fatality rate.

COMMUNICABLE DISEASES

Chicken Pox And Measles

00:00:35

Epidemiology & Clinical Features :

	Chicken pox	measles
Etiology	Varicella zoster virus	ssRNA morbilliform virus : Paramyxoviridae family
Peak age	1 - 10 years (m/c : 3 - 6 yrs)	6 months - 3 years
Incubation period	10 - 21 days (median : 14 - 16 days)	10 - 14 days
mode of transmission	Airborne > Droplet	
Period of communicability	← 1 - 2 days Rash onset 4 - 5 days →	← 4 days Rash onset 4 days →
Isolation period	6 - 7 days after onset of rash/Till scab (Non-infective) formation	6 days after onset of rash
2° attack rate	80 - 85%	
Case fatality rate	< 1%	-
Rash and other features	Vesicular rash : 1 st on trunk	• High grade fever • maculopapular rash : Post-auricular region → Face • 3 Cs : (m/c) - Conjunctivitis - Coryza - Cough • Diarrhea • malnutrition • Vitamin A deficiency
Complications	• In neonates/immunocompromised : Pneumonia • In early pregnancy : Congenital varicella syndrome ↓ Symptoms : - microcephaly - LBW - microphthalmia - Limb atrophy - micrognathia - Deafness	• Otitis media (m/c) • SSPE : Sub acute sclerosing panencephalitis (Rare, specific) • Pneumonia (m/c fatal complication)

Vaccine :

Chicken pox :

- Strain : OKA (Live vaccine).
- Dosage : 0.5 mL SC at 12 - 18 months.
- 2 doses given
 - < 12 y : 1 - 3 months apart.
 - > 12 y : 1 - 1.5 months apart.
- Duration of protection : 10 years.

measles :

----- Active space -----

- Strain : Edmonston - Zagreb (Live vaccine).
- Dosage : 0.5 mL, SC, right upper arm.
- Reconstituent : Distilled water (Use within 6 hrs).
 - No open vial policy.
- Stabilizer : Sorbitol, gelatin (Possible allergens).
- Antibiotic : **Neomycin > Erythromycin**.
- Schedule (UIP) : Given as MR vaccine.
 - 2 doses
 - 1 : At 9 months.
 - 2 : At 16 - 24 months.
 - max age : 5 yrs (Catch - up).

Post - Exposure Prophylaxis :

Chickenpox : (In case of outbreak, **isolation done first**.)

	Post-exposure prophylaxis	Administration
Immunocompetent	monovalent vaccine : OKA strain	within 5 days of exposure
Pregnant/ Immunocompromised	Varicella Zoster Immune Globulin (VZIG) : Dose : 12.5 u/kg $\xrightarrow{3 \text{ weeks}}$ Repeat dose in severe cases	within 72 hours of exposure

measles :

	Post - exposure prophylaxis	Administration
Immunocompetent	measles vaccine : • Effect : Within 11 - 12 days • Duration of protection : Lifelong	within 3 days of exposure
Pregnant/ Immunocompromised	Immunoglobulin : 0.5 mL/kg	within 4 - 6 days of exposure

Note :

vaccine & immunoglobulin not given together.

Campaign dose/mv-0 dose :

- **Extra dose** of measles vaccine given around **6 - 9 months** of age.
- Indications :
 - measles outbreak.
 - Conflict zones/Refugee camps.
 - Known contact of measles in day care centres.
- **MR-1** : Given as per schedule.
 - Gap maintained : **> 4 weeks**.

----- Active space -----

Rubella

00:14:50

- AKA 3 day fever/German measles.
- mild, self-limiting disease.

Epidemiology :

50 - 60% of cases : Subclinical.

Peak age :

- Developing countries : 3 - 10 years.
- Developed countries : > 15 years.

Incubation period : 10 - 21 days.

Period of communicability : 7 days before & after onset of rash.

Trend of infection : Outbreaks seen every 6 - 9 years.

Post - Exposure Prophylaxis :

Rubella vaccine :

- RA 27/3 (Live vaccine).
- Administration : Single dose after 12 months of age.
- Duration of protection : 14 - 16 years.

Rubella immunoglobulin : Pregnant females with known exposure.

Indication during outbreaks :

1. Non-pregnant women of reproductive age.
 - Avoid pregnancy 1 - 3 months post vaccination.
2. Adolescent females.
3. Infants.

Diphtheria

00:19:16

Epidemiology :

Etiology :

Corynebacterium diphtheriae : Gram +ve, non-motile



Exotoxin release.

Incubation period : 2 - 6 days.

Presentation : Asymptomatic > Symptomatic cases.

Period of communicability : 2 - 4 weeks after onset of disease.

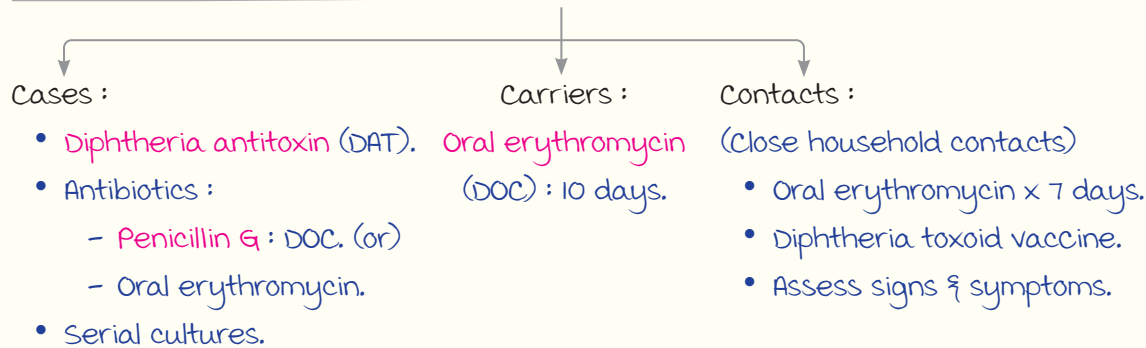
Isolation criteria : until 2 negative throat cultures (24 hours apart).

Carrier stages :

- Healthy.
- Convalescent.
- Incubatory.

Prevention & Control :

----- Active space -----



DAT doses :

Clinical presentation	DAT dose (IU)
Pharyngeal/laryngeal disease	20,000 - 40,000
Nasopharyngeal disease	40,000 - 60,000
Extensive disease	80,000 - 100,000

Diphtheria toxoid vaccine indications :

- > 3 doses + last dose < 5 yrs → No extra vaccine.
- > 3 doses + last dose > 5 yrs → Single dose diphtheria toxoid.
- unknown status/< 3 doses → Diphtheria toxoid + complete diphtheria series.
(2 primary + 2 booster)

Hepatitis B

00:23:22

Epidemiology :

15 - 20% of Indian population : HBsAg +ve.

Causative organism :

ds DNA virus (Dane particle).

- Heat labile.
- Susceptible to hypochlorite disinfectants.

Reservoir : Human.

Incubation period : 1 - 6 months.

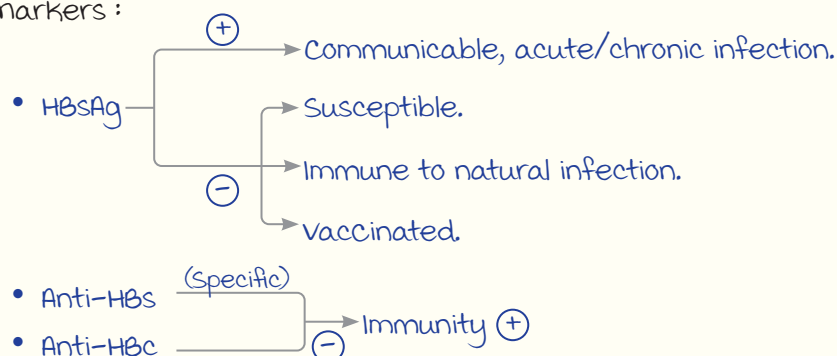
Period of communicability : Till

HBsAg +ve in blood.

mode of transmission : Parenteral.

Presentation : Acute/chronic hepatitis.

markers :



----- Active space -----

Hepatitis B Vaccine :

Recombinant vaccine.

Dose :

- At birth : 0.5 mL Im.
 - Site : Left anterolateral aspect of thigh.
 - Given within 24 hrs → Prevents mother to child transmission.
- Adult : 1 mL (Lifelong).

Contraindication : Anaphylaxis.

Cold chain storage : Freeze sensitive, hence

- Top shelf of ILR.
- Lowest shelf of fridge.

Schedule : 0, 1, 6 months. → Resume if interrupted.

Hepatitis B Immunoglobulin :

Indication : Accidental needle injuries → Check HBsAg in HCP

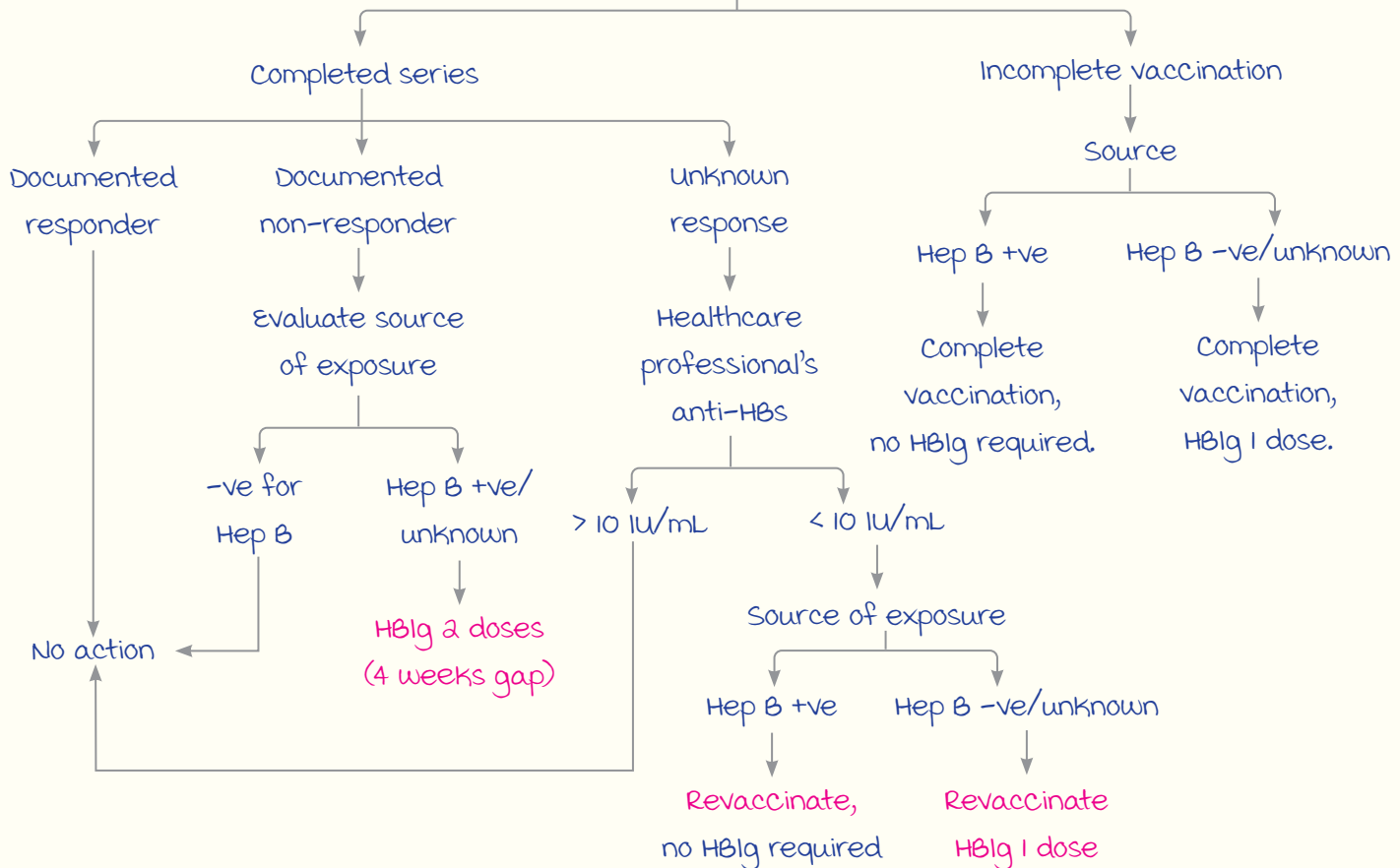
- +ve : No vaccination.
- -ve : Give vaccine.

Timing : Immediately (< 48 hours).

Dose :

- 0.05 - 0.07 mL/kg body weight.
- Given in 2 - 3 doses.

CDC guidelines for responding to HBV exposure :



Responder : Anti-HBs +ve > 10 IU.

Non-responder : ↓↓ Anti-HBs after a series of vaccines.

WHO's 5Cs for Hepatitis B prevention :

----- Active space -----

- Consent.
- Confidentiality.
- Correct test results.
- Counselling.
- Connection to preventive, treatment & care services.

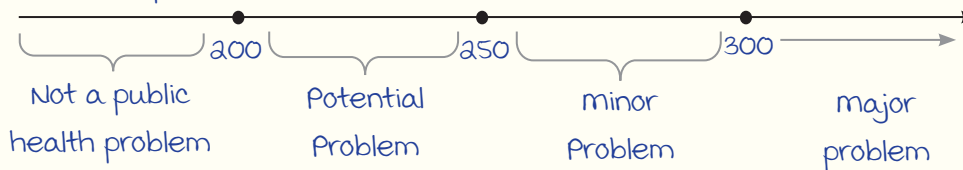
Soil Transmitted Helminthic Infections

00:33:55

	Ascariasis (m/c)	Hookworm
Agent	<i>Ascaris lumbricoides</i>	• <i>Ancylostoma duodenale</i> • <i>Necator americanus</i>
Size	15 - 30 cm	10 - 13 cm
Infective form	Egg	Larva
Transmission	Ingestion	Penetration of skin
Soil	Hard, clay soil	Soft, porous soil
Predominant feature	Diarrhea	Anemia

Chandler's index : No. of hookworm eggs per gram of stool.

- Evaluation of public health burden.



National Deworming Days :

- Albendazole is given on 10th February & 10th August.
- Dose :
 - 1 - 2 years of age : 200 mg.
 - > 2 years of age : 400 mg.
- Prophylaxis for hookworm infection

Rabies

00:36:43

Epidemiology :

Causative organism : ssRNA virus (Rhabdoviridae family).

- Bullet shaped.

Types	Characteristic	Incubation period
Fixed virus	used in vaccine formulation	6 days
Street virus	Causes rabies disease	variable (Days - years)

----- Active space -----

Categories of Animal Bite :

Category	Type of exposure	management		
		Wound mx	ARV	Rabies IG
I	Touching/feeding animals, animal licks on intact skin .	↑ ↓	-	-
II	- Nibbling of uncovered skin - minor scratches or abrasions Bleed ⊖		↑	-
III	- Single/multiple transdermal bites/scratches with bleeding. - Contamination of mucous membranes or broken skin with saliva from animal licks. - Exposures due to direct contact with all wild animals.		↓	↑

Prophylaxis :

Post-exposure prophylaxis :

- Indication :
 - All wild animal bites.
 - Pet animal bites.
- PEP not given for the following infected sources :
 - Human bites.
 - Bat bites.
 - Small rodents (Squirrel/rats).
 - Insect bites.
 - Drinking milk from cattle.
- Regimen :

	ESSEN regime	modified/updated Thai Red Cross/2020a Regime
Regime (Days)	0, 3, 7, 14, 28	0, 3, 7, 28
Number of visits	5	4
Route	Im (1 dose/visit)	ID (2 doses/visit)

Pre-exposure & re-exposure prophylaxis 🌐 :

	Pre-exposure prophylaxis	Re-exposure prophylaxis
Route	Im or ID	
Dose	1 per visit	
Regime (Days)	0, 7, 21 or 28	0, 3 (No R.I.G.)

Wound mx :

- Wash with soap under running water for 15 minutes.
- Delayed sutures (> 24 hrs).

Indication for re-exposure prophylaxis :

Previously immunized (Last vaccine > 3 months ago).

----- Active space -----

Summary :

Prophylaxis	Visits	Doses	Route
Re-exposure	2	2	ID or Im
Pre-exposure	3	3	ID or Im
Updated Thai regimen	4	8	ID
Essen regimen	5	5	Im

Rabies Immunoglobulin :

Indication :

- < 7 days of exposure.
- Not given for re-exposure.

Dose :

- Equine RIG : 40 IU/kg
- Human RIG : 20 IU/kg (Safer, efficient)

Route :

maximum infiltrated in/around the wound



Remaining given Im at the nearest site

Tetanus and Rickettsia

00:47:37

Tetanus :

Post exposure prophylaxis :

Clean wound :

- Sterile instrument injury.
- Brought within 6 hours.

Category	Clean wound	Unclean wound
A : Full immunization < 5 years ago.	Only wound care	
B : Full immunization b/w 5 - 10 years	Wound care + TT single dose	
C : Full immunization > 10 years ago.	• Wound care + • TT single dose	• Wound care + • TT single dose + • Human tetanus immunoglobulin (hTlg).
D : Unknown/never taken immunization.	Wound care + TT complete course	Wound care + TT complete course + hTlg.

----- Active space -----

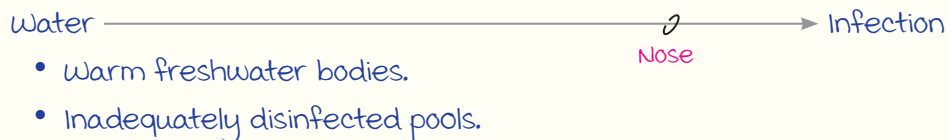
Rickettsia :

	Causative agent	Vector	Reservoir
Epidemic typhus	<i>R. prowazekii</i>	Louse	Rodents >> Human
Endemic (murine) typhus	<i>R. typhi</i>	Flea	Rodents
Scrub typhus	<i>Rickettsia/Orientia tsutsugamushi</i>	mite	Rodents
Mediterranean spotted fever (Indian tick typhus)	<i>R. conorii</i>	Tick	Dogs
Rocky mountain spotted fever	<i>R. rickettsii</i>	Tick	Dogs, rodents
Rickettsial pox	<i>R. akari</i>	mite	mice
Q fever (Rashes ☹)	<i>Coxiella burnetii</i>	None >> Ticks	Sheep, cattle, goat
Trench fever	<i>Bartonella quintana</i>	Louse	Human

Naegleria Fowleri

00:52:28

Thermophilic, free - living amoeba.

Transmission :**Stages in Life Cycle :**

- **Trophozoite stage** (m/c) :
 - Active & infective stage.
 - Only stage found in humans.
 - Flagellate stage.
 - Cystic stage.
- } Temporary.

Clinical Features :

- Fever.
- Nausea.
- Vomiting.
- Neck stiffness.
- Seizures.
- Photophobia.
- Altered mental status.

Diagnosis :

- **Polymerase chain reaction** (Definitive).
- CSF wet mount microscopy.

Management :

- **Amphotericin-B** (DOC).
- miltefosine.
- Rifampicin.
- Symptomatic management of ↑ ICP/ meningitis.

NUTRITION

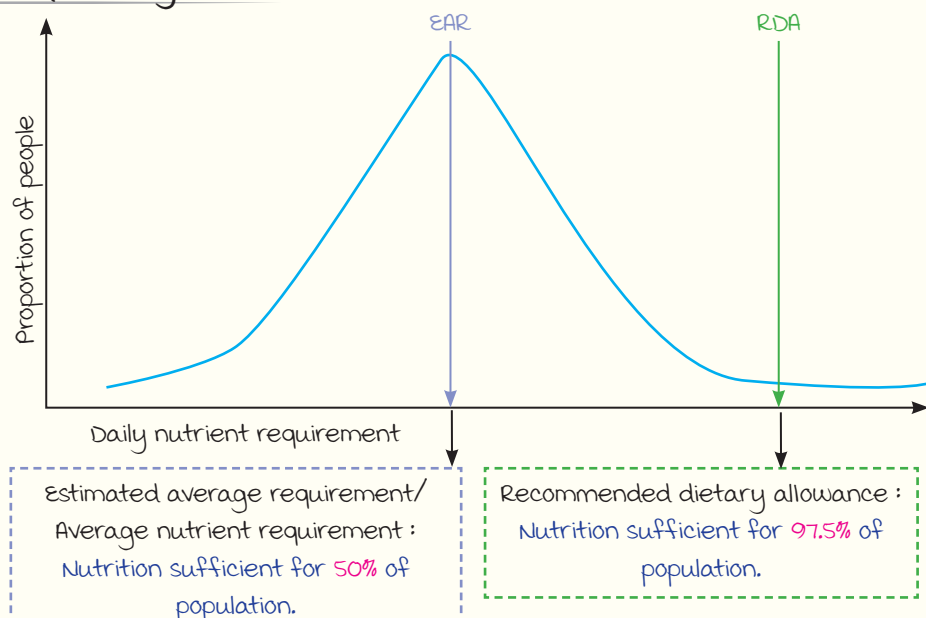
----- Active space -----

Based on National Institute of Nutrition guidelines (Updated 2024).

Nutritional Requirements

00:00:51

EAR, RDA & Dietary Goals :



Proximate principles (Energy providing food products)	Energy provided (kcal/g)	Dietary goals (Recommended daily intake : % of total diet)
Carbohydrates	4	50 - 70
Proteins	4	10 - 15
Fats	9	10 - 30 (Ideal : 20)

Note :

Other energy providing food products :

- Fibre : 2 kcal/g.
- Alcohol : 7 kcal/g.

Reference male & Female 🌐 :

	Reference male	Reference female
Weight (kg)	65	55
Age (Years)	19 - 39	
Height (meters)	1.77	1.62
BMI (kg/m ²) Normal : 18.5 - 22.9	20.75	20.95
Activity	• Occupation : 8 h • Sleep : 8 h • Sitting/moving around : 4 - 6 h • Walking (Active recreation) : 2 - 3 h	

----- Active space -----

RDA update (2020) :

Calorie requirements :

	Sedentary worker	moderate worker	Heavy worker
male	2100 Kcal	+600	+700
Female	1660 Kcal	+500	+600

micronutrients  :

	male	Non-pregnant female	Pregnant female	Lactating female
Iron (mg/day)	19	29	27	23
Calcium (mg/day)	1000			1200
Iodine (mcg/day)	150 - 160		250	280
Vitamin A (mcg/day)	1000	850	900	950
Folic acid (mcg/day)	300	222	570	330
Protein (g/day)	54	45	- T2 : +9.5 - T3 : +22	0 - 6 m : +17 > 6 m : +15

Postmenopausal Ca^{2+} requirement : 1200 mg/day.

Nutritional requirements for children :

Age	Amount at each meal
6 months	2 - 3 tablespoons
7 - 8 months	2/3 rd of a 250 mL bowl
9 - 11 months	3/4 th of a 250 mL bowl
12 - 24 months	250 mL bowl

Proteins

00:12:00

Protein Indicators :

Digestible indispensable amino acid score (DIAAS) :

Best indicator for protein quality.

Net protein utilization (NPU)  :

- m/c indicator used.
- $$\text{NPU} = \frac{\text{DC} \times \text{BV}}{\text{Total protein ingested}} = \frac{\text{Amount of nitrogen retained for body mass}}{\text{Total protein ingested}} \times 100.$$

Digestibility coefficient (DC) :

$$\text{DC} = \frac{\text{Amount of amino acid absorbed}}{\text{Amount of protein ingested}} \times 100.$$

Biological value (BV) :

$$BV = \frac{\text{Nitrogen retained for body mass} \times 100}{\text{Amount of amino acid absorbed}}$$

----- Active space -----

Protein energy ratio :

- $\frac{\text{Amount of energy from protein} \times 100}{\text{Total energy in diet}}$
- max in fish.

Limiting Amino Acids :

	Limiting amino acid
Cereals	Threonine & lysine
maize	Tryptophan & lysine
Pulses	Cystein & methionine

Pellagra :

- Etiology : Pellagrogenic foods :
 - Jowar
 - Sorghum
 - maize
- Diagram illustrating the conversion of tryptophan to niacin:
- ```

graph LR
 T[↓ Tryptophan] --> N[↓ Niacin]
 L[↑ Leucine] -.-> T
 L -.-> N
 style L stroke-dasharray: 5 5
 style N stroke:#f00

```
- Symptoms (4D) :
    - Diarrhea.
    - Dermatitis : Casal's necklace.
    - Dementia.
    - Death.

**Fatty Acids (FA)** 

00:19:41

|                                       | Saturated fatty acids                                                                          | Unsaturated fatty acids                                                                                         |                              |                                                                                     |
|---------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------|
|                                       |                                                                                                | monounsaturated FA (MUFA)                                                                                       | Polyunsaturated FA (PUFA)    |                                                                                     |
|                                       |                                                                                                |                                                                                                                 | Omega-3                      | Omega-6                                                                             |
| Examples                              | <ul style="list-style-type: none"> <li>Palmitic acid</li> <li>Stearic acid</li> </ul>          | Oleic acid                                                                                                      | α-linolenic acid             | Linoleic acid                                                                       |
| Rich sources                          | <ul style="list-style-type: none"> <li>Animal sources</li> <li>Coconuts (exception)</li> </ul> | <ul style="list-style-type: none"> <li>Olives</li> <li>Groundnuts</li> <li>Mustard</li> <li>Avocados</li> </ul> | Flax seeds                   | <ul style="list-style-type: none"> <li>Safflowers</li> <li>Grapeseed oil</li> </ul> |
| Long chain derivatives & rich sources | -                                                                                              | -                                                                                                               | Eicosapentaenoic acid : Fish | Arachidonic acid : milk, egg                                                        |

----- Active space -----

## Vitamins & Minerals

00:24:35

Hidden hunger :

- micronutrient deficiency.
- m/c deficiency : **Iron**.

### Vitamin Deficiencies

| Vitamin             | Name             | Deficiency related conditions                      |
|---------------------|------------------|----------------------------------------------------|
| Vit A               | Retinol          | Night blindness, xerophthalmia                     |
| Vit B <sub>1</sub>  | Thiamine         | Beriberi, Wernicke's encephalopathy                |
| Vit B <sub>2</sub>  | Riboflavin       | Ariboflavinosis                                    |
| Vit B <sub>3</sub>  | Niacin           | Pellagra                                           |
| Vit B <sub>5</sub>  | Pantothenic acid | Burning foot syndrome                              |
| Vit B <sub>6</sub>  | Pyridoxine       | Peripheral neuropathy                              |
| Vit B <sub>7</sub>  | Biotin           | Dermatitis                                         |
| Vit B <sub>9</sub>  | Folic acid       | megaloblastic anaemia                              |
| Vit B <sub>12</sub> | Cyanocobalamin   | Pernicious anemia                                  |
| Vit C               | Ascorbic acid    | Scurvy                                             |
| Vit D               | Cholecalciferol  | Rickets (Children), osteomalacia (Adults)          |
| Vit E               | Tocopherol       | Dermatitis, infertility                            |
| Vit K               | Phylloquinone    | Hemorrhagic disease of newborn, bleeding disorders |

### Vitamin A :

Sources :

- **Halibut fish** (Richest).
- Palm oil.
- Dates.
- Orange & yellow fruits/vegetables (Carrots, papaya, pumpkin).
- Green leafy vegetables.

### Deficiency/xerophthalmia

Associated with :

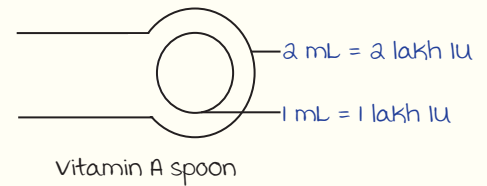
- Infections (**measles**).
- malnutrition.

| WHO stages           | Grade | Features                                               |
|----------------------|-------|--------------------------------------------------------|
| Night blindness      | Xn    | First symptom                                          |
| Conjunctival xerosis | X1a   | First clinical sign                                    |
| Bitot spots          | X1b   | Epidemiological marker ( <b>Prevalence &lt; 0.5%</b> ) |
| Corneal xerosis      | X2    | -                                                      |
| Corneal ulcers       | X3a   | Irreversible eye damage                                |
| Keratomalacia        | X3b   |                                                        |

National vitamin A deficiency prophylaxis program :

- Dose :
  - 6 months to 1 year : 1 lakh IU.
  - 1 to 5 years : 2 lakh IU/6 months.
- Total dose : 17 lakh IU (9 doses).
- Storage : Amber colored bottle (D/t heat & light sensitivity).
- Administration : At **anganwadi centre** > ASHA worker.

• measurement :



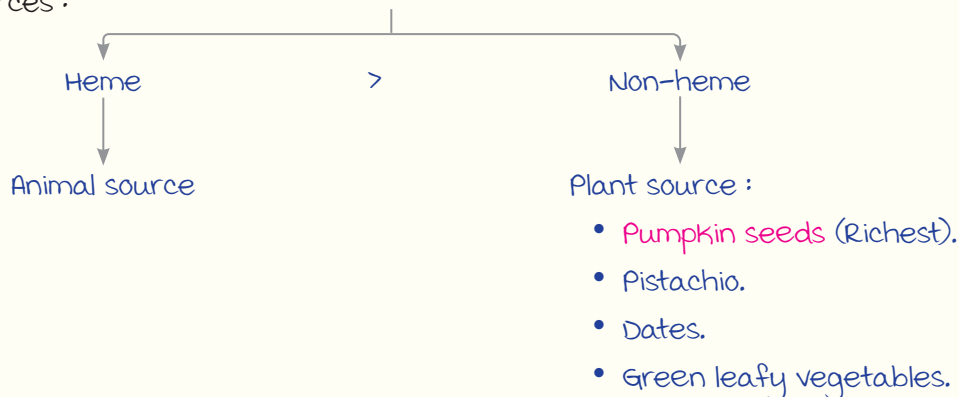
Rx for documented vitamin A deficiency :

Dose :

- < 6 months : 50,000 IU
  - 6 months to 1 year / < 8 kg : 1 lakh IU
  - 1 to 5 years : 2 lakh IU
- 3 doses on day 0, 1 & 14.

**Iron** 🌐 :

Sources :



Absorption :

Ferric iron (Ingested)  $\xrightarrow{+ \text{Vit C}}$  Ferrous iron (Absorbed).  
 (-) Phytates, tannin,  $\text{Ca}^{2+}$ , oxalates, fiber

**Richest Sources :**

| Vitamin/mineral         | Richest source                                |
|-------------------------|-----------------------------------------------|
| Vitamin A               | Halibut fish                                  |
| Vitamin C               | Indian gooseberry (Amla)                      |
| Iodine                  | Sea weed > Sea food                           |
| Iron                    | Animal sources > Plant source (Pumpkin seeds) |
| Vitamin B <sub>1</sub>  | Organ meat, nuts, outer covering of grains    |
| Vitamin B <sub>12</sub> | Organ meat, milk (No plant sources)           |
| Vitamin D               | Cod fish                                      |
| Calcium                 | Buffalo milk, meat                            |

----- Active space -----

**Eggs and Milk**

00:36:27

Egg :

Nutrition per 60 g of egg :

| Component      | Amount  |
|----------------|---------|
| Total calories | 60 kcal |
| Fat            | 6 g     |
| Protein        | 6 g     |
| Calcium        | 30 mg   |
| Cholesterol    | 240 mg  |
| Iron           | 1.8 mg  |

Features :

- NPU : 97 (max).
- Avidin : Protein in raw egg → Biotin deficiency.
- Source of :
  - Natural cholesterol (Richest).
  - DHA.
  - PUFA.

**Breastmilk vs. Buffalo milk :**

| Component | In comparison to buffalo milk | Features                                           |
|-----------|-------------------------------|----------------------------------------------------|
| Energy    | 50%                           | -                                                  |
| Fats      | 50%                           | Rich in DHA & omega-6 FA → Promote myelination     |
| Protein   | 25%                           | Rich in cysteine & taurine → Promote axonal growth |
| Lactose   | High                          | Richest source                                     |
| Iron      | -                             | Adequate source till 6 months of age               |
| Vitamin C | High                          | ↑ Iron absorption                                  |

methods of Pasteurization :

Disinfection of milk.

|             | vat/Holder                                | High temperature short time        | ultra high temperature (most sophisticated)                                                                                      |
|-------------|-------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Temperature | 63 - 64°C                                 | 73°C                               | <ul style="list-style-type: none"> <li>• 1<sup>st</sup> phase : 125°C</li> <li>• 2<sup>nd</sup> phase : High pressure</li> </ul> |
| Cooling     | Hold for ~30 minutes<br>↓<br>Slow cooling | Rapid cooling (within few minutes) | Rapid cooling (within few seconds)                                                                                               |
| use         | Rural areas, small quantities             | urban areas                        | most large scale suppliers                                                                                                       |

Note :

- Germinated food products : ↑ Protein, easily digested.
- Fermented food products : ↑ Na, easily digested.
- Amylase rich food (ARF) :
  - ↑ Carbohydrate digestion.
  - Example : mango, banana, sprouts.

## Food Intoxicants

00:45:59

----- Active space -----

| Disease         | Toxin                                              | Food product                        | Preventive method                                                                                                               |
|-----------------|----------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Neurolathyrism  | Beta oxalyl amino alanine (BOAA)                   | Khesari dal (Lathyrus sativus)      | <ul style="list-style-type: none"> <li>Vitamin C</li> <li>Steeping &amp; parboiling</li> </ul>                                  |
| Endemic ascites | Pyrolizidine (Crotalaria seeds)                    | Adulterated millets                 | Sieving                                                                                                                         |
| Epidemic dropsy | Sanguinarine (Argemone mexicana)                   | Adulterated mustard oil             | <ul style="list-style-type: none"> <li>Nitric acid test</li> <li>Chromatography test (most sensitive)</li> </ul> On mustard oil |
| Aflatoxicosis   | Aflatoxin (Aspergillus flavus : Storage fungus)    | Stored grains (Groundnuts)          | <ul style="list-style-type: none"> <li>Low humidity</li> <li>Improving ventilation</li> </ul>                                   |
| Ergotism        | Ergot alkaloid (Claviceps purpurea : Field fungus) | Rye, sorghum, bajra (Black swollen) | Floating plant in 20% salt solution                                                                                             |

### Neurolathyrism :

- Seen in central India.
- UMN disease.
- Joints affected : **Knee** > Hip > Ankle.



Lathyrus sativus

### Stages :

| Stage | Name         | Clinical features           |
|-------|--------------|-----------------------------|
| I     | Asymptomatic | -                           |
| II    | No stick     | Gait disturbances (limping) |
| III   | One stick    | Crossed gait                |
| IV    | Two stick    | Inability to walk           |
| V     | Crawlers     | Inability to stand          |



Neurolathyrism

### Epidemic Dropsy :

#### Clinical features :

- Glaucoma.
- Pedal edema.
- Congestive cardiac failure.
- Ascites.
- Pleural effusion.
- Diarrhea.



Argemone mexicana



Aflatoxicosis (Greenish)

----- Active space -----

**Food Legislation & Guidelines**

00:55:35

Food Safety and Standards Authority of India (FSSAI) :  
Regulation of orally consumed food products.

AGMARK :  
Legislate agricultural (Raw) products.

Food & Agricultural Organisation (FAO) :  
Codex Alimentarius : International standards, guidelines &  
codes for big chain restaurants.



# ENVIRONMENT

----- Active space -----

## Air Pollution

00:00:26

### Types of Air Pollutants :

#### Primary :

- CO.
- $SO_a$  : marker for industrial air pollution.
- methane.
- CFC : Only man - made gas with greenhouse effect.
- Other gases.

#### Secondary :

- Dust (Particulate matter) : measured by GRIT index.
- Smoke : measured by soiling index.

### Air Comfort Parameters :

| Parameters                   | measured by                   |
|------------------------------|-------------------------------|
| Temperature                  | Thermometer                   |
| Humidity                     | Psychrometer, Hygrometer      |
| Velocity/movement            | Anemometer                    |
| Low air velocity/circulation | Kata thermometer > Anemometer |
| Cooling power                | Kata thermometer              |
| Radiant heat                 | Globe thermometer             |

effective air temperature  
corrected effective air temperature (CEAT)  
(best parameter)



Anemometer



Kata thermometer (Red bulb)

### Air Quality Index (AQI) :

measured by CPCB (Central Pollution Control Board).

- under ministry of forest, environment & climate change.

#### Parameters measured :

- Smoke.
- Particulate matter :  $< 2.5 \mu m (Pm_{2.5})$  &  $< 10 \mu m (Pm_{10})$ .
- $NO_a$ .
- $SO_a$ .
- CO.
- Ozone/ $O_3$ .
- Ammonia/ $NH_3$ .
- Lead (Pb).

#### Grading :

- $< 100$  : Good.
- $100-200$  : moderate.
- $> 200$  : Poor.
- $> 400$  : Severe.

Note :  $CO_a$  is not a part of the AQI index.

----- Active space -----

## Water Quality

00:07:47

### Parameters :

#### Physical :

- Colour :  $\leq 15$  Hazen units (AKA True colour units).
- Temperature : Room temperature.
- Odour : Absent.
- Turbidity :  $\leq 5$  nephelometric units.

#### Biological :

- General : Coliform count  
- (N) : Zero.
- Recent contamination : Fecal streptococci.
- Remote contamination : Clostridia perfringens.

#### Chemical :

Chlorination :  
(Cl : Disinfection)

Hardness

### Water Chlorination :

Chlorine demand :

- Amount of bleaching powder required to disinfect 455 L of water.
- measured by Horrocks apparatus.

mechanism of action :

At breakpoint chlorination, there is formation of :

#### Free radicals :

- Kill bacteria  $\rightarrow$  Disinfection.
- Ions : Hypochlorous > Hypochlorite.

Residual chlorine compounds.

Free chlorine :  
Evaporates.

Chlorine Estimation : measures residual chlorine compounds.

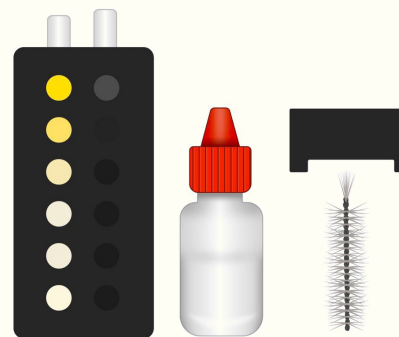
#### Orthotolidine test :

measures free residual chlorine (FRC) & combined residual chlorine (CRC) together.

< Ortho toluidine-4 arsenate test :  
measures FRC & CRC separately.

Permissible Levels of Chlorine :

|                    | Cl <sup>-</sup> Level |
|--------------------|-----------------------|
| Drinking water     | > 0.5 ppm             |
| Impending outbreak | > 0.7 ppm             |
| Swimming pools     | > 1 ppm               |



Chloroscope : measures chlorine levels in water.

Organisms unaffected by chlorination (Regular levels) :

----- Active space -----

- Sporing organisms.
- Protozoal cysts.
- Helminths.
- Ova.
- molluscs.
- Cyclops.
- Cercariae.
- Hepatitis A.
- Poliovirus.

### Hardness of water :

Types → Temporary : Ca & mg bicarbonates (m/c).  
 → Permanent : Ca & mg → Sulphates (m/c), chlorides.  
 → Rare : Nitrates & phosphates.

Levels of hardness : 1 mEq of hardness = 50 mg  $\text{CaCO}_3$

|           | Category                         |
|-----------|----------------------------------|
| < 1 mEq   | Soft                             |
| 1 - 3 mEq | moderately hard (drinking water) |
| 3 - 6 mEq | Hard                             |
| > 6 mEq   | Very hard                        |

Treatment :

- Boiling
  - Addition of lime
  - Addition of sodium carbonate
  - Permutit process / Ion-exchange process : uses zeolites. (Best)
- } used for temporary hardness.

### Sand Filter

00:17:26

|                              | Rapid sand filter (m/c)                                   | Slow sand filter                                                                                         |
|------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Purity                       | 99%                                                       | 99.999%                                                                                                  |
| Space required               | Lesser                                                    | more                                                                                                     |
| Chemical added               | Alum                                                      | Not added                                                                                                |
| Sand size                    | 0.4 - 0.7 mm                                              | 0.2 - 0.3 mm                                                                                             |
| Pre filtration sedimentation | Yes, 2 - 6 hrs                                            | No                                                                                                       |
| main filtration              | Sand bed                                                  | Schmutzdecke (Biological/<br>zooglean) layer :<br>Aerates water<br>↓<br>Aerobic digestion<br>of bacteria |
| Cleaning                     | Backwashing :<br>• Technical process<br>• Frequently done | Scraping :<br>• manual process<br>• Once every few months                                                |

----- Active space -----

**Housing & Light**

00:19:55

**Overcrowding :**

2 persons of ages > 9 yrs, of opposite gender sharing the same room  
(Exception : Husband & wife).

Space criteria :

- 1 person : < 70 sq.ft.
- 2 persons : < 110 sq.ft.
- For every additional person : Add 40 sq.ft.

**Acceptable Norm of Persons Per Room :**

| No. of persons | Ideal rooms |
|----------------|-------------|
| upto 2         | 1           |
| 3 - 4          | 2           |
| 5              | 3           |
| 7              | 4           |

**Light :**

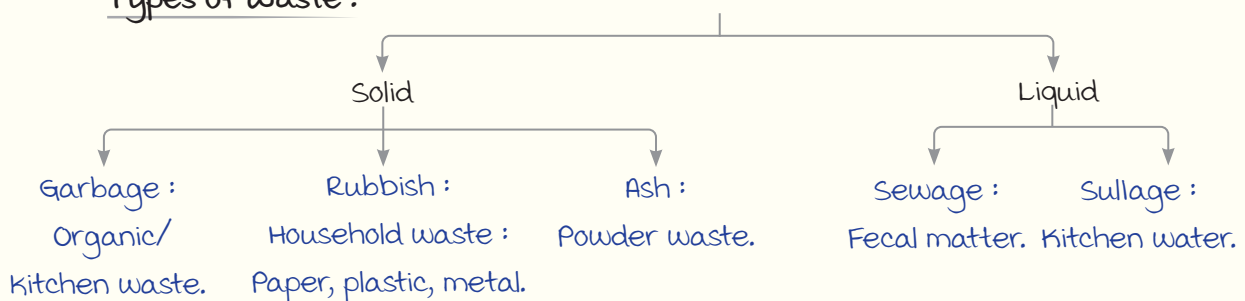
minimum amount of light required : > 100 lux.

Lux : Amount of light reaching a surface.

Lumen : Flow of light.

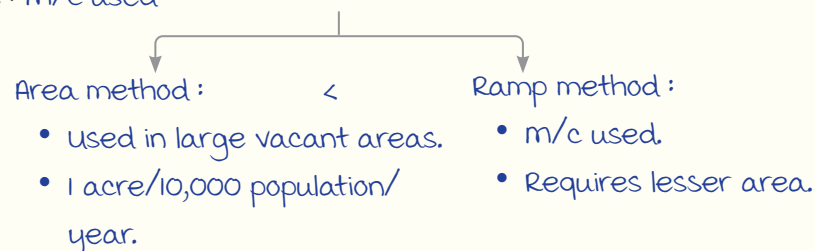
**Sewage & Waste**

00:21:27

**Types of Waste :****Solid Waste Disposal :**

Dumping : Non sanitary method.

Landfills : m/c used



Burning : Leads to air pollution.

Incineration : Burning at high temperature ( $800 - 1000^{\circ}\text{C}$ )  $\rightarrow$  Ash.

Pyrolysis :

Burning at  $400^{\circ}\text{C}$  without oxygen. (No combustion)  $\rightarrow$  Bio-char (Solid)  
 Bio-oil (Liquid)  
 Syngas (Gas)  $\rightarrow$  Reused for energy production.

Inertization : mixing of chemical waste with cement.

Pulverization : mutilation/shredding of solid/metallic waste.

Bangalore method :

- Layering method used in rural areas.
- municipal garbage is layered with night soil (Fecal matter, cow dung, earth)  $\rightarrow$  Anaerobic digestion (Time consuming)  $\rightarrow$  Compost.

### Sewage Treatment Plant :

Biological oxygen demand (BOD) : Amount of  $\text{O}_2$  required for microbial metabolism of organic compounds at  $20^{\circ}\text{C}$  for 100 gms of sewage in 5 days.

| BOD                      | Sewage   |
|--------------------------|----------|
| $< 100 \text{ mg/L}$     | Weak     |
| $100 - 500 \text{ mg/L}$ | Moderate |
| $> 500 \text{ mg/L}$     | Strong   |

Primary treatment : Removal of large impurities.

- Screening
- Grit chamber
- Primary sedimentation

Secondary treatment :

Aeration :

- most important step.
- Aerobic digestion.

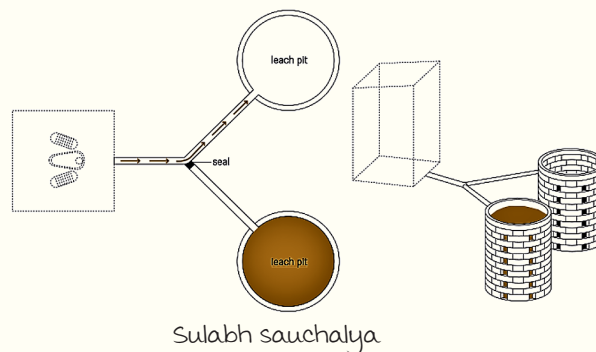
### Types of Latrines :

Sanitary latrine :

- AKA RCA latrine.
- Water seal (+).
- Connected to a sewer system.

Trench method :

- Burial method
- used in temporary camps.



Sulabh sauchalya : a pit latrine

- Low volume flush system  $\rightarrow$   $\uparrow$  Aerobic & anaerobic digestion  $\rightarrow$  manure.
- Not connected to a sewer system

----- Active space -----

----- Active space -----

**Bio Medical Waste (BMW) Management**

00:32:03

- Guidelines given by Central pollution control board (CPCB).
  - under ministry of environment, forest, climate change.
- BMW does not cover :
  - Radioactive waste
  - E-waste
  - Solid waste
  - Hazardous chemicals
  - Lead acid batteries
  - Hazardous waste
  - Hazardous microorganisms (Genetically engineered/Bioterrorism agents)

**Categories of BMW 🌐 :**

| Category | Type of waste                                                                                                        | Disposal method        | Bag/bin                                       |
|----------|----------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------|
| Yellow   | <ul style="list-style-type: none"> <li>Infectious waste</li> <li>Laboratory waste</li> </ul>                         | Incineration           | Non chlorinated plastic bag (0.5 μ thickness) |
| Red      | Rubber, plastic, tubing                                                                                              | Recycle, return, reuse | Non chlorinated plastic bag (0.5 μ thickness) |
| Blue     | <ul style="list-style-type: none"> <li>Broken glass/vials/ampoules</li> <li>metallic orthopaedic implants</li> </ul> | Recycle, return, reuse | Cardboard box/<br>Plastic bin                 |
| White    | Sharps, metals                                                                                                       | Recycle, return, reuse | Puncture proof translucent plastic bin        |

**Yellow category :**

- Anatomical/human waste :
  - Placenta.
  - Fetus (if dead).
  - Liquid waste : urine/serum.
- Soiled waste. (Cotton, dressings)
- Linen, bedding.
- medicines & consumables :
  - Expired/discarded.
  - Cytotoxic. (including bottle)
- Blood bags (exception to rubber)
- Culture : media, broths etc.
- Other liquids :
  - X-ray developing fluid.
  - Lab : NS, reagents.
  - Floor washing liquid.
- unopened vaccine vials.
- vacutainer with blood.

📦 : Pre-treatment required.

**Red category :**

- Foley's catheter.
- urine bag.
- IV bottles.
- Gloves.
- Syringe without needle.
- Ryle's tube.
- Cannulae without needle/trocar.
- vacutainer without blood.

**Blue category :**

- Empty vial.
- metallic implants.
- Broken glass/ampoules.

**White category :**

- Scalpel.
- Blades.
- Sutures.
- Needles/trocars.
  - From cannulae, syringes.
  - LP needles.

## Occupational Health

00:41:18

----- Active space -----

### Pneumoconiosis :

Restrictive lung diseases.

Size of particulate matter :

| Size ( $\mu$ ) | effect                             |
|----------------|------------------------------------|
| > 10           | Not airborne                       |
| 5 - 10         | Lodged in upper respiratory tract  |
| 3 - 5          | Lodged in lower respiratory tract  |
| 1 - 3          | Lodged in alveoli (most dangerous) |
| < 1            | Diffusible into circulation        |

} Cause pneumoconiosis.

Types of pneumoconiosis :

| Disease      | Particle                  | Source Industries / sectors                                                                                                                             |
|--------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Silicosis    | Silica                    | <ul style="list-style-type: none"> <li>Sand stone</li> <li>Granite</li> <li>Pottery &amp; ceramic</li> <li>Gold</li> <li>mica</li> <li>Steel</li> </ul> |
| Baritosis    | Barium                    | Photography, printing, barium diagnostic works                                                                                                          |
| Byssinosis   | Cotton dust               | Textile industries                                                                                                                                      |
| Bagassosis   | Sugar cane dust (Bagasse) | Cane sugar factories, paper & card board                                                                                                                |
| Tobaccosis   | Tobacco dust              | Tobacco factories                                                                                                                                       |
| Farmers lung | mouldy hay (Grain dust)   | Agricultural industry                                                                                                                                   |
| Anthraxis    | Coal dust                 | -                                                                                                                                                       |

Silicosis vs. Asbestosis :

|                       | Silicosis                     | Asbestosis                                                                                                                                                                                                                                                                                             |
|-----------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cause                 | Silica dust                   | Asbestos fibers → Chrysotile : Curly white fibers<br>→ Amphibole : <ul style="list-style-type: none"> <li>Spiculated, needle like</li> <li>more dangerous</li> </ul> → Amosite : Brown fibers<br>→ Crocidolite : <ul style="list-style-type: none"> <li>Blue fibers</li> <li>most dangerous</li> </ul> |
| Zone of lung affected | upper zone                    | Lower zone                                                                                                                                                                                                                                                                                             |
| Pathology             | Alveolitis → Nodular fibrosis | Peribronchial inflammation                                                                                                                                                                                                                                                                             |
| X-ray                 | Snow storm appearance         | Ground glass appearance                                                                                                                                                                                                                                                                                |
| Association           | TB                            | Bronchial Ca > mesothelioma (Rare, specific)                                                                                                                                                                                                                                                           |

Bagassosis :

- D/t inhalation of *Thermoactinomyces sacchari* (fungus) that grows on bagasse.

↓  
Hypersensitivity pneumonitis

- Not a notifiable disease.

----- Active space -----

**Chemical carcinogens**  :

| Chemicals                   | Cancers      |
|-----------------------------|--------------|
| Benzidine                   | Bladder      |
| Benzene                     | Leukemia     |
| Asbestos                    | mesothelioma |
| vinyl chloride              | Liver        |
| wood dust, nickel, chromium | Nasal sinus  |

**Employee State Insurance Scheme (ESI)** 

00:49:01

- under the ministry of Labour.
- Includes all organizations.
  - Except : Defence, mines, railways & central government.

**Contributions :**

- Employee : 0.75% of wage
  - Employer : 3.25% of wage
- } 4% wage paid monthly to ESI.

**Eligibility :**

- monthly salary  $\leq$  Rs. 21,000/month.
- If physically handicapped  $\leq$  Rs. 25,000/month.

**Benefits :**

- Direct : By ESI hospitals/clinics.
- Indirect : By empanelment scheme (Associated hospitals).

| Benefit                                | Features                                                                                                                                                                                                                                         |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sickness benefit                       | <ul style="list-style-type: none"> <li>• upto 70% wage (Paid by ESI)</li> <li>• Leave upto 91 days</li> </ul>                                                                                                                                    |
| Extended sickness benefit              | <ul style="list-style-type: none"> <li>• upto 80% wage</li> <li>• Leave upto 2 yrs</li> </ul>                                                                                                                                                    |
| Enhanced sickness benefit              | <ul style="list-style-type: none"> <li>• 14 days leave for tubectomy</li> <li>• 7 days leave for vasectomy</li> </ul> <p>} with 100% wage</p>                                                                                                    |
| maternity benefit<br>(Only for female) | <ul style="list-style-type: none"> <li>• Any medical condition affecting pregnancy : 4 weeks leave</li> <li>• miscarriage/abortion : 6 weeks leave</li> <li>• Delivery : 26 wks leave (Not &gt; 8 wks before date of confinement/EDD)</li> </ul> |
| Dependent benefit                      | -                                                                                                                                                                                                                                                |
| Disability benefit                     | -                                                                                                                                                                                                                                                |
| Funeral expense                        | Rs. 15,000 in cash to the family                                                                                                                                                                                                                 |

Shramnik Kalyan Yojana :

----- Active space -----

- unemployment security in case of :
  - Retrenchment (Lay-offs)
  - Forceful closure of company.
  - Non occupational injury.
- Benefit : upto 50% wage for 2 years (If still unemployed).

Factories Act (1948)

00:55:54

Permissible Age for Work :

| Age                             | Employment                                                                                                                                                                                   |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| < 14 yrs                        | Cannot be employed (Article 24)                                                                                                                                                              |
| 15 - 19 yrs<br>(All tender age) | Appropriate work (Article 39) : <ul style="list-style-type: none"> <li>• No overtime</li> <li>• No night duty</li> <li>• No hazardous work</li> <li>• 1 extra leave every 14 days</li> </ul> |

Permissible Hours for work (Including adults) :

- ≤ 48 hrs/week.
- ≤ 60 hrs/week, including overtime.

Facilities :

mandatory facilities :

| No. of Employees | Facility        |
|------------------|-----------------|
| > 50             | Crèche          |
| > 250            | Canteen         |
| > 500            | Welfare officer |
| > 1000           | Safety officer  |

Space : ≥ 500 cu. ft/worker.

----- Active space -----

## MISCELLANEOUS TOPICS IN COMMUNITY MEDICINE

### Health Indicators

00:00:31

|                                     | Physical quality of life indicator (PQLI) : Obsolete | Human development index (HDI)              | Global hunger index (GHI)                                                                                            |
|-------------------------------------|------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Survival                            | Life expectancy at 1 year                            | Life expectancy at birth                   | under-5 mortality rate                                                                                               |
| Education                           | Literacy rate                                        | mean school years<br>Expected school years | -                                                                                                                    |
| Living standard/<br>Economic status | Infant mortality rate                                | Gross national income per capita           | Undernourishment in children < 5 yrs (Wasting & stunting)<br>Inadequate food supply (Adult & child undernourishment) |
| Given by                            | WHO                                                  | UNDP                                       | -                                                                                                                    |
| Value & Rank                        | -                                                    | HDI = 0.685<br>Rank = 130                  | GHI = 25.8 (Serious hunger)<br>Rank = 102/123                                                                        |

### Human Development Index :

Geometric mean of three dimensions.

|           | Parameter                        | minimum value | maximum value |
|-----------|----------------------------------|---------------|---------------|
| Life      | Life expectancy at birth         | 20 yrs        | 85 yrs        |
| Knowledge | Expected years of schooling      | 0 yrs         | 18 yrs        |
|           | mean years of schooling          | 0 yrs         | 5 yrs         |
| Income    | Gross national income per capita | Variable      |               |

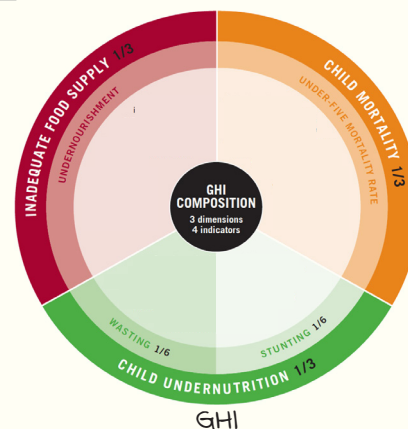
Dimension Index (DI) =  $\frac{\text{Actual value} - \text{minimum value}}{\text{maximum value} - \text{minimum value}}$

↓

- Fixed for every country.
- Helps in comparing health indicators of different countries.

### Global Hunger Index (GHI) :

$$GHI = \frac{\frac{\% \text{ of undernourished} + \frac{\% \text{ of Wasting, stunting}}{2} + \frac{\% \text{ of U}_5\text{MR}}{3}}{3}}$$



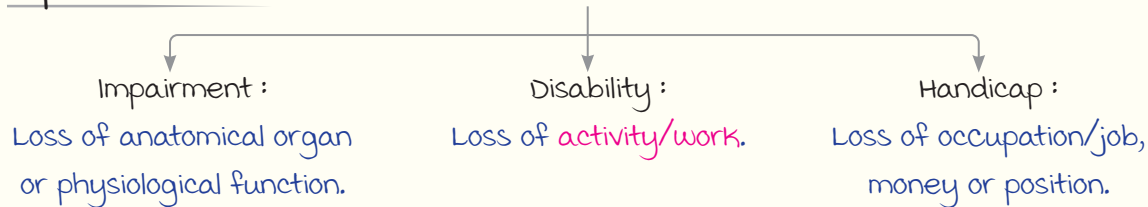
multidimensional Poverty Index :

----- Active space -----

- Given by UNDP.
- Negative indicator.
- Uses 10 variables.

Disability Indicators

00:08:24

Important Terms :Indicators :

|                 | Disability adjusted life years (DALY)                              | Health adjusted life expectancy (HALE)                   | Quality adjusted life years (QALY)                             |
|-----------------|--------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------|
| Description     | Years lost to life (YLL)<br>+<br>Years lived with disability (YLD) | Number of years in good health, free from disability     | Gain in number of quality years of life due to an intervention |
| Best measure of | Burden of disease in a community                                   | Healthy life expectancy & health status of the community | Effectiveness of an intervention                               |

Sullivan Index :

- Positive indicator ; Specific to a person & a disease.
- measured by = Life expectancy - Duration of disability.
- Independent of age structure of population.

Commonly Used Indicators :

| Assessment criteria                               | Indicator         |
|---------------------------------------------------|-------------------|
| Overall general indicator for country development | IMR               |
| United nations indicator for nation's development | U <sub>s</sub> MR |
| Life expectancy                                   | HDI               |
| Hunger & malnutrition                             | GHI > MPI         |
| Overall maternal/MCH care                         | mMR               |
| Overall health services & facilities              | PNMR > IMR        |
| Disability rate                                   | DALY              |
| Burden of a disease                               |                   |
| Effectiveness of an intervention                  | QALY              |
| Healthy life years or health status of community  | HALE              |

----- Active space -----

## Healthcare, Socio-economic Indicators & Utilization Rates

00:17:10

### Healthcare Service/Delivery/Policy Indicators :

- Doctor – population ratio.
- Population – bed ratio.
- Population per trained birth assistant.

### Socio-economic/Development Indicators :

- Dependency ratio.
- Literacy rate.
- Family size.
- Per capita calorie intake.
- Housing (Number of persons per room).

### Utilization Rates :

- Immunization coverage rate : All due vaccines taken till 1 yr of age.
- Proportion of pregnant women receiving antenatal care.
- Bed occupancy rate =  $\frac{\text{Admissions}}{\text{No. of beds} \times \text{Days}} \times 100$ .
- Average length of stay.
- Bed turnover ratio.

## Health Communication

00:19:45

### Types :

- One way : Didactic.
- Two way : Socratic.

### Knowledge vs. Propaganda :

- Knowledge : understanding of information.
- Propaganda : Passive installation of information.

### Barriers to Communication :

- Physiological.
- Psychological.
- Environmental.
- Cultural.

### Uses of Communication :

- Information.
- Education & knowledge.
- Motivation.
- Persuasion & propaganda (Passive)
- Counselling
- Health promotion & development.

## methods of Communication :

----- Active space -----

| Personal                                                                                             | Group                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | mass                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Letters</li> <li>message</li> <li>Personal letters</li> </ul> | <ul style="list-style-type: none"> <li><b>Focused group discussion</b> : Discussion among 6 - 12 members in a group with a moderator</li> <li><b>Panel discussion</b> : Discussion               <ul style="list-style-type: none"> <li>- In front of audience : On a theme</li> <li>- With a moderator : By experts</li> </ul> </li> <li><b>Symposium</b> : Series of lectures in front of audience by experts</li> <li>Lecture</li> <li><b>Workshop</b> : Hands-on training, skill development</li> <li>Roleplay</li> <li>Demonstration</li> <li>Conference</li> <li><b>DELPHI Technique</b> : Discussion among groups to solve major problems.</li> </ul> | <ul style="list-style-type: none"> <li>media</li> <li>TV</li> <li>Radio</li> <li>Bulk mail</li> <li>Posters</li> <li>Advertisement</li> <li>Folkways</li> </ul> |

## Techniques of Communication :

GATHER : **Interview** technique.

Greet → Ask → Tell → Help → Educate → Review.

Cafeteria technique :

- To promote **family planning**.
- Choice of contraceptive lies with the individual.

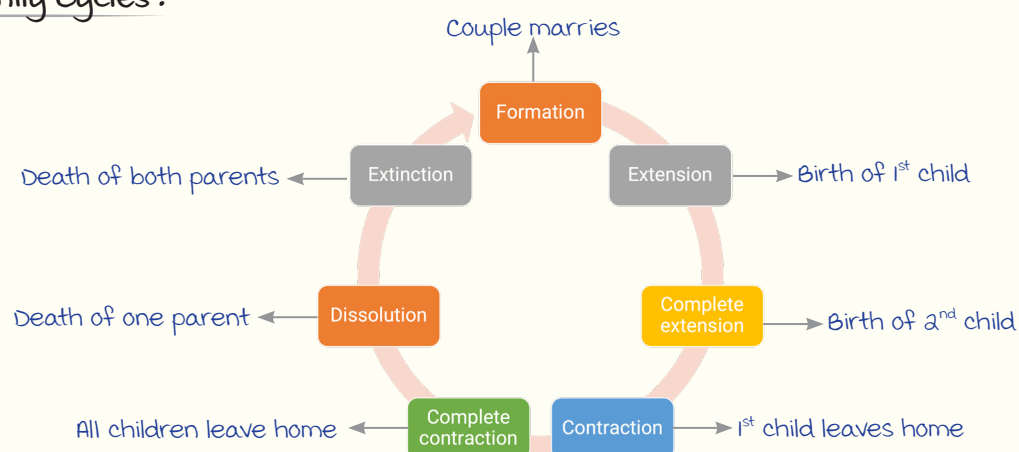
SPIKES protocol : Delivering **bad news**.

Setting → Perception of condition → Invitation from patient to give information → Knowledge : Give information → Explore emotions & sympathise → Strategy & Summary

## Family, Socio-economic Scales & Social Medicine

00:27:10

### Family Cycles :



----- Active space -----

Types of Families :

| Name         | Description                                                                                                                                                                                |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proton       | Single individual staying alone                                                                                                                                                            |
| Electron     | One parent family                                                                                                                                                                          |
| Nuclear      | Single married couple with/without children                                                                                                                                                |
| Atomic       | Nuclear family with any other family members but no other married couples                                                                                                                  |
| molecular    | Two married couples of any different generations with/without unmarried people of any generation                                                                                           |
| Joint        | <ul style="list-style-type: none"> <li>• <math>\geq 2</math> married couples of a single generation (OR)</li> <li>• <math>\geq 3</math> married couples of multiple generations</li> </ul> |
| 3 generation | Children live with parents and grandparents.                                                                                                                                               |

Socioeconomic Scales :

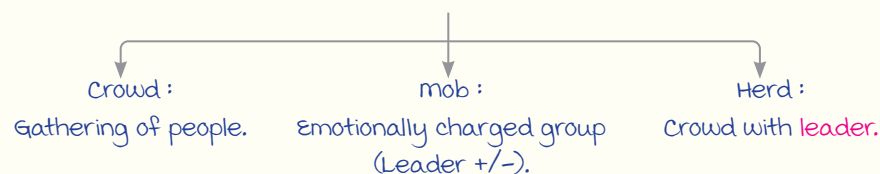
|                           | Area      | Criteria                                                                                                                          |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| modified Kuppuswamy scale | urban     | <ul style="list-style-type: none"> <li>• Income of family</li> <li>• Occupation</li> <li>• Education</li> </ul> of head of family |
| Uday Pareekh scale        | Rural     | 10 variables                                                                                                                      |
| BG Prasad scale           | Universal | Income only : 5 classes                                                                                                           |

Social medicine :

Opinion vs. Belief :

|             | Opinion           | Belief                        |
|-------------|-------------------|-------------------------------|
| Description | Point of view     | Views derived from close ones |
| Basis       | Based on evidence | Inherited/Learned             |
| Permanence  | Temporary         | Permanent                     |

Crowd vs. mob vs. Herd :



Acculturation : mixing of two cultures.

Folkways : Right ways of doing things (Specific to places/cultures).

mores : Stringent customs.

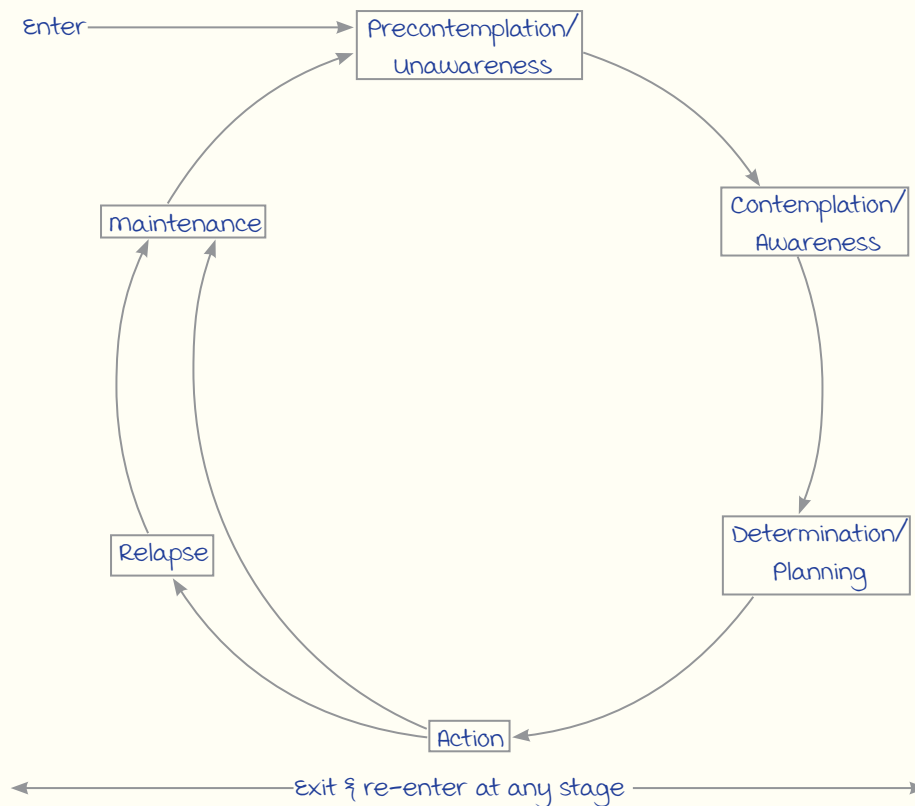
Attitude vs. Habits :

|                          | Attitude  | Habit     |
|--------------------------|-----------|-----------|
| Permanence               | Permanent | Temporary |
| Influenced by motivation | No        | Yes       |

Behaviour change communication model 🌐:

motivation works at level (Best at unawareness → Awareness).

----- Active space -----



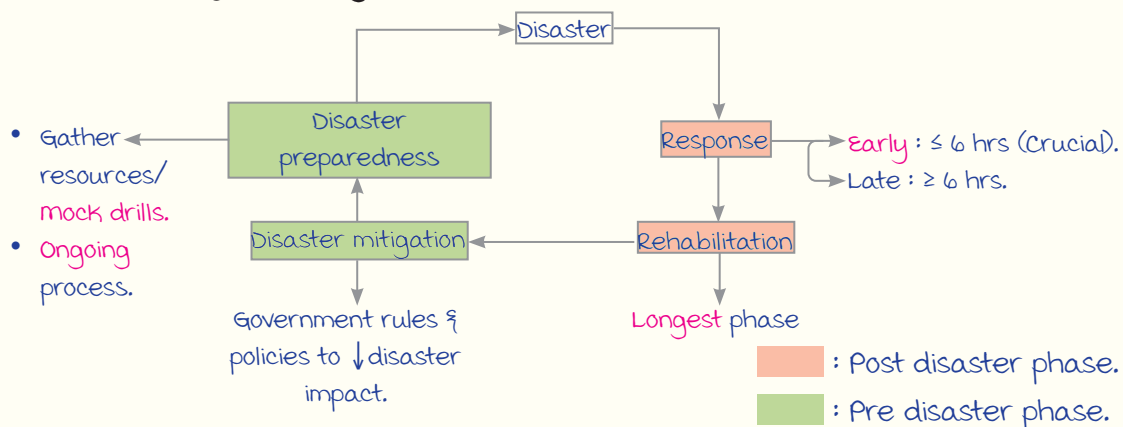
## Disaster Medicine

00:35:34

### National Disaster Response Force (NDRF) :

- managed by national disaster management authority (NDMA).
  - under ministry of home affairs.
- Chairman : Prime minister.

### Disaster management Cycle 🌐 :



----- Active space -----

Haddon matrix :

Injury prevention during RTA/vehicular accidents using epidemiological tools.

Key Points :

m/c type of disaster : Hydrological (Floods) > Geological.

m/c disease post disaster : Acute gastroenteritis.

Long - term sequelae post disaster : malnutrition (↓Economy : Vicious cycle)

m/c vaccine preventable disease outbreak : measles.

vaccine coverage to be ensured : measles.

vaccine for general population : Not given.

vaccine for health professionals : TT, Hep B.

Triage :

Red : Immediate attention (mins - hrs).

Yellow : urgent attention (Hrs - 1 day).

Green : Routine care (Days).

Black : Dead persons.

Priority.



# RECENT UPDATES IN COMMUNITY MEDICINE & HEALTH CARE PLANNING

----- Active space -----

## Management Designs

00:00:25

### Qualitative Designs :

- Organisational design.
- Personnel management.
- Communication.
- Information systems.
- **management by objectives.**

### Quantitative Designs :

Cost - benefit analysis : **Based on monetary gains.**

Cost - effective analysis :

- Based on number of lives affected.
- used in :
  - National health programmes (NHP).
  - Education.
  - Health.

Input - output analysis :

- Determines efficiency & effectivity of the process.
- **efficiency** : used in ideal state/ laboratories.
- **effectivity** :
  - used in real/practice situations.
  - Example : **NHP.**

PERT (Program evaluation & review technique) :

**Sequence** of events.

CPM (Critical path method) :

- Critical path : Path in a sequence taking most time.
- use : To assess **minimum time** to complete the event.

Work sampling :

- Assess the type of activities & time needed to perform the same.
- use :
  - Standardize jobs.
  - Determine manpower requirement.

## Health Care Planning

00:06:44

### Basic Terminology :

|                | Goals      | Targets   | Objectives |
|----------------|------------|-----------|------------|
| Time           | Long term  | mid term  | Short term |
| Characteristic | Open ended | milestone | End points |
| Quantifiable   | ⊖          | ⊕         | ⊕          |

----- Active space -----

**Health Care Committees :**

| Health care committees                                              | Contribution                                                                                                                                                                                                                            |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bhore committee (1946)                                              | <ul style="list-style-type: none"> <li>• Social physicians</li> <li>• Rural doctors</li> <li>• 1 PHC/40,000 population</li> <li>• 3 million population plan</li> </ul>                                                                  |
| Mudaliar committee<br>(AKA <b>Survey &amp; planning committee</b> ) | Planned for medical services in IAS pattern                                                                                                                                                                                             |
| Chadach committee                                                   | <ul style="list-style-type: none"> <li>• Integration of malaria programme with other programmes</li> <li>• <b>Basic health worker formation</b></li> </ul>                                                                              |
| Mukherjee committee.                                                | <ul style="list-style-type: none"> <li>• Delinking malaria programme</li> <li>• Family planning programme</li> </ul>                                                                                                                    |
| Junglawalla committee                                               | Equal pay for equal work                                                                                                                                                                                                                |
| Kartar Singh committee                                              | <b>multipurpose worker formation</b>                                                                                                                                                                                                    |
| Srivastava committee                                                | <ul style="list-style-type: none"> <li>• Reorientation of medical education</li> <li>• Strengthened :               <ul style="list-style-type: none"> <li>- Rural health care system</li> <li>- Referral system</li> </ul> </li> </ul> |
| Bajaj committee                                                     | manpower & finance recommendation                                                                                                                                                                                                       |

**Niti Aayog :**

- Planning commission/Think tank of India.
- Replaced health care committees in 2015.

**Legislative Lists** 

00:13:44

| List     | Name            | managed by                 | Example                                                                                                                                                                          |
|----------|-----------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| List I   | Union list      | Central government         | <ul style="list-style-type: none"> <li>• Defence</li> <li>• Foreign affairs</li> <li>• Citizenship</li> <li>• Ports, reserve banks, institutes of national importance</li> </ul> |
| List II  | State list      | State governments          | <ul style="list-style-type: none"> <li>• Public health, sanitation, hospitals</li> <li>• Fisheries, agriculture, industries</li> </ul>                                           |
| List III | Concurrent list | Central + state government | <ul style="list-style-type: none"> <li>• <b>medical profession, medical education</b></li> <li>• vital statistics</li> <li>• Trade commerce, civil procedure</li> </ul>          |

## International Health Agencies

00:15:02

----- Active space -----

### World Health Organization :

Established on 7<sup>th</sup> of April, 1948 (World health day).

Headquarters : Geneva, Switzerland.

Functions :

- Promote health.
- Prevention of disease.
- Promote family planning & research.

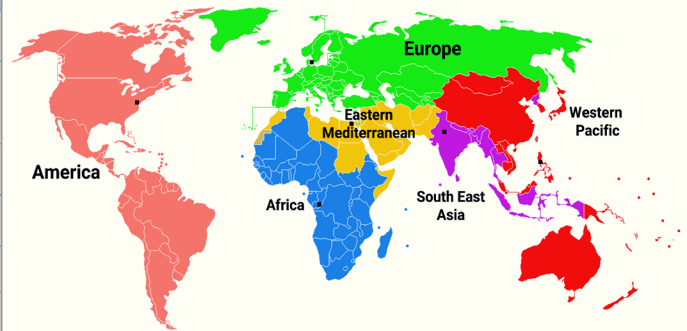
member states : 194.

Executive board :

- 34 members (1/3<sup>rd</sup> renewed annually).
- Term : 3 years.
- Functions : Planning, execution & implementation of international health policies.

WHO regions 🌐 :

| Region                | Notable countries  | Headquarters        |
|-----------------------|--------------------|---------------------|
| Southeast Asia        | India              | New Delhi, India    |
| Eastern mediterranean | Pakistan           | Alexandria, Egypt   |
| Western Pacific       | China, Philippines | Manila, Philippines |
| Africa                | -                  | Brazzaville, Congo  |
| America               | -                  | Washington DC, USA  |
| Europe                | Russia             | Copenhagen, Denmark |



### Other Organizations 🌐 :

| Organization                                                                                    | Assistance in India                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UNICEF (United Nations International Children's Emergency Fund)<br>Headquarters : New York, USA | <ul style="list-style-type: none"> <li>• MCH programmes</li> <li>• UIP, ICDS, IMNCI &amp; child health</li> <li>• Slums, urban child health, rural health</li> </ul>                                                                      |
| USAID                                                                                           | <ul style="list-style-type: none"> <li>• Family planning</li> <li>• TB programme</li> </ul>                                                                                                                                               |
| Ford Foundation                                                                                 | <ul style="list-style-type: none"> <li>• Established national institute of health &amp; family welfare (NIHFW)</li> <li>• Rural health</li> <li>• Toilets &amp; sanitation (RCA latrine)</li> <li>• Drainage system in Kolkata</li> </ul> |
| Rockefeller Foundation                                                                          | <ul style="list-style-type: none"> <li>• Established all India institute of hygiene &amp; public health, Kolkata</li> <li>• Rural health</li> <li>• Toilets &amp; sanitation</li> </ul>                                                   |
| SIDA (Swedish international development agency)                                                 | TB programme                                                                                                                                                                                                                              |
| DANIDA (Danish international development agency)                                                | Blindness programme                                                                                                                                                                                                                       |
| Colombo plan                                                                                    | <ul style="list-style-type: none"> <li>• AIIMS (Funding grant : New Zealand)</li> <li>• Cobalt therapy unit from Canada</li> </ul>                                                                                                        |

----- Active space -----

Notifiable Diseases for Outbreaks :

- Yellow fever.
- Plague.
- Influenza.
- Polio.
- COVID/SARS.

Newer Policies by GOI

00:20:14

Ayushman Bharat Scheme (2018) :

- Implementation :
    - Date : 23<sup>rd</sup> September 2018.
    - Name : Pradhan mantri Jan Arogya Yojana (PM-JAY).
  - upto 5 lakh health insurance/family.
  - 50 crore population (10 crore families)
  - No limit on members, age, type of illness
- } World's largest health insurance scheme.

Pm vay vandana yojana (2024) :

- upto 5 lakh health insurance.
- For senior citizens : Age > 70 yrs.

Awards & Incentives 🌐 :

| Scheme                                      |          | Purpose                                                     |
|---------------------------------------------|----------|-------------------------------------------------------------|
| National quality assurance programme (NQAP) | SaQushal | Safety & quality-self assessment tool for health facilities |
|                                             | musQan   | Quality child friendly health services (< 12 yrs)           |
| Kayakalp                                    |          | Sanitation & cleanliness in hospitals                       |
| Nirmal gram                                 |          | Zero open defecation & providing toilets in the village     |
| Swachh bharat                               |          | Cleanliness, sanitation & solid waste management            |

Schemes for Rural Health :

| Scheme                                    | Purpose                                          |
|-------------------------------------------|--------------------------------------------------|
| Ujala                                     | Affordable LED bulbs for all                     |
| Ujjwala yojana (under petroleum ministry) | Safer fuels (LPG cylinders)                      |
| Ujjawala                                  | Prevention & rescue of human trafficking victims |



Mission Shakti :

----- Active space -----

Under ministry of women & child development.

Sambal : For women's security.

- Nari adalat.
- Women helplines.
- One stop centres.
- Beti bachao scheme.

Samarthya : For women's **empowerment**.

- Pmmvy.
- Ujjawala.
- Swadhar greh.

Indradhanush Scheme :

7 coloured bedsheets for 7 days in ESI hospitals.

Miscellaneous Topics

00:25:20

Essential medicines :

- monitored by CDSCO (Central drugs standard control organization).
- Head office : New Delhi (6 zonal offices).

|                               | Features                                                                                                                                                  |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counterfeit medicine          | made with intention to deceive                                                                                                                            |
| Sub-standard medicine         | <ul style="list-style-type: none"> <li>• AKA <b>out of specification medicines</b>.</li> <li>• Authorized, but not meet the quality standards.</li> </ul> |
| Falsified medical products    | Deliberately <b>misrepresent</b> identity/source of composition.                                                                                          |
| Unregistered medical products | Not evaluated/approved.                                                                                                                                   |

AMRIT vs. AMRUT Scheme :

|         | AMRIT                                                             | AMRUT                                                      |
|---------|-------------------------------------------------------------------|------------------------------------------------------------|
| Purpose | For affordable medicines & reliable <b>implants</b> for treatment | Atal mission of rejuvenation & <b>urban transformation</b> |

Jai Vigyan :

To promote understanding of **rheumatic fever** :

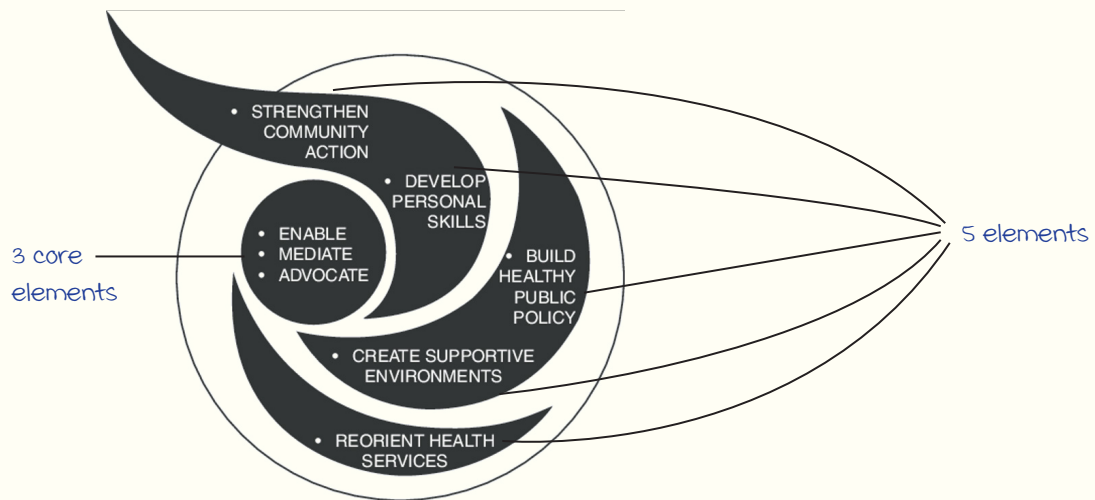
- Prevalence of RF.
- Pathology of RHD & RF.
- Prevalence of streptococcal infection.
- vaccine development for RHD/RF.

Health budget 2023 - 2025 🌐 :

- National programme for **elimination of sickle cell anemia**.  
- Purpose : Screening in **tribal areas**.
- Dedicated multidisciplinary courses for **medical devices**.
- Competency based medical education (**CBME**).
- GDP % spending increased to 2.5%.

----- Active space -----

## Ottawa Charter for Public Health 🌐 :



## Sustainable Development Goals (2015 – 2030) 🌐 :

- Formulated by United Nations.
- 17 goals :
  - **Goal 3** : Health related
  - Targets of goal 3 by 2030 :
    - a. 3.1 : **mmr** < 70
    - b. 3.2 : **NMR** < 12/1000 live births & **usmr** < 25/1000 live births.
    - c. 3.3 : End epidemics of **AIDS, TB, malaria** & neglected **tropical diseases**.
    - d. 3.4 : Reduce 1/3<sup>rd</sup> of **premature deaths** from NCD.

## ICD-II Classification :

- 3 volumes (Numbered in arabic).
- **4 categories** with **2 subcategories**.
- 26 chapters.

### New chapters/updates in ICD-II :

- **Sleep wake** disorders
- **Immune** mediated disorders
- Sexual health
- Traditional medicine
- HIV : Chronic condition
- **Gaming & hoarding** as separate conditions
- updated criteria for PTSD

### Linked classifications :

- Oncology.
- External causes of injury.
- Functional, disability & health.
- Primary care.